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Quantum Soundness of the Classical Low Individual Degree Test

MIPStarRE Project

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Chapter 1

Overview

Theorem 1.1 (Raz–Safra)**.** *Suppose Provers* A *and* B *pass the* $k = 2$ *surface-versus-point low-degree test with probability* $1 - \varepsilon$*. Then there exists a degree-*d *polynomial* $g \colon \mathbb{F}_q^m \to \mathbb{F}_q$ *such that*

$$\mathbb{P}_{u \sim \mathbb{F}_q^m}[g(u) = a] \geq 1 - \varepsilon - \mathrm{poly}(m) \cdot \mathrm{poly}(d/q).$$

Here a *denotes the prover's point answer.*

Remark 1.2. The current Lean declaration is a wrapper around the quoted classical Raz–Safra theorem: it takes the specialized auxiliary hypothesis at the ambient point-answer function, error parameter, and chosen slack bound, whose premise is the explicit pass predicate . The surface test itself is modeled using parameterized genuine 2-dimensional surfaces, i.e. ordered frames with linearly independent directions, together with a hidden parameter specifying the queried point on the surface, so the surface prover does not see that point directly. Its conclusion currently returns the repository-wide Lean type , which encodes individual degree at most d rather than a separate total-degree-at-most-d object. Accordingly, this result is not marked as formalized.

Proof. See Raz–Safra [RS97]. □

Theorem 1.3 (Polishchuk–Spielman)**.** *Suppose Provers* A *and* B *pass the low individual degree test with probability* $1 - \varepsilon$*. Then there exists a polynomial* $g \colon \mathbb{F}_q^m \to \mathbb{F}_q$ *with individual degree* d *such that*

$$\mathbb{P}_{u \sim \mathbb{F}_q^m}[g(u) = a] \geq 1 - \mathrm{poly}(m) \cdot \big(\mathrm{poly}(\varepsilon) + \mathrm{poly}(d/q)\big).$$

Here a *denotes the prover's point answer.*

Remark 1.4. The current Lean declaration keeps the external Polishchuk–Spielman dependency explicit through a caller-chosen slack-bound parameter together with the auxiliary hypothesis . The repository no longer fixes any convenience placeholder formula for the schematic term $\mathrm{poly}(m) \cdot (\mathrm{poly}(\varepsilon) + \mathrm{poly}(d/q))$ appearing in the overview statement above: any concrete rate must instead be supplied by the specialized external hypothesis. Accordingly, this result is not marked as formalized.

Proof. See Polishchuk–Spielman [PS94]. □

Remark 1.5 (A degree-$(d+1)$ strategy that passes with probability $1 - \frac{1}{m}$). Fix the polynomial $h(x_1, \dots, x_m) = x_1^{d+1}$. Consider the following classical strategy for the degree-d low individual degree test. On a point query $u \in \mathbb{F}_q^m$, Prover A answers $a = h(u)$. On an axis-parallel line query $\ell = \{u + xe_i : x \in \mathbb{F}_q\}$, Prover B returns the restriction $h|_\ell$ when $i \neq 1$, since h is then constant along ℓ, and returns the zero polynomial when $i = 1$.

The verifier certainly accepts whenever $i \neq 1$, which happens with probability $1 - \frac{1}{m}$. Thus the strategy passes the degree-d test with probability at least $1-\varepsilon$ for $\varepsilon = \frac{1}{m}$. However, Prover A is answering according to a polynomial whose individual degree in the first variable is $d+1$. By Schwartz–Zippel, any polynomial g of individual degree at most d agrees with h on at most a $\frac{d+1}{q}$ fraction of all inputs, so

$$\mathbb{P}_{u \sim \mathbb{F}_q^m}[g(u) = a] \leq 1 - m\varepsilon + \frac{d+1}{q}.$$

This shows that the m-dependent slack in Theorems 1.3 and 1.6 is unavoidable up to polynomial factors.

Theorem 1.6 (Main theorem, informal)**.** *If $d > 0$ and a two-prover projective strategy passes the (m, q, d) low individual degree test with probability $1 - \varepsilon$, then there are global polynomial measurements whose evaluations agree with the point answers except with error* $\mathrm{poly}(m)(\mathrm{poly}(\varepsilon) + \mathrm{poly}(d/q))$.

Proof. This is the informal form of Theorem 2.12. ☐

Chapter 2

The low individual degree test

2.1 The game and its strategies

Definition 2.1 (Roles)**.** A role is an element of $\{\mathrm{A}, \mathrm{B}\}$. For a role $r \in \{\mathrm{A}, \mathrm{B}\}$, write $\bar{r}$ for the other element of $\{\mathrm{A}, \mathrm{B}\}$.

Definition 2.2 (Low individual degree test)**.** Fix integers $m, d \geq 0$ and a prime power q. The (m, q, d) low individual degree test has three equally likely subtests. In the axis-parallel lines test, choose a uniformly random role $r \in \{\mathrm{A}, \mathrm{B}\}$, a uniformly random point $u \in \mathbb{F}_q^m$, and a uniformly random coordinate $i \in \{1, \dots, m\}$. Let $\ell = \{u + te_i : t \in \mathbb{F}_q\}$. Player r receives ℓ and returns a degree-d univariate polynomial $f \colon \ell \to \mathbb{F}_q$; Player $\bar{r}$ receives u and returns a field element $a \in \mathbb{F}_q$. The verifier accepts when $f(u) = a$. In the self-consistency test, both provers receive the same uniformly random point $u \in \mathbb{F}_q^m$ and must return the same field element. In the diagonal lines test, choose a uniformly random role $r \in \{\mathrm{A}, \mathrm{B}\}$, a uniformly random point $u \in \mathbb{F}_q^m$, a uniformly random index $i \in \{1, \dots, m\}$, and a uniformly random direction vector $v \in \mathbb{F}_q^m$ whose last $m - i$ coordinates are 0. Let $\ell = \{u + tv : t \in \mathbb{F}_q\}$. Player r receives ℓ and returns a degree-md univariate polynomial $f \colon \ell \to \mathbb{F}_q$; Player $\bar{r}$ receives u and returns a field element $a \in \mathbb{F}_q$. The verifier accepts when $f(u) = a$.

Lemma 2.3 (Branch-average form of the classical test)**.** *For every classical strategy, the acceptance probability of the low individual degree test is*

$$\frac{1}{3}\left(\frac{P_{\mathrm{axis}}^{\mathrm{line,point}} + P_{\mathrm{axis}}^{\mathrm{point,line}}}{2} + P_{\mathrm{self}} + \frac{P_{\mathrm{diag}}^{\mathrm{line,point}} + P_{\mathrm{diag}}^{\mathrm{point,line}}}{2}\right),$$

where the five terms are the acceptance probabilities of the two axis-parallel role orderings, the self-consistency branch, and the two diagonal role orderings.

Definition 2.4 (Symmetric projective strategy)**.** A symmetric projective strategy for the (m, q, d) low individual degree test consists of a permutation-invariant bipartite state $|\psi\rangle \in \mathcal{H} \otimes \mathcal{H}$, a projective point measurement $A^u = \{A_a^u\}_{a \in \mathbb{F}_q}$ for each $u \in \mathbb{F}_q^m$, a projective axis-parallel line measurement $B^\ell = \{B_f^\ell\}$ for each axis-parallel line ℓ, and a projective diagonal-line measurement $L^\ell = \{L_f^\ell\}$ for each line ℓ.

Definition 2.5 (General projective strategy)**.** A general projective strategy for the (m, q, d) low individual degree test consists of a bipartite state $|\psi\rangle \in \mathcal{H}_{\mathrm{A}} \otimes \mathcal{H}_{\mathrm{B}}$ together with point, axis-parallel line, and diagonal-line projective measurements for each prover separately. We write these measurements as $A^{\mathrm{A}}, B^{\mathrm{A}}, L^{\mathrm{A}}$ on the first prover and $A^{\mathrm{B}}, B^{\mathrm{B}}, L^{\mathrm{B}}$ on the second.

Lemma 2.6 (Bipartite strategy block identities)**.** *The block operators obtained from a bipartite projective strategy are positive semidefinite, and the role-indexed blocks multiply according to the Kronecker delta on roles.*

Remark 2.7. Throughout this work, the term strategy refers to a projective strategy, and the term symmetric strategy refers to a symmetric projective strategy.

Definition 2.8 (Good strategy)**.** A strategy is $(\varepsilon, \delta, \gamma)$-good if it passes the axis-parallel lines test with probability at least $1 - \varepsilon$, the self-consistency test with probability at least $1 - \delta$, and the diagonal lines test with probability at least $1 - \gamma$.

Proposition 2.9 (Good-strategy characterization)**.** *Using notation which will be introduced in Chapter 3 below, a symmetric strategy is $(\varepsilon, \delta, \gamma)$-good if and only if it satisfies the following three conditions. For ℓ and u as in the axis-parallel lines test,*

$$A_a^u \otimes I \simeq_\varepsilon I \otimes B^\ell_{[f(u)=a]}, \quad \textit{and} \quad A_a^u \otimes I \simeq_\delta I \otimes A_a^u.$$

And for ℓ and u as in the diagonal lines test,

$$A_a^u \otimes I \simeq_\gamma I \otimes L^\ell_{[f(u)=a]}.$$

Definition 2.10 (Last-direction line notation)**.** For $u \in \mathbb{F}_q^m$, the last-direction axis-parallel line in $\mathbb{F}_q^{m+1}$ is

$$\ell_u = \{(u, x) \mid x \in \mathbb{F}_q\}.$$

If (ψ, A, B, L) is a symmetric strategy for the $(m+1, q, d)$ low individual degree test, we write B_f^u for the operator $B_f^{\ell_u}$. For a function $f\colon \ell_u \to \mathbb{F}_q$, we also write $f(x)$ for $f(u, x)$.

Definition 2.11 (Restricted diagonal lines test)**.** Consider the (m, q, d) low individual degree test. For $j \in \{1, \dots, m\}$, the j-restricted diagonal lines test is the diagonal lines test conditioned on $i = j$. In particular, the m-restricted diagonal lines test is the diagonal lines test in which the sampled line is a uniformly random line in $\mathbb{F}_q^m$.

2.2 Formal soundness target

Theorem 2.12 (Quantum soundness of the low individual degree test)**.** *Let $(\psi, A^{\mathrm{A}}, B^{\mathrm{A}}, L^{\mathrm{A}}, A^{\mathrm{B}}, B^{\mathrm{B}}, L^{\mathrm{B}})$ be a projective strategy that passes the (m, q, d) low individual degree test with probability at least $1 - \varepsilon$. Assume $d > 0$. Let $k > 0$ with $k \geq 400\,md$ and set*

$$\nu = 100000k^2m^4 \left(\varepsilon^{1/40000} + (d/q)^{1/40000} + e^{-k/(2560000m^2)}\right).$$

Then there exist projective measurements $G^{\mathrm{A}}, G^{\mathrm{B}} \in \mathrm{PolyMeas}(m, q, d)$ with the following properties:

1. *on average over $u \sim \mathbb{F}_q^m$,*

$$A_a^{\mathrm{A},u} \otimes I \simeq_\nu I \otimes G^{\mathrm{B}}_{[g(u)=a]}, \qquad I \otimes A_a^{\mathrm{B},u} \simeq_\nu G^{\mathrm{A}}_{[g(u)=a]} \otimes I;$$

2.

$$G_g^{\mathrm{A}} \otimes I \simeq_\nu I \otimes G_g^{\mathrm{B}}.$$

Remark 2.13 (Conditional form of the soundness theorem). The conditional theorem currently assumes a repaired bridge from the role-register residual to the final projective-completion hypotheses. This is not a hypothesis of Theorem 2.12; it is the remaining step needed to derive the paper theorem from the pass condition.

Remark 2.14 (k-bound boundary for the public theorem). The paper statement uses the weaker hypothesis $k \geq md$, but the proof invokes the Section 6 pasting theorem with side condition $k \geq 400\,md$. This side condition is not implied by $k \geq md$. The Lean and blueprint public statement therefore records the stronger large-k assumption required by the pasting step rather than hiding the side-condition gap. The helper `mainFormal_trivial_witness` remains only for already-saturated branches where $\nu \geq 1$ makes the three $\simeq_\nu$ conclusions vacuous; it is not a replacement for a proof in the interval $md \leq k < 400\,md$.

Chapter 3

Preliminaries

3.1 Polynomials and measurements

For complex numbers $\alpha, \beta \in \mathbb{C}$, write $\alpha \approx_\varepsilon \beta$ when $|\alpha - \beta| \leq \varepsilon$. In particular,

$$\text{if } \alpha \approx_\varepsilon \beta \text{ and } \beta \approx_\delta \gamma, \text{ then } \alpha \approx_{\varepsilon+\delta} \gamma. \tag{3.1}$$

Fix a prime power $q = p^t$. Write $\omega = e^{2\pi i/p}$, and let $\mathrm{tr} : \mathbb{F}_q \to \mathbb{F}_p$ denote the finite-field trace.

Definition 3.1 (Finite field trace)**.** The finite-field trace is the map $\mathrm{tr} : \mathbb{F}_q \to \mathbb{F}_p$ defined by

$$\mathrm{tr}[x] = \sum_{\ell=0}^{t-1} x^{p^\ell}.$$

Lemma 3.2. *Let $a \in \mathbb{F}_q$. Then*

$$\mathbb{E}_{x \sim \mathbb{F}_q} \omega^{\mathrm{tr}[x \cdot a]} = \begin{cases} 1 & \text{if } a = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If $a = 0$, then $\mathrm{tr}[x \cdot a] = 0$ for every x, so the average is 1. If $a \neq 0$, choose $y \in \mathbb{F}_q$ such that $\mathrm{tr}[a \cdot y] \neq 0$, and set

$$C = \mathbb{E}_{x \sim \mathbb{F}_q} \omega^{\mathrm{tr}[x \cdot a]}.$$

Translation invariance of the uniform distribution gives

$$C = \mathbb{E}_{x \sim \mathbb{F}_q} \omega^{\mathrm{tr}[(x+y) \cdot a]} = C \cdot \omega^{\mathrm{tr}[y \cdot a]}.$$

Since $\omega^{\mathrm{tr}[y \cdot a]} \neq 1$, it follows that $C = 0$. □

Lemma 3.3. *Let $v \in \mathbb{F}_q^m$. Then*

$$\mathbb{E}_{u \sim \mathbb{F}_q^m} \omega^{\mathrm{tr}[u \cdot v]} = \begin{cases} 1 & \text{if } v = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. By linearity of the trace and independence of the coordinates,

$$\mathbb{E}_{u \sim \mathbb{F}_q^m} \omega^{\mathrm{tr}[u \cdot v]} = \prod_{i=1}^{m} \mathbb{E}_{u_i \sim \mathbb{F}_q} \omega^{\mathrm{tr}[u_i \cdot v_i]}.$$

Proposition 3.2 shows that the i-th factor is 1 when $v_i = 0$ and 0 otherwise, so the product is 1 exactly when $v = 0$. □

Definition 3.4 (Low individual degree polynomials)**.** Let $\mathcal{P}(m, q, d)$ be the set of polynomials $g \in \mathbb{F}_q[x_1, \dots, x_m]$ whose degree in each variable is at most d, viewed as functions $\mathbb{F}_q^m \to \mathbb{F}_q$ via evaluation. (Over finite fields, distinct low-degree polynomials can induce the same function; we identify elements of $\mathcal{P}(m, q, d)$ with their polynomial representatives, not their functional equivalence classes.)

Lemma 3.5 (Monotonicity of individual-degree classes)**.** *Here, "individual degree d" means degree at most d in each coordinate. In particular,*

$$\mathcal{P}(m, q, d) \subseteq \mathcal{P}(m, q, d + 1).$$

Lemma 3.6 (Schwartz–Zippel lemma [Sch80, Zip79])**.** *Let $g, h : \mathbb{F}_q^m \to \mathbb{F}_q$ be two distinct polynomials of total degree d. Then*

$$\mathbb{P}_{x\sim\mathbb{F}_q^m}[g(x) = h(x)] \leq \frac{d}{q}.$$

Proof. Apply the Schwartz–Zippel lemma to the nonzero polynomial $g-h$, which has total degree at most d. ☐

Lemma 3.7 (Schwartz–Zippel for individual degree)**.** *If $g, h \in \mathcal{P}(m, q, d)$ are distinct, then*

$$\mathbb{P}_{u\sim\mathbb{F}_q^m}[g(u) = h(u)] \leq \frac{md}{q}.$$

Proof. The difference $g-h$ is a nonzero polynomial of total degree at most md. Apply Lemma 3.6. ☐

Lemma 3.8 (Polynomial agreement bound on coded points)**.** *Let g, g' : $\mathrm{Polynomial\,params}$ be two distinct full polynomial outcomes with individual degrees at most d. Then*

$$\mathbb{E}_{u\sim\mathrm{Point\,params}}[\mathbf{1}[g(u) = g'(u)]] \leq \frac{md}{q}.$$

This packages the md/q loss term used in the `mainFormal` self-consistency cascade (`inductive_step.tex`, lines 119–133) and in `comMain` (`commutativity-G.tex`).

Proof. Transport along the equivalence between coded $\mathbb{F}_q$ points and the underlying scalar function space, then apply Lemma 3.7. ☐

Lemma 3.9 (Axis-line polynomial agreement bound)**.** *Let f, h : $\mathrm{AxisLinePolynomial\,params}$ be two line-polynomial outcomes with distinct underlying polynomials. Then*

$$\mathbb{E}_{t\sim\mathbb{F}_q}[\mathbf{1}[f(t) = h(t)]] \leq \frac{md}{q}.$$

The native univariate root-counting bound is d/q; the statement deliberately pads this to the paper's ambient md/q loss for Section 6.1.

Proof. Reindex coded line parameters to the scalar field, count the roots of the nonzero univariate polynomial $f - h$, and use $m \geq 1$ to pad d/q to md/q. ☐

Definition 3.10 (Measurements and submeasurements)**.** Let $\mathcal{H}$ be a Hilbert space and let $\mathcal{A}$ be a set of outcomes. A submeasurement on $\mathcal{A}$ is a family $A = \{A_a\}_{a\in\mathcal{A}}$ of Hermitian positive semidefinite operators on $\mathcal{H}$ such that $\sum_a A_a \leq I$. It is a measurement when $\sum_a A_a = I$, and it is projective when each A_a is an idempotent projection. (In the Lean formalization, outcome sets are assumed finite throughout.)

Definition 3.11 (Low-degree polynomial measurements)**.** We write $\mathrm{PolySub}(m, q, d)$ for the submeasurements indexed by $\mathcal{P}(m, q, d)$, and $\mathrm{PolyMeas}(m, q, d)$ for the corresponding measurements.

Definition 3.12 (Post-processing)**.** Let $A = \{A_a\}_{a\in\mathcal{A}}$ be a family of operators and let $f\colon \mathcal{A} \to \mathcal{B}$. The post-processed family $A_{[f(a)=b]}$ is defined by

$$A_{[f(a)=b]} = \sum_{a\,:\,f(a)=b} A_a.$$

Proposition 3.13 (Post-processing preserves measurements)**.** *Let $A = \{A_a\}_{a\in\mathcal{A}}$ be a family of operators and let $f\colon \mathcal{A} \to \mathcal{B}$. Then*

$$\sum_a A_a = \sum_b A_{[f(a)=b]}.$$

Consequently, if $\{A_a\}$ is a submeasurement, respectively a measurement, then $\{A_{[f(a)=b]}\}$ is again a submeasurement, respectively a measurement.

Proof. Each $a \in \mathcal{A}$ contributes exactly once to the right-hand side, namely in the summand indexed by $b = f(a)$. $\square$

Remark 3.14 (Post-processing notation)*.* In the notation $A_{[f(a)=b]}$, the measurement outcome is the variable to which the function is applied. For example, if $G = \{G_g\} \in \mathrm{PolySub}(m, q, d)$ and $u \in \mathbb{F}_q^m$, then

$$G_{[g(u)=a]} = \sum_{g\,:\,g(u)=a} G_g.$$

Here g is the measurement outcome, and u is the evaluation point.

Definition 3.15 (Completion of a submeasurement)**.** Let $A = \{A_a^x\}_{a\in\mathcal{A}}$ be a submeasurement. Its completion is the measurement $\widehat{A}$ with outcome set $\widehat{\mathcal{A}} = \mathcal{A} \cup \{\perp\}$ given by $\widehat{A}_a^x = A_a^x$ for $a \in \mathcal{A}$ and $\widehat{A}_\perp^x = I - \sum_a A_a^x$.

3.2 Consistency and state-dependent distance

Definition 3.16 (Consistency)**.** Let $|\psi\rangle \in \mathcal{H}_\mathrm{A} \otimes \mathcal{H}_\mathrm{B}$. Let $A = \{A_a^x\}$ and $B = \{B_a^x\}$ be submeasurements with the same answer set, and let $\mathcal{D}$ be a distribution on the question set. We write

$$A_a^x \otimes I \simeq_\delta I \otimes B_a^x$$

when

$$\mathbb{E}_{x\sim\mathcal{D}} \sum_{a\neq b} \langle\psi| A_a^x \otimes B_b^x |\psi\rangle \le \delta. \tag{3.2}$$

Lemma 3.17 (Characterization of a good symmetric strategy)**.** *A symmetric projective strategy (ψ, A, B, L) is $(\varepsilon, \delta, \gamma)$-good if and only if the following hold on the relevant test distributions:*

$$A_a^u \otimes I \simeq_\varepsilon I \otimes B_{[f(u)=a]}^\ell, \qquad A_a^u \otimes I \simeq_\delta I \otimes A_a^u, \qquad A_a^u \otimes I \simeq_\gamma I \otimes L_{[f(u)=a]}^\ell.$$

Proof. Each subtest accepts exactly when the two measurement outcomes agree. Because the relevant families are measurements, Lemma 3.18 rewrites those acceptance probabilities as consistency bounds. $\square$

Lemma 3.18 (Consistency for measurements)**.** *If $A = \{A_a^x\}$ and $B = \{B_a^x\}$ are measurements, then*

$$A_a^x \otimes I \simeq_\delta I \otimes B_a^x \qquad \textit{if and only if} \qquad \mathbb{E}_x \sum_a \langle\psi| A_a^x \otimes B_a^x |\psi\rangle \geq 1 - \delta.$$

Proof. Expand the off-diagonal sum by writing $\sum_{a \neq b} A_a^x \otimes B_b^x = \sum_b (I - A_b^x) \otimes B_b^x$. □

Definition 3.19 (State-dependent distance)**.** Let $|\psi\rangle \in \mathcal{H}$, let $A = \{A_a^x\}$ and $B = \{B_a^x\}$ be families of operators on $\mathcal{H}$, and let $\mathcal{D}$ be a distribution on the question set. We write

$$A_a^x \approx_\delta B_a^x$$

when

$$\mathbb{E}_{x \sim \mathcal{D}} \sum_a \|(A_a^x - B_a^x)|\psi\rangle\|^2 \leq \delta.$$

In particular, if $A_a^x \otimes I \simeq_\varepsilon I \otimes C_a^x$, the question is what hypothesis on A and B allows one to conclude that

$$B_a^x \otimes I \simeq_\varepsilon I \otimes C_a^x. \tag{3.3}$$

Lemma 3.20 (Transferring consistency through state-dependent distance)**.** *Let $A = \{A_a^x\}$ and $B = \{B_a^x\}$ be measurements and let $C = \{C_a^x\}$ be a submeasurement. If $A_a^x \otimes I \simeq_\delta I \otimes C_a^x$ and $A_a^x \otimes I \approx_\varepsilon B_a^x \otimes I$, then*

$$B_a^x \otimes I \simeq_{\delta + \sqrt{\varepsilon}} I \otimes C_a^x.$$

Proof. Rewrite the inconsistency with C as total mass minus diagonal overlap. The total mass is unchanged because A and B are measurements, and the diagonal overlap changes by at most $\sqrt{\varepsilon}$ by Cauchy–Schwarz. □

Lemma 3.21 (Consistency implies state-dependent distance for measurements)**.** *If A and B are measurements and $A_a^x \otimes I \simeq_\delta I \otimes B_a^x$, then*

$$A_a^x \otimes I \approx_{2\delta} I \otimes B_a^x.$$

If both measurements are projective, the converse also holds: if $A_a^x \otimes I \approx_{2\delta} I \otimes B_a^x$ then $A_a^x \otimes I \simeq_\delta I \otimes B_a^x$.

Proof. Expand the squared norm of $(A_a^x \otimes I - I \otimes B_a^x)|\psi\rangle$ and use the diagonal-overlap formula from Lemma 3.18; projectivity turns the diagonal inequality into an equality, giving the converse. □

Remark 3.22. The implication above does not extend to submeasurements. For example, if $A_a^x = 0$ for all a, then $A_a^x \otimes I \simeq_0 I \otimes B_a^x$, but

$$\mathbb{E}_x \sum_a \|(A_a^x \otimes I - I \otimes B_a^x)|\psi\rangle\|^2 = \mathbb{E}_x \sum_a \|(I \otimes B_a^x)|\psi\rangle\|^2,$$

which is nonzero unless $(I \otimes B_a^x)|\psi\rangle = 0$ for all x and a.

3.3 Consistency from state-dependent distance

Theorem 3.23 (Closeness of inner products)**.** *Let $\{A_a^x\}$, $\{B_a^x\}$, and $\{C_{a,b}^x\}$ be matrices. Suppose that $A_a^x \approx_\gamma B_a^x$ and that for all x,*

$$\sum_a \Big(\sum_b C_{a,b}^x\Big)\Big(\sum_b C_{a,b}^x\Big)^\dagger \le I.$$

Then

$$\mathbb{E}_x \sum_{a,b} \langle\psi|C_{a,b}^x A_a^x|\psi\rangle \approx_{\sqrt{\gamma}} \mathbb{E}_x \sum_{a,b} \langle\psi|C_{a,b}^x B_a^x|\psi\rangle. \tag{3.4}$$

Similarly, suppose that $(A_a^x)^\dagger \approx_\gamma (B_a^x)^\dagger$ and that for all x,

$$\sum_a \Big(\sum_b C_{a,b}^x\Big)^\dagger\Big(\sum_b C_{a,b}^x\Big) \le I.$$

Then

$$\mathbb{E}_x \sum_{a,b} \langle\psi|A_a^x C_{a,b}^x|\psi\rangle \approx_{\sqrt{\gamma}} \mathbb{E}_x \sum_{a,b} \langle\psi|B_a^x C_{a,b}^x|\psi\rangle. \tag{3.5}$$

Proof. To prove (3.4), write the difference as

$$\mathbb{E}_x \sum_a \langle\psi|\Big(\sum_b C_{a,b}^x\Big)(A_a^x - B_a^x)|\psi\rangle.$$

Cauchy–Schwarz bounds its magnitude by

$$\Big(\mathbb{E}_x \sum_a \langle\psi|\Big(\sum_b C_{a,b}^x\Big)\Big(\sum_b C_{a,b}^x\Big)^\dagger|\psi\rangle\Big)^{1/2}\Big(\mathbb{E}_x \sum_a \langle\psi|(A_a^x - B_a^x)^\dagger(A_a^x - B_a^x)|\psi\rangle\Big)^{1/2},$$

and the two hypotheses bound these factors by 1 and $\sqrt{\gamma}$ respectively. For (3.5), rewrite the difference as

$$\mathbb{E}_x \sum_{a,b} \langle\psi|(C_{a,b}^x)^\dagger((A_a^x)^\dagger - (B_a^x)^\dagger)|\psi\rangle,$$

and apply (3.4) to the adjoint families. □

Theorem 3.24. *Let $A = \{A_a^x\}$, $B = \{B_a^x\}$, and $C = \{C_a^x\}$ be submeasurements such that $A_a^x \approx_\delta B_a^x$. Then*

$$\mathbb{E}_x \sum_a \langle\psi|A_a^x C_a^x|\psi\rangle \approx_{\sqrt{\delta}} \mathbb{E}_x \sum_a \langle\psi|B_a^x C_a^x|\psi\rangle.$$

Proof. The difference has magnitude

$$\Big|\mathbb{E}_x \sum_a \langle\psi|(A_a^x - B_a^x)C_a^x|\psi\rangle\Big|.$$

By Cauchy–Schwarz this is at most

$$\Big(\mathbb{E}_x \sum_a \langle\psi|(A_a^x - B_a^x)^2|\psi\rangle\Big)^{1/2}\Big(\mathbb{E}_x \sum_a \langle\psi|(C_a^x)^2|\psi\rangle\Big)^{1/2}.$$

The first factor is at most $\sqrt{\delta}$ by hypothesis, and the second is at most 1 because C is a submeasurement. □

Theorem 3.25. *Let $\{A_a^x\}$, $\{B_a^x\}$, and $\{C_{a,b}^x\}$ be matrices. Suppose that $A_a^x \approx_\delta B_a^x$ and that for all x and a,*

$$\sum_b (C_{a,b}^x)^\dagger (C_{a,b}^x) \leq I.$$

Then

$$C_{a,b}^x A_a^x \approx_\delta C_{a,b}^x B_a^x.$$

Proof. The error term is

$$\mathbb{E}_x \sum_{a,b} \langle\psi|(A_a^x - B_a^x)^\dagger (C_{a,b}^x)^\dagger C_{a,b}^x (A_a^x - B_a^x)|\psi\rangle.$$

Using $\sum_b (C_{a,b}^x)^\dagger (C_{a,b}^x) \leq I$, this is bounded by

$$\mathbb{E}_x \sum_a \langle\psi|(A_a^x - B_a^x)^\dagger (A_a^x - B_a^x)|\psi\rangle \leq \delta.$$

$\square$

Lemma 3.26 (Triangle inequality for vectors squared)**.** *For vectors $|\psi_1\rangle, \dots, |\psi_k\rangle$,*

$$\left\||\psi_1\rangle + \dots + |\psi_k\rangle\right\|^2 \leq k(\||\psi_1\rangle\|^2 + \dots + \||\psi_k\rangle\|^2).$$

Proof. First,

$$(x_1 + \dots + x_k)^2 \leq k(x_1^2 + \dots + x_k^2) \tag{3.6}$$

for all real numbers $x_1, \dots, x_k$. Apply (3.6) to $x_i = \||\psi_i\rangle\|$ and combine it with the triangle inequality. $\square$

Lemma 3.27 (Triangle inequality for $\approx_\delta$)**.** *Suppose $(A_i)_a^x \approx_{\delta_i} (A_{i+1})_a^x$ for $i = 1, \dots, k$. Then*

$$(A_1)_a^x \approx_{k(\delta_1 + \dots + \delta_k)} (A_{k+1})_a^x.$$

Proof. Expand $(A_1 - A_{k+1})|\psi\rangle$ as a telescoping sum and apply Lemma 3.26. $\square$

Remark 3.28 (Lean finite-step triangle helpers). The general operator-family theorem above is complemented in Lean by reusable binary and three-step helper lemmas for operator expectations, ‘qSDD‘, and ‘SDDRel‘. The three-step ‘SDDRel‘ helper is the specialization used in the Step 6 projectivization chain in Theorem 2.12; this remark is only a cross-reference and makes no separate proof-level completeness claim.

Lemma 3.29 (Triangle inequality for $\simeq_\delta$)**.** *Suppose $A_a^x \otimes I \simeq_\varepsilon I \otimes B_a^x$, $C_a^x \otimes I \simeq_\delta I \otimes B_a^x$, and $C_a^x \otimes I \simeq_\gamma I \otimes D_a^x$, where all four families are measurements. Then*

$$A_a^x \otimes I \simeq_{\varepsilon + 2\sqrt{\delta+\gamma}} I \otimes D_a^x.$$

Proof. Convert the two consistencies involving C to state-dependent distance, compose them by the triangle inequality, and transfer the result back to consistency by Lemma 3.20. $\square$

Lemma 3.30 (Data processing for consistency)**.** *If $A = \{A_a^x\}$ and $B = \{B_a^x\}$ are measurements such that $A_a^x \otimes I \simeq_\delta I \otimes B_a^x$, and f is a map on the answer set, then*

$$A_{[f(a)=b]}^x \otimes I \simeq_\delta I \otimes B_{[f(a)=b]}^x.$$

Proof. Merging outcome classes can only decrease the off-diagonal mass. □

Lemma 3.31 (Consistency controls a submeasurement)**.** *Let $A = \{A_a^x\}$ be a submeasurement and $B = \{B_a^x\}$ a measurement such that $A_a^x \otimes I \simeq_\gamma I \otimes B_a^x$. Then*

$$A_a^x \otimes I \approx_\gamma A_a^x \otimes B_a^x \approx_\gamma A^x \otimes B_a^x, \tag{3.7}$$

where $A^x = \sum_a A_a^x$. In particular,

$$A_a^x \otimes I \approx_{4\gamma} A^x \otimes B_a^x.$$

Proof. For the first approximation in (3.7), bound

$$\mathbb{E}_x \sum_a \langle\psi|\big(A_a^x \otimes (I - B_a^x)\big)^2|\psi\rangle$$

by the inconsistency between A and B. The second approximation is identical, applied to

$$\mathbb{E}_x \sum_a \langle\psi|\big((A^x - A_a^x) \otimes B_a^x\big)^2|\psi\rangle.$$

Then compose the two bounds with Lemma 3.27. □

Lemma 3.32 (Switch-sandwich)**.** *Suppose $A = \{A_a^x\}$ is a projective submeasurement satisfying*

$$A_a^x \otimes I \approx_\delta I \otimes A_a^x. \tag{3.8}$$

Then for every operator B with $0 \le B \le I$,

$$\mathbb{E}_x \sum_a \langle\psi|A_a^x B A_a^x \otimes I|\psi\rangle \approx_{2\sqrt{\delta}} \mathbb{E}_x \sum_a \langle\psi|B \otimes A_a^x|\psi\rangle \approx_{\sqrt{\delta}} \mathbb{E}_x \sum_a \langle\psi|B A_a^x \otimes I|\psi\rangle. \tag{3.9}$$

Proof. We prove the first approximation in (3.9) in two steps. First,

$$\mathbb{E}_x \sum_a \langle\psi|A_a^x B A_a^x \otimes I|\psi\rangle \approx_{\sqrt{\delta}} \mathbb{E}_x \sum_a \langle\psi|A_a^x B \otimes A_a^x|\psi\rangle. \tag{3.10}$$

This is obtained by applying Cauchy–Schwarz to the difference and using $B^2 \le B \le I$ together with (3.8). For the second step, move the remaining left copy of A_a^x across the bipartition in the same way; projectivity then collapses the sandwich and yields the first approximation in (3.9). The second approximation in (3.9) is the same argument with the rightmost factor A_a^x omitted. □

3.4 Strong self-consistency

Lemma 3.33 (Completeness transfer to a projective approximation)**.** *Let $A = \{A_a^x\}$ be a submeasurement and $P = \{P_a^x\}$ a projective submeasurement such that $A_a^x \otimes I \approx_\varepsilon P_a^x \otimes I$. Then*

$$\langle\psi|A \otimes I|\psi\rangle \ge \langle\psi|P \otimes I|\psi\rangle - 2\sqrt{\varepsilon}.$$

Proof. Since P is projective,

$$\langle\psi|P \otimes I|\psi\rangle = \mathbb{E}_x \sum_a \langle\psi|(P_a^x)^2 \otimes I|\psi\rangle.$$

Apply Proposition 3.24 twice to replace $(P_a^x)^2$ first by $A_a^x P_a^x$ and then by $(A_a^x)^2$, and conclude with $(A_a^x)^2 \le A_a^x$. □

Definition 3.34 (Strong self-consistency)**.** Let $|\psi\rangle$ be permutation-invariant and let $A = \{A_a^x\}$ be a submeasurement. We say that A is δ-strongly self-consistent when

$$\mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes A_a^x|\psi\rangle \geq \langle\psi|A \otimes I|\psi\rangle - \delta.$$

Lemma 3.35 (Strong self-consistency implies ordinary self-consistency)**.** *If A is δ-strongly self-consistent, then $A_a^x \otimes I \simeq_\delta I \otimes A_a^x$. If A is a measurement, the converse also holds.*

Proof. Rewrite the off-diagonal mass as the difference between the total mass of A and the diagonal overlap in the strong self-consistency inequality. When A is a full measurement, the total mass is the identity, so this inequality becomes an equality; hence the ordinary self-consistency defect and the strong self-consistency defect coincide, giving the converse. □

Lemma 3.36 (Strong self-consistency and state-dependent distance)**.** *If A is δ-strongly self-consistent, then*

$$A_a^x \otimes I \approx_{2\delta} I \otimes A_a^x.$$

If A is projective, the converse also holds.

Proof. Expanding the squared norm gives

$$\begin{aligned}
&\mathbb{E}_x \sum_a \|(A_a^x \otimes I - I \otimes A_a^x)|\psi\rangle\|^2 \\
&= 2 \cdot \Big(\mathbb{E}_x \sum_a \langle\psi|(A_a^x)^2 \otimes I|\psi\rangle - \mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes A_a^x|\psi\rangle\Big) \\
&\leq 2 \cdot \Big(\mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes I|\psi\rangle - \mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes A_a^x|\psi\rangle\Big). \qquad (3.11)
\end{aligned}$$

The strong self-consistency bound controls the last line by 2δ. If A is projective, then (3.11) is an equality, which gives the converse. □

Lemma 3.37 (Post-processing preserves strong self-consistency)**.** *If A is δ-strongly self-consistent and f is a function on its answer set, then*

$$A^x_{[f(a)=b]} \otimes I \approx_{2\delta} I \otimes A^x_{[f(a)=b]}.$$

Proof. The same computation as in Lemma 3.36 gives

$$\begin{aligned}
&\mathbb{E}_x \sum_b \|(A^x_{[f(a)=b]} \otimes I - I \otimes A^x_{[f(a)=b]})|\psi\rangle\|^2 \\
&\leq 2 \cdot \Big(\langle\psi|A \otimes I|\psi\rangle - \mathbb{E}_x \sum_b \langle\psi|A^x_{[f(a)=b]} \otimes A^x_{[f(a)=b]}|\psi\rangle\Big). \qquad (3.12)
\end{aligned}$$

The post-processed diagonal term dominates $\mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes A_a^x|\psi\rangle$, so strong self-consistency bounds (3.12) by 2δ. □

Lemma 3.38 (Completeness transfer for a strongly self-consistent approximation)**.** *Let A be a δ-strongly self-consistent submeasurement and let B be a submeasurement such that $A_a^x \otimes I \approx_\varepsilon B_a^x \otimes I$. Then*

$$\langle\psi|B \otimes I|\psi\rangle \geq \langle\psi|A \otimes I|\psi\rangle - \delta - 2\sqrt{\varepsilon}.$$

Proof. Bound $\langle\psi|B \otimes I|\psi\rangle$ from below by $\mathbb{E}_x \sum_a \langle\psi|B_a^x \otimes B_a^x|\psi\rangle$. Then apply Proposition 3.24 twice to replace this by $\mathbb{E}_x \sum_a \langle\psi|A_a^x \otimes A_a^x|\psi\rangle$, and finish with strong self-consistency of A. □

Lemma 3.39 (Self-consistency implies data-processing closeness)**.** *Let A be a δ-strongly self-consistent submeasurement and let P be a projective submeasurement such that $P^x_a \otimes I \approx_\varepsilon A^x_a \otimes I$. Then for every function f,*

$$P^x_{[f(a)=b]} \otimes I \approx_{8\delta+8\sqrt{\varepsilon}} A^x_{[f(a)=b]} \otimes I.$$

Proof. First prove

$$P^x_{[f(a)=b]} \otimes I \approx_{2\delta+4\sqrt{\varepsilon}} I \otimes A^x_{[f(a)=b]}. \tag{3.13}$$

Expanding the squared norm and using projectivity gives

$$\begin{aligned}&\mathbb{E}_x \sum_b \|(P^x_{[f(a)=b]} \otimes I - I \otimes A^x_{[f(a)=b]})|\psi\rangle\|^2 \\ &\leq \langle\psi|P \otimes I|\psi\rangle + \langle\psi|I \otimes A|\psi\rangle - 2 \cdot \mathbb{E}_x \sum_b \langle\psi|P^x_{[f(a)=b]} \otimes A^x_{[f(a)=b]}|\psi\rangle. \end{aligned} \tag{3.14}$$

Proposition 3.33 bounds the first term by $\langle\psi|A \otimes I|\psi\rangle + 2\sqrt{\varepsilon}$, and Proposition 3.24 together with strong self-consistency bounds the third term below by $\langle\psi|A \otimes I|\psi\rangle - \delta - \sqrt{\varepsilon}$. Substituting into (3.14) yields (3.13). Now combine (3.13) with

$$A^x_{[f(a)=b]} \otimes I \approx_{2\delta} I \otimes A^x_{[f(a)=b]}$$

from Lemma 3.37, and conclude by Lemma 3.27. □

Lemma 3.40 (Self-consistency moves one copy to the same side)**.** *Let $A = \{A_a\}$ be a ζ-strongly self-consistent submeasurement. Then*

$$\sum_a \langle\psi|A_a^2 \otimes I|\psi\rangle \geq \sum_a \langle\psi|A_a \otimes I|\psi\rangle - \zeta.$$

Proof. Apply Cauchy–Schwarz to the two-sided overlap $\sum_a \langle\psi|A_a \otimes A_a|\psi\rangle$ and compare it with the strong self-consistency lower bound. □

Lemma 3.41 (Bounding the missing mass of a close submeasurement)**.** *Let $A = \{A_a\}$ be a measurement that is ζ-strongly self-consistent, and let $B = \{B_a\}$ be a submeasurement such that $A_a \otimes I \approx_\delta B_a \otimes I$. Writing $B = \sum_a B_a$, one has*

$$\|(I - B) \otimes I|\psi\rangle\|^2 \leq 2\sqrt{\delta} + \zeta.$$

Proof. Since $0 \leq I - B \leq I$, the left-hand side is at most $1 - \langle\psi|B \otimes I|\psi\rangle$. Compare $\sum_a \langle\psi|B_a^2 \otimes I|\psi\rangle$ with $\sum_a \langle\psi|A_a B_a \otimes I|\psi\rangle$ and then with $\sum_a \langle\psi|A_a^2 \otimes I|\psi\rangle$ using the $\approx_\delta$ hypothesis, and finish with Lemma 3.40. □

Lemma 3.42 (Completing a close submeasurement to a full measurement)**.** *Let $A = \{A_a\}$ be a measurement that is ζ-strongly self-consistent, let $B = \{B_a\}$ be a submeasurement such that $A_a \otimes I \approx_\delta B_a \otimes I$, and let C be the measurement obtained by adding the missing mass $I - B$ to one distinguished answer a^*. Then*

$$A_a \otimes I \approx_{2\delta+4\sqrt{\delta}+2\zeta} C_a \otimes I.$$

Proof of 3.42. By Lemma 3.26,

$$\sum_a \|(A_a - C_a) \otimes I|\psi\rangle\|^2 \leq 2\delta + 2 \cdot \|(I - B) \otimes I|\psi\rangle\|^2.$$

It remains to bound the residual term. Since $0 \le I - B \le I$,

$$\|(I-B)\otimes I|\psi\rangle\|^2 \le 1 - \sum_a \langle\psi|B_a^2 \otimes I|\psi\rangle. \tag{3.15}$$

By Proposition 3.24,

$$\sum_a \langle\psi|B_a^2 \otimes I|\psi\rangle \approx_{\sqrt{\delta}} \sum_a \langle\psi|(A_a B_a)\otimes I|\psi\rangle \approx_{\sqrt{\delta}} \sum_a \langle\psi|A_a^2 \otimes I|\psi\rangle.$$

Proposition 3.43 gives

$$\sum_a \langle\psi|A_a^2 \otimes I|\psi\rangle \ge \langle\psi|A \otimes I|\psi\rangle - \zeta = 1 - \zeta.$$

Hence (3.15) implies

$$\|(I-B)\otimes I|\psi\rangle\|^2 \le 2\sqrt{\delta} + \zeta.$$

Substituting this into the first bound proves the claim. □

Theorem 3.43. *This is the same statement as Lemma 3.40: for a ζ-strongly self-consistent submeasurement A,*

$$\sum_a \langle\psi|(A_a)^2 \otimes I|\psi\rangle \ge \sum_a \langle\psi|A_a \otimes I|\psi\rangle - \zeta.$$

Proof. Immediate from Lemma 3.40. □

Chapter 4

Making measurements projective

Theorem 4.1 (Naimark dilation)**.** *Let $|\psi\rangle$ be a state in $\mathcal{H}_{\mathrm{A}} \otimes \mathcal{H}_{\mathrm{B}}$. Let $A = \{A^x_a\}$ be a sub-measurement acting on $\mathcal{H}_{\mathrm{A}}$ and $B = \{B^y_b\}$ be a sub-measurement acting on $\mathcal{H}_{\mathrm{B}}$. Then there exist Hilbert spaces $\mathcal{H}_{\mathrm{A}_{\mathsf{aux}}}$ and $\mathcal{H}_{\mathrm{B}_{\mathsf{aux}}}$, a state $|\mathsf{aux}\rangle \in \mathcal{H}_{\mathrm{A}_{\mathsf{aux}}} \otimes \mathcal{H}_{\mathrm{B}_{\mathsf{aux}}}$, and measurements $\widehat{A} = \{\widehat{A}^x_a\}$ and $\widehat{B} = \{\widehat{B}^y_b\}$ acting on $\mathcal{H}_{\mathrm{A}} \otimes \mathcal{H}_{\mathrm{A}_{\mathsf{aux}}}$ and $\mathcal{H}_{\mathrm{B}} \otimes \mathcal{H}_{\mathrm{B}_{\mathsf{aux}}}$, respectively, such that, if $|\widehat{\psi}\rangle = |\psi\rangle \otimes |\mathsf{aux}\rangle$, then for all x, y, a, b,*

$$\langle\psi|A^x_a \otimes B^y_b|\psi\rangle = \langle\widehat{\psi}|\widehat{A}^x_a \otimes \widehat{B}^y_b|\widehat{\psi}\rangle.$$

In addition, $|\mathsf{aux}\rangle$ is a product state.

Proof.

For each question x, choose a local auxiliary state $|\mathsf{aux}_{\mathrm{A},x}\rangle$ of dimension one larger than the number of outcomes of A^x, and apply Lemma 4.6 to obtain a projective sub-measurement $\widetilde{A}^x$. Do the same for each question y on Bob's side, obtaining auxiliary states $|\mathsf{aux}_{\mathrm{B},y}\rangle$ and projective sub-measurements $\widetilde{B}^y$.

Let

$$|\mathsf{aux}\rangle = \Big(\bigotimes_x |\mathsf{aux}_{\mathrm{A},x}\rangle\Big) \otimes \Big(\bigotimes_y |\mathsf{aux}_{\mathrm{B},y}\rangle\Big), \qquad |\widehat{\psi}\rangle = |\psi\rangle \otimes |\mathsf{aux}\rangle,$$

and let $\widehat{A}^x_a$ and $\widehat{B}^y_b$ be obtained by letting $\widetilde{A}^x_a$ or $\widetilde{B}^y_b$ act on the relevant auxiliary factor and the identity act on every other auxiliary factor. Because the auxiliary state is a product state and the two dilations act on disjoint tensor factors, the correlation for fixed (x, y) reduces to the compression identity from Lemma 4.6 on the x- and y-registers. Hence

$$\langle\widehat{\psi}|\widehat{A}^x_a \otimes \widehat{B}^y_b|\widehat{\psi}\rangle = \langle\psi|A^x_a \otimes B^y_b|\psi\rangle.$$

$\square$

Theorem 4.2 (Orthogonalization lemma)**.** *Let $|\psi\rangle$ be a permutation-invariant state, and let $A = \{A_a\}$ be a sub-measurement with strong self-consistency*

$$\sum_a \langle\psi|A_a \otimes A_a|\psi\rangle \geq \sum_a \langle\psi|A_a \otimes I|\psi\rangle - \zeta.$$

Then there exists a projective sub-measurement $P = \{P_a\}$ such that

$$A_a \otimes I \approx_{100\zeta^{1/4}} P_a \otimes I.$$

Remark 4.3 (Lean auxiliary declarations: residual domination). The following are Lean auxiliary declarations, not hypotheses in Theorem 4.2. Let $A = \{A_a\}_{a\in\mathcal{A}}$ be a sub-measurement, and complete it to $\widehat{A}$ on $\mathcal{A} \cup \{\perp\}$ by setting $\widehat{A}_\perp = I - A_{\mathrm{tot}}$. Let $\widehat{P}$ be a projective sub-measurement on the completed outcome set. If the completed projective family puts at least the original residual mass on the fresh outcome,

$$\widehat{A}_\perp \le \widehat{P}_\perp,$$

then the restriction $P_a = \widehat{P}_a$, $a \in \mathcal{A}$, satisfies

$$P_{\mathrm{tot}} \le A_{\mathrm{tot}}.$$

Consequently, for any bipartite state ψ,

$$\langle \psi, I \otimes P_{\mathrm{tot}} \rangle \le \langle \psi, I \otimes A_{\mathrm{tot}} \rangle.$$

This isolates the order-theoretic invariant needed to pass from the option-completed orthonormalization output to the monotone-total route used in the self-improvement theorem. The formal development also records the corresponding strengthened orthonormalization statement: if the repair hypothesis for the option-completed measurement carries this residual-domination invariant, then the orthonormalized restriction satisfies both the usual state-dependent-distance conclusion and the displayed total comparisons.

Remark 4.4 (Lean right-register completion helpers). The inductive-step application of the orthogonalization lemma uses completion on both tensor factors. The formal development records the analytic completion statement on the left register; the shared preliminary helper compares right- and left-tensor placements at the level of the underlying state-dependent-distance form. Its submeasurement and indexed wrappers transport state-dependent distance to the right register under the permutation-invariance hypothesis on the state. The remaining two declarations combine this transport with the orthonormalize-and-complete output so that the line-156 projectivization handoff can later be produced from the two completed measurements without restating the tensor-factor bookkeeping.

Remark 4.5 (Lean line-169 projectivization match-mass declarations). Lean exposes the quantitative obstruction at the projectivization handoff used in the inductive step. The existing line-156 handoff gives only the generic consequence of Proposition 3.20: from $G^{\mathrm{A}} \simeq_{\zeta_1} G^{\mathrm{B}}$ and $G^{\mathrm{A}} \approx_{\zeta_2} Q^{\mathrm{A}}$ one obtains $Q^{\mathrm{A}} \simeq_{\zeta_1+\sqrt{\zeta_2}} G^{\mathrm{B}}$, and similarly on the other tensor factor. The declarations `leftConsistency_with_triangleSub_loss` and `rightConsistency_with_triangleSub_loss` are checked wrappers for this honest square-root-loss route.

Lean also proves a more local repaired transport that keeps the comparison at the pre-completion stage. If P^{A} is the orthonormalized projective sub-measurement with $G^{\mathrm{A}} \approx_\epsilon P^{\mathrm{A}}$, then Proposition 3.24 bounds the diagonal match-mass loss by $\sqrt{\epsilon}$, and canonical completion can only add nonnegative diagonal match mass. Thus the declarations `leftConsistency_of_completion_and_sdd` and `rightConsistency_of_completion_and_sdd` prove $Q^{\mathrm{A}} \simeq_{\zeta_1+\sqrt{\epsilon}} G^{\mathrm{B}}$ and its Bob-side mirror from the pre-projective ζ_1 consistency plus the pre-completion $\approx_\epsilon$ comparison. In particular, substituting the checked orthonormalization error $\epsilon = 100\zeta_1^{1/4}$ yields the repaired line-169 bound $\zeta_1 + 10\zeta_1^{1/8}$ recorded by `leftConsistency_with_orthonormalization_loss` and its mirror. This is much sharper than the direct completion-level $\zeta_1 + \sqrt{\zeta_2}$ route, but it is still not the paper's exact ζ_1.

The paper-tight line-169 estimate at exactly ζ_1 therefore requires more than state-dependent-distance closeness. Lean records the missing construction-level invariant as `ProjectivizationMatchMassMonotonicit` replacing G^{A} by the completed projective measurement Q^{A}, and role-reversing on Bob's side, must not decrease the diagonal match mass against the opposite pre-projective measurement. Its two consistency theorems prove that this primitive match-mass monotonicity is sufficient to recover the exact line-169 `ConsRel` links. Lean also proves that the canonical completion step itself can only increase this match mass: once the repaired projective sub-measurements preserve the relevant diagonal match mass, completing them at a distinguished outcome produces the required invariant for the completed projective measurements. The remaining upstream projectivization work is therefore the corresponding preservation property for the orthonormalization repair output. The pointwise-domination constructor records the degenerate no-change situation in which the source measurement is already dominated by the projective submeasurement. The expectation-level constructors refine the non-degenerate residual without requiring operator domination: it is enough to prove, for each outcome, that the diagonal match-mass contribution against the fixed partner measurement is preserved. The QXP-specialized constructor records the same requirement for the canonical projectors $P_a = \widehat{X}^\dagger T_a \widehat{X}$ produced by the rank-reduction repair.

4.0.1 Naimark dilation

Lemma 4.6. *Let $A = \{A_a\}$ be a sub-measurement with k distinct outcomes $a \in \mathcal{A}$, and let $|\mathsf{aux}\rangle \in \mathbb{C}^{k+1}$ be any state. Then there exists a projective sub-measurement $\widehat{A} = \{\widehat{A}_a\}$ such that for each outcome a,*

$$(I \otimes \langle\mathsf{aux}|) \cdot \widehat{A}_a \cdot (I \otimes |\mathsf{aux}\rangle) = A_a.$$

Proof. Consider an orthonormal basis for $\mathbb{C}^{k+1}$ consisting of vectors $|a\rangle$ for $a \in \mathcal{A}$ and the vector $|\perp\rangle$. Let U be any unitary such that for each vector $|\phi\rangle$,

$$U \cdot (|\phi\rangle \otimes |\mathsf{aux}\rangle) = \sum_{a \in \mathcal{A}} ((A_a)^{1/2}|\phi\rangle) \otimes |a\rangle + ((I - A)^{1/2}|\phi\rangle) \otimes |\perp\rangle,$$

where $A = \sum_a A_a$. Then

$$U \cdot (I \otimes |\mathsf{aux}\rangle) = \sum_{a \in \mathcal{A}} (A_a)^{1/2} \otimes |a\rangle + (I - A)^{1/2} \otimes |\perp\rangle,$$

so for any $a \in \mathcal{A}$,

$$(I \otimes \langle a|) \cdot U \cdot (I \otimes |\mathsf{aux}\rangle) = (A_a)^{1/2}.$$

Define

$$\widehat{A}_a = U^\dagger \cdot (I \otimes |a\rangle\langle a|) \cdot U.$$

Each $\widehat{A}_a$ is a projector, and

$$\begin{aligned}(I \otimes \langle\mathsf{aux}|) \cdot \widehat{A}_a \cdot (I \otimes |\mathsf{aux}\rangle) &= (I \otimes \langle\mathsf{aux}|) \cdot U^\dagger \cdot (I \otimes |a\rangle) \cdot (I \otimes \langle a|) \cdot U \cdot (I \otimes |\mathsf{aux}\rangle) \\ &= (A_a)^{1/2} \cdot (A_a)^{1/2} = A_a.\end{aligned}$$

□

Remark 4.7 (Naimark dilation does not preserve $\approx_\delta$). Let $|\psi\rangle$ be any state in $\mathbb{C}^d \otimes \mathbb{C}^d$, and consider the two-outcome measurements $A = \{A_0, A_1\}$ and $B = \{B_0, B_1\}$ with

$$A_0 = A_1 = B_0 = B_1 = \tfrac{1}{2} I.$$

For each outcome $a \in \{0, 1\}$, the operators $A_a \otimes I$ and $I \otimes B_a$ are both equal to $\frac{1}{2} I \otimes I$, so $A_a \otimes I \approx_0 I \otimes B_a$ on the state $|\psi\rangle$.

Now apply the Naimark construction with a qubit auxiliary space on each side and auxiliary state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Since A and B are genuine measurements rather than sub-measurements, the extra $|\perp\rangle$ direction from 4.6 is unnecessary in this example. The unitary from 4.6 can then be taken to be the identity, so the dilated measurements are

$$\widehat{A}_a = I \otimes |a\rangle\langle a|, \qquad \widehat{B}_a = I \otimes |a\rangle\langle a|.$$

If $|\widehat{\psi}\rangle = |\psi\rangle \otimes |+\rangle \otimes |+\rangle$, then a direct calculation gives

$$\sum_{a\in\{0,1\}} \left\|(\widehat{A}_a \otimes I - I \otimes \widehat{B}_a)|\widehat{\psi}\rangle\right\|^2 = 1.$$

Thus on the state $|\widehat{\psi}\rangle$ the dilated family satisfies $\widehat{A}_a \otimes I \approx_\delta I \otimes \widehat{B}_a$ only for $\delta \geq 1$, even though the original family satisfied the same relation with $\delta = 0$. This is why we use Theorem 4.2 rather than simply applying Theorem 4.1: Naimark preserves the outcome probabilities, but it need not preserve the state-dependent $\approx_\delta$ estimates needed later.

4.0.2 Orthogonalization lemma

Lemma 4.8 (Orthogonalization lemma for measurements)**.** *Let $|\psi\rangle$ be a state, and let $A = \{A_a\}$ and $B = \{B_a\}$ be measurements such that*

$$A_a \otimes I \simeq_\zeta I \otimes B_a$$

on state $|\psi\rangle$. Here $|\psi\rangle$ is not assumed to be permutation-invariant. Then there exists a projective sub-measurement $P = \{P_a\}$ such that

$$A_a \otimes I \approx_{84\zeta^{1/4}} P_a \otimes I.$$

Remark 4.9 (Lean auxiliary declarations for the orthogonalization lemma). The current Lean declarations for Lemma 4.8 are conditional. They assume explicit spectral-truncation and repair hypotheses for the $Q/X/\widehat{X}/P$ part of the paper proof. The declaration `orthonormalizationMainLemma` uses the general repair hypothesis on the product space, whereas `orthonormalizationMainLemma_local` and `orthonormalizationMeasurement` use the locality-preserving repair hypothesis, requiring the repaired family to have the form $P_a \otimes I$. These hypotheses are not present in the paper lemma and must be proved from its assumptions. Consequently these conditional declarations do not yet prove the paper lemma.

Lemma 4.10 (Locality-preserving projectivization repair)**.** *Paper origin: `references/ldt-paper/orthonormalizat` lines 534–860 (rank reduction and the Q-side setup, including 4.24 and 4.25) and 862–1194 (the $X/\widehat{X}/P$ algebra: 4.26, 4.29, 4.30, 4.31, 4.32, 4.33, plus the final triangle-inequality assembly producing the $84\zeta^{1/4}$ bound). Together these are the late repair stage of the orthogonalization-lemma proof.*

Let $|\psi\rangle$ be a state on a bipartite space, let $A = \{A_a\}$ be a measurement, and let $\zeta \geq 0$. Given the rounded almost-projective family produced by Lemma 4.14 on the left-lifted family $\{A_a \otimes I\}$, there exists a projective sub-measurement $P = \{P_a\}$ on the underlying space such that the lifted family $\{P_a \otimes I\}$ satisfies

$$A_a \otimes I \approx_{\mathsf{roundingToProjectiveError}(\zeta)} P_a \otimes I.$$

The locality-preserving form – output $P_a \otimes I$ rather than an arbitrary lifted family – is the form recorded in Remark 4.9; it is transported through the $Q/X/\widehat{X}/P$ algebra via the rectangular polar decomposition for the sigma-range embedding.

Proof. The paper's argument transports the rounded family produced by 4.14 through the $Q/X/\widehat{X}/P$ algebra (4.26, 4.29, 4.30, 4.31, 4.32, 4.33) and then descends to the locality-preserving lifted form. The proof is not yet complete (tracked by issue 1032). □

Remark 4.11 (Lean conditional declaration: local orthogonalization). This is the local Lean declaration recorded in Remark 4.9. Let $|\psi\rangle$ be a unit vector, let $A = \{A_a\}$ be a measurement, and let $0 \le \zeta \le 1$. Assume bipartite strong self-consistency

$$\sum_a \langle\psi|A_a \otimes A_a|\psi\rangle \ge \sum_a \langle\psi|A_a \otimes I|\psi\rangle - \zeta,$$

together with a spectral-truncation witness for the left-lifted family $\{A_a \otimes I\}$ and a locality-preserving repair witness producing projective families of the form $\widehat{P}_a = P_a \otimes I$. Then there exists a projective sub-measurement $P = \{P_a\}$ on the same space as A such that

$$A_a \otimes I \approx_{84\zeta^{1/4}} P_a \otimes I.$$

Since A is a measurement, $\sum_a A_a = I$, so bipartite strong self-consistency of A with itself is equivalent to bipartite consistency of A with itself. Feed this consistency hypothesis, together with the spectral-truncation and locality-preserving repair witnesses, into the measurement variant of 4.8 with $B = A$. The repair witness chooses the repaired family of the form $P_a \otimes I$, so the conclusion descends to a local projective sub-measurement $P = \{P_a\}$ on the space of A with error bound $84\zeta^{1/4}$.

Remark 4.12 (Lean conditional declaration: measurement-level orthogonalization). This is the measurement-level Lean declaration recorded in Remark 4.9. Let $|\psi\rangle$ be a unit vector, let $A = \{A_a\}$ be a measurement, and let $0 \le \zeta \le 1$. Assume bipartite strong self-consistency of A together with the spectral-truncation and locality-preserving repair witnesses for the left-lifted family $\{A_a \otimes I\}$. Then there exists a projective sub-measurement $P = \{P_a\}$ such that

$$A_a \otimes I \approx_{100\zeta^{1/4}} P_a \otimes I.$$

Using the bound stated in Remark 4.11, weaken the error using $84 \le 100$ and $\zeta^{1/4} \ge 0$. No completion step is needed because A is already a measurement.

Proof of 4.2 assuming 4.8.

Let $A = \{A_a\}$ be a sub-measurement with outcomes in $\mathcal{A}$. Write $A = \sum_a A_a$. The strong self-consistency assumption implies

$$\langle\psi|A \otimes A|\psi\rangle \ge \langle\psi|A \otimes I|\psi\rangle - \zeta,$$

so

$$\langle\psi|A \otimes (I - A)|\psi\rangle \le \zeta.$$

Rearranging once more gives

$$\langle\psi|(I - A) \otimes (I - A)|\psi\rangle \ge \langle\psi|(I - A) \otimes I|\psi\rangle - \zeta.$$

Complete A to the measurement $\widehat{A} = \{\widehat{A}_a\}_{a\in\widehat{\mathcal{A}}}$ on $\widehat{\mathcal{A}} = \mathcal{A} \cup \{\perp\}$ by setting $\widehat{A}_a = A_a$ for $a \in \mathcal{A}$ and $\widehat{A}_\perp = I - A$. Adding the two preceding inequalities shows that $\widehat{A}$ is 2ζ-strongly self-consistent. By 3.35, this is equivalent to

$$\widehat{A}_a \otimes I \simeq_{2\zeta} I \otimes \widehat{A}_a.$$

Apply 4.8 to obtain a projective sub-measurement $\widehat{P}$ such that

$$\widehat{A}_a \otimes I \approx_{84(2\zeta)^{1/4}} \widehat{P}_a \otimes I.$$

Discard the extra outcome and set $P_a = \widehat{P}_a$ for $a \in \mathcal{A}$. Then

$$A_a \otimes I \approx_{84(2\zeta)^{1/4}} P_a \otimes I,$$

and $84 \cdot 2^{1/4} \leq 100$, so the claimed bound follows. $\square$

Proof of 4.8. The statement is trivial when $\zeta > 1/4$, so assume

$$\zeta \leq 1/4. \tag{4.1}$$

By 3.18,

$$\sum_a \langle\psi|A_a \otimes B_a|\psi\rangle \geq 1 - \zeta.$$

Apply Cauchy–Schwarz to

$$\sum_a \langle\psi|(A_a \otimes I) \cdot (I \otimes B_a)|\psi\rangle$$

to obtain

$$\sum_a \langle\psi|(A_a)^2 \otimes I|\psi\rangle \geq (1-\zeta)^2 \geq 1 - 2\zeta.$$

Because A is a measurement, this rewrites as

$$\sum_a \langle\psi|(A_a - (A_a)^2) \otimes I|\psi\rangle \leq 2\zeta. \tag{4.2}$$

Lemma 4.13. *For any $x \in [0,1]$,*

$$(x - \mathsf{trunc}_\delta(x))^2 \leq \frac{1}{\delta} \cdot (x - x^2).$$

Proof. If $x \geq 1-\delta$, then $\mathsf{trunc}_\delta(x) = 1$. Since $\delta \leq 1/2$ below, one also has $x \geq \delta$, so

$$(x - \mathsf{trunc}_\delta(x))^2 = (1-x)^2 \leq (1-x)\frac{x}{\delta} = \frac{1}{\delta}(x - x^2).$$

If $x < 1-\delta$, then $\mathsf{trunc}_\delta(x) = 0$, and similarly

$$(x - \mathsf{trunc}_\delta(x))^2 = x^2 \leq x\frac{1-x}{\delta} = \frac{1}{\delta}(x - x^2).$$

$\square$

Lemma 4.14 (Rounding to projectors)**.** *There exists a set of projective matrices $\{R_a\}$ such that*

$$A_a \otimes I \approx_{2\sqrt{\zeta}} R_a \otimes I.$$

and

$$R := \sum_a R_a \leq (1 + 2\sqrt{\zeta}) \cdot I.$$

Proof. Write the eigendecomposition of A_a as

$$A_a = \sum_i \lambda_{a,i} |u_{a,i}\rangle\langle u_{a,i}|.$$

Choose

$$0 < \delta \leq 1/2, \tag{4.3}$$

define $\mathsf{trunc}_\delta : [0,1] \to \{0,1\}$ by $\mathsf{trunc}_\delta(x) = 1$ when $x \geq 1-\delta$ and 0 otherwise, and set

$$R_a = \mathsf{trunc}_\delta(A_a) = \sum_i \mathsf{trunc}_\delta(\lambda_{a,i}) |u_{a,i}\rangle\langle u_{a,i}|.$$

By 4.13,

$$(A_a - R_a)^2 \leq \frac{1}{\delta}(A_a - (A_a)^2).$$

Summing expectations and using 4.2 gives

$$\sum_a \|(A_a - R_a) \otimes I|\psi\rangle\|^2 \leq \frac{2\zeta}{\delta},$$

so $A_a \otimes I \approx_{2\zeta/\delta} R_a \otimes I$.

For every $x \in [0,1]$, one has $\mathsf{trunc}_\delta(x) \leq x/(1-\delta)$, hence

$$R_a \leq \frac{1}{1-\delta} A_a.$$

Summing over a and using that A is a measurement,

$$R = \sum_a R_a \leq \frac{1}{1-\delta} I \leq (1+2\delta)I,$$

where the last estimate uses $\delta \leq 1/2$. Setting $\delta = \sqrt{\zeta}$ gives the claim, and 4.1 ensures $\delta \leq 1/2$. $\square$

Lemma 4.15 (Combinatorial averaging for the $r > d$ branch)**.** *Let $f : \alpha \to \mathbb{R}$ be a function on a finite index set α with $|\alpha| = r$, and let $L \subseteq \alpha$ with $|L| = d$ and $d < r$. Write $S = L^{\complement}$ and assume $f(s) \leq f(l)$ for every $s \in S$ and $l \in L$. Then the pairwise double-counting inequality*

$$|L| \sum_{s \in S} f(s) \leq |S| \sum_{l \in L} f(l)$$

holds, and hence also

$$r \sum_{s \in S} f(s) \leq |S| \sum_{x \in \alpha} f(x).$$

If in addition $r \leq (1 + 2\sqrt{\zeta})\, d$, $\sum_{x \in \alpha} f(x) \leq 1 + 2\sqrt{\zeta}$, and $0 \leq \zeta \leq 1/4$, then

$$\sum_{s \in S} f(s) \leq 4\sqrt{\zeta}.$$

Proof. Choose L to be a set of d indices on which the sum of f is maximal. If some $s \in S$ and $l \in L$ satisfied $f(l) < f(s)$, replacing l by s would increase this sum, so every small index is bounded by every large index. Summing the inequalities $f(s) \leq f(l)$ over all pairs $(s,l) \in S \times L$ gives

$$|L| \sum_{s \in S} f(s) \leq |S| \sum_{l \in L} f(l).$$

Since α is the disjoint union of L and S, adding $|S| \sum_{s\in S} f(s)$ to both sides yields

$$r \sum_{s\in S} f(s) \le |S| \sum_{x\in\alpha} f(x).$$

The hypotheses $|L| = d$ and $r \le (1+2\sqrt{\zeta})d$ imply $|S|/r \le 2\sqrt{\zeta}$. Dividing by r and using $\sum_{x\in\alpha} f(x) \le 1+2\sqrt{\zeta}$ gives

$$\sum_{s\in S} f(s) \le 2\sqrt{\zeta}(1+2\sqrt{\zeta}).$$

Finally, $\zeta \le 1/4$ implies $\sqrt{\zeta} \le 1/2$, and hence $2\sqrt{\zeta}(1+2\sqrt{\zeta}) \le 4\sqrt{\zeta}$. □

Lemma 4.16 (Rank reduction)**.** *There exists a set of projection matrices $\{Q_a\}$ such that*

$$A_a \otimes I \approx_{12\sqrt{\zeta}} Q_a \otimes I.$$

and

$$Q := \sum_a Q_a \le (1+2\sqrt{\zeta}) \cdot I.$$

Furthermore,

$$\sum_a \operatorname{rank}(Q_a) \le d.$$

Proof. Let $\{R_a\}$ be given by 4.14. Write $r_a = \operatorname{rank}(R_a)$ and $r = \sum_a r_a$. If $r \le d$, take $Q_a = R_a$. Otherwise,

$$r = \sum_a r_a = \sum_a \operatorname{Tr}(R_a) = \operatorname{Tr}(R) \le (1+2\sqrt{\zeta})d. \tag{4.4}$$

Let $|v_{a,1}\rangle, \dots, |v_{a,r_a}\rangle$ be an orthonormal basis for the range of R_a, so

$$R_a = \sum_{i=1}^{r_a} |v_{a,i}\rangle\langle v_{a,i}|.$$

Define the overlaps

$$o_{a,i} = \langle\psi|(|v_{a,i}\rangle\langle v_{a,i}| \otimes I)|\psi\rangle.$$

Their total satisfies

$$\sum_a \sum_{i=1}^{r_a} o_{a,i} = \langle\psi|R \otimes I|\psi\rangle \le 1+2\sqrt{\zeta}. \tag{4.5}$$

Let Large be the set of the d largest overlaps $o_{a,i}$, breaking ties arbitrarily, and let Small be the complement. Then

$$|\mathsf{Small}| = r - d \le 2\sqrt{\zeta}\, d \le 2\sqrt{\zeta}\, r, \tag{4.6}$$

by 4.4. Averaging the overlaps over the r indices gives

$$\sum_{(a,i)\in\mathsf{Small}} o_{a,i} \le \frac{|\mathsf{Small}|}{r} \sum_{a,i} o_{a,i} \le 4\sqrt{\zeta}, \tag{4.7}$$

using 4.6, 4.5, and 4.1.

For each a, let $\mathsf{Large}_a = \{i : (a,i) \in \mathsf{Large}\}$ and define

$$Q_a = \sum_{i\in\mathsf{Large}_a} |v_{a,i}\rangle\langle v_{a,i}|.$$

Then $\sum_a \operatorname{rank}(Q_a) = |\mathsf{Large}| \leq d$, and $Q_a \leq R_a$ for every a, so

$$Q = \sum_a Q_a \leq \sum_a R_a = R \leq (1 + 2\sqrt{\zeta})I.$$

Also,

$$R_a - Q_a = \sum_{i \in \mathsf{Small}_a} |v_{a,i}\rangle\langle v_{a,i}|$$

is a projector, hence

$$\sum_a \|(R_a - Q_a) \otimes I|\psi\rangle\|^2 = \sum_{(a,i)\in\mathsf{Small}} o_{a,i} \leq 4\sqrt{\zeta}$$

by 4.7. Therefore $R_a \otimes I \approx_{4\sqrt{\zeta}} Q_a \otimes I$. Combining this with 4.14 and the triangle inequality for $\approx_\delta$ gives

$$A_a \otimes I \approx_{12\sqrt{\zeta}} Q_a \otimes I.$$

□

Henceforth let $Q = \{Q_a\}$ be the family given by 4.16.

Lemma 4.17 (Completeness of Q)**.**

$$\langle\psi|Q \otimes I|\psi\rangle \geq 1 - 11\zeta^{1/4}.$$

Proof. Since each Q_a is projective,

$$\langle\psi|Q \otimes I|\psi\rangle = \sum_a \langle\psi|(Q_a)^2 \otimes I|\psi\rangle \approx_{5\zeta^{1/4}} \sum_a \langle\psi|(Q_a \cdot A_a) \otimes I|\psi\rangle. \tag{4.8}$$

By Cauchy–Schwarz,

$$\sum_a \langle\psi|(Q_a)^2 \otimes I|\psi\rangle \leq 1 + 2\sqrt{\zeta} \quad \text{and} \quad \sum_a \langle\psi|(Q_a - A_a)^2 \otimes I|\psi\rangle \leq 12\sqrt{\zeta}$$

from 4.16. A second Cauchy–Schwarz estimate yields

$$\sum_a \langle\psi|(Q_a \cdot A_a) \otimes I|\psi\rangle \approx_{4\zeta^{1/4}} \sum_a \langle\psi|(A_a)^2 \otimes I|\psi\rangle. \tag{4.9}$$

Combining 4.8 and 4.9 with 4.2 gives

$$\langle\psi|Q \otimes I|\psi\rangle \geq \sum_a \langle\psi|A_a \otimes I|\psi\rangle - 2\zeta - 9\zeta^{1/4} = 1 - 11\zeta^{1/4},$$

since $\zeta \leq \zeta^{1/4}$ under 4.1. □

Lemma 4.18 (Completeness of $\sqrt{Q}$)**.**

$$\langle\psi|\sqrt{Q} \otimes I|\psi\rangle \geq 1 - 12\zeta^{1/4}.$$

Proof. Diagonalize $Q = \sum_i \nu_i |u_i\rangle\langle u_i|$. Because $Q \leq (1 + 2\sqrt{\zeta})I$ by 4.16, each ν_i is at most $1 + 2\sqrt{\zeta}$. Therefore

$$\sqrt{Q} = \sum_i \sqrt{\nu_i}|u_i\rangle\langle u_i| \geq \frac{1}{\sqrt{1 + 2\sqrt{\zeta}}} Q.$$

The scalar estimate

$$\frac{1}{\sqrt{1+2\sqrt{\zeta}}} \geq 1-\sqrt{\zeta}$$

gives $\sqrt{Q} \geq (1-\sqrt{\zeta})Q$. Hence 4.17 implies

$$\langle\psi|\sqrt{Q}\otimes I|\psi\rangle \geq (1-\sqrt{\zeta})(1-11\zeta^{1/4}) \geq 1-12\zeta^{1/4}.$$

□

Lemma 4.19 (Q is almost projective)**.**

$$\sum_a (Q_a \cdot Q \cdot Q_a - Q_a) \leq 4\sqrt{\zeta}\cdot I.$$

Proof. Since $Q \leq (1+2\sqrt{\zeta})I$ by 4.16,

$$\begin{aligned}
\sum_a Q_a \cdot Q \cdot Q_a - \sum_a Q_a &\leq (1+2\sqrt{\zeta})\sum_a Q_a \cdot I \cdot Q_a - \sum_a Q_a \\
&= (1+2\sqrt{\zeta})Q - Q \\
&= 2\sqrt{\zeta}\,Q \\
&\leq 2\sqrt{\zeta}(1+2\sqrt{\zeta})I \\
&\leq 4\sqrt{\zeta}\,I,
\end{aligned}$$

using 4.1 in the last step. □

Definition 4.20 (Matrix decomposition of Q_a)**.** For each a, let

$$m_a = \mathrm{rank}(Q_a),$$

and let $|v_{a,1}\rangle, \dots, |v_{a,m_a}\rangle$ be an orthonormal basis for the range of Q_a, so that

$$Q_a = \sum_{i=1}^{m_a} |v_{a,i}\rangle\langle v_{a,i}|.$$

Let

$$m = \sum_a m_a,$$

and let $\{|a,i\rangle\}$ be an orthonormal basis of $\mathbb{C}^m$, indexed by all a and all $1 \leq i \leq m_a$. For each a, define

$$X_a = \sum_{i=1}^{m_a} |a,i\rangle\langle v_{a,i}|, \qquad T_a = \sum_{i=1}^{m_a} |a,i\rangle\langle a,i|,$$

and define

$$X = \sum_a X_a = \sum_a \sum_{i=1}^{m_a} |a,i\rangle\langle v_{a,i}|. \tag{4.10}$$

Let $T = \{T_a\}$. Then T is a projective measurement on $\mathbb{C}^m$.

Definition 4.21 (SVD of X)**.** The paper presents this step by choosing a singular value decomposition

$$X = U \cdot \Sigma_{m \times d} \cdot V^\dagger.$$

The formalization has two closely related auxiliary declarations. The reduced datum records the downstream consequences of the choice: the matrices X and $\widehat{X}$, the identities $X^\dagger X = Q$, $\widehat{X}\widehat{X}^\dagger = I$, and $X^\dagger \widehat{X} = \sqrt{Q}$, together with the $Q_a = X^\dagger T_a X$ restatement. The rectangular-SVD lemmas take explicit matrices $U, V, \Sigma, I_{m\times d}$ and derive the two stored $\widehat{X}$ identities from the matrix calculation below.

Remark 4.22. If the family $\{Q_a\}$ were a projective measurement, then the vectors $|v_{a,i}\rangle$, over all a and all $1 \le i \le m_a$, would be orthonormal. Then (4.10) would already be a singular value decomposition of X, with $\Sigma = I_{m\times d}$. Equivalently,

$$X = U \cdot I_{m\times d} \cdot V^\dagger.$$

This motivates replacing Σ by I in the definition of $\widehat{X}$ and P.

Definition 4.23 (Definition of P)**.** The paper defines

$$\widehat{X} = U \cdot I_{m\times d} \cdot V^\dagger.$$

We take $\widehat{X}$ to be the reduced matrix supplied by the stored datum. For each a, define

$$\widehat{X}_a = T_a \cdot \widehat{X}, \qquad P_a = \widehat{X}_a^\dagger \cdot \widehat{X}_a.$$

Lemma 4.24. *For each a,*

$$X_a = T_a \cdot X.$$

Proof. Multiply the definition of X by T_a on the left and use that T_a selects exactly the basis vectors $|a,i\rangle$:

$$T_a \cdot X = \Big(\sum_{i=1}^{m_a} |a,i\rangle\langle a,i|\Big)\Big(\sum_b \sum_{j=1}^{m_b} |b,j\rangle\langle v_{b,j}|\Big) = \sum_{i=1}^{m_a} |a,i\rangle\langle v_{a,i}| = X_a.$$

□

Lemma 4.25 (Q_a restated)**.** *For each a,*

$$Q_a = X_a^\dagger \cdot X_a = X^\dagger \cdot T_a \cdot X = X_a^\dagger \cdot X.$$

Proof. The identity $X_a^\dagger X_a = Q_a$ follows by expanding the basis sums. The remaining equalities come from 4.24 and the projectivity of T:

$$X_a^\dagger \cdot X_a = (X^\dagger \cdot T_a)(T_a \cdot X) = X^\dagger \cdot T_a \cdot X = X_a^\dagger \cdot X.$$

□

Lemma 4.26 (X squared)**.**

$$X^\dagger \cdot X = Q.$$

Proof. Use that T is a measurement and then 4.25:

$$X^\dagger \cdot X = \sum_a X^\dagger \cdot T_a \cdot X = \sum_a Q_a = Q.$$

The additional paper identities involving U, V, Σ are the usual SVD computations. The Lean development also records the spectral calculation which will identify the mixed positive-Gram-image product with $\sqrt{Q}$: the positive eigenvalues give the nonzero columns, while the zero eigenspace has zero coefficient in the square-root expansion and is killed by X. The conjugate right-eigenvector rows over the positive spectrum are also orthonormal. Equivalently, any row-space vector orthogonal to the positive Gram images is killed by $X^\dagger$. ☐

Lemma 4.27. *For each a,*

$$X_a^\dagger \cdot (X \cdot X^\dagger - I_{m\times m})^2 \cdot X_a = Q_a \cdot Q \cdot Q_a - Q_a.$$

Proof. Expand the square and apply 4.26 and 4.25 term by term:

$$X_a^\dagger (X \cdot X^\dagger \cdot X \cdot X^\dagger) X_a = Q_a \cdot Q \cdot Q_a,$$

while

$$X_a^\dagger (X \cdot X^\dagger) X_a = (X_a^\dagger \cdot X)(X^\dagger \cdot X_a) = Q_a \cdot Q_a = Q_a,$$

and $X_a^\dagger I_{m\times m} X_a = X_a^\dagger X_a = Q_a$. Substituting these identities into

$$(X \cdot X^\dagger - I_{m\times m})^2 = X \cdot X^\dagger \cdot X \cdot X^\dagger - 2X \cdot X^\dagger + I_{m\times m}$$

gives the claim. ☐

Lemma 4.28 (P_a restated)**.** *For each a,*

$$P_a = \widehat{X}^\dagger \cdot T_a \cdot \widehat{X} = \widehat{X}_a^\dagger \cdot \widehat{X}.$$

Proof. Since T is projective,

$$P_a = \widehat{X}_a^\dagger \cdot \widehat{X}_a = (\widehat{X}^\dagger \cdot T_a)(T_a \cdot \widehat{X}) = \widehat{X}^\dagger \cdot T_a \cdot \widehat{X} = \widehat{X}_a^\dagger \cdot \widehat{X}.$$

☐

Lemma 4.29 ($\widehat{X}$ squared)**.**

$$\widehat{X} \cdot \widehat{X}^\dagger = I_{m\times m}.$$

Proof. The paper proves this by multiplying out $\widehat{X} = U \cdot I_{m\times d} \cdot V^\dagger$ and using the unitarity of U and V. In Lean, the dimension hypothesis $m \le d$ already gives a rectangular matrix with orthonormal rows on the sigma auxiliary space; the chosen matrix `sigmaFinXHatCoisometry` and its specification record this existence result as a hypothesis for the future construction. The formal lemma `orthonormal_normalized_image_of_adjoint_comp_eigenvectors` records the singular-vector normalization calculation, while `normalized_matrix_image_rows_mul_conjTranspose` combines the corresponding row-coisometry identity, and `positive_gram_spectrum_image_rows_transpose_mixed` records the adjoint action $X^\dagger(\lambda^{-1/2} X v) = \lambda^{1/2} v$ on the positive spectral part of the Gram operator. The formal lemma `exists_unitary_with_positive_gram_spectrum_rows` records the next finite-dimensional step: after choosing distinct row positions for the positive spectrum, these normalized image rows extend to a square unitary matrix on the row space. The companion lemma `exists_rectangular_coisometry_extending_orthonormal_rows` records the analogous rectangular completion for prescribed right rows when the dimension bound permits it. The formal lemma `rectangularSvd_xHat_coisometry` records the corresponding matrix calculation, and `exists_xHat_of_rectangularSvd` gives the resulting choice of $\widehat{X}$ together with the mixed square-root identity below. □

Lemma 4.30 (X times $\widehat{X}$)**.**

$$X^\dagger \cdot \widehat{X} = \sqrt{Q}.$$

Proof. The paper obtains this from the SVD formula for X and the definition of $\widehat{X}$. The formal lemma `rectangularSvd_xHat_mixed_of_sqrtQ` is the corresponding matrix calculation in the form whose square-root target is the operator Q supplied by the QXP construction. The formal lemma `rectangularSvd_middle_eq_sqrt_of_square` records the spectral step: a positive middle factor whose square is Q is the positive square root of Q. The reduced datum then records this mixed-product identity as a primitive consequence, which is exactly the part needed in 4.31 and 4.33. The companion paper identity for $X\widehat{X}^\dagger$ is not used downstream. □

Lemma 4.31 (Squared difference)**.**

$$(X - \widehat{X}) \cdot (X - \widehat{X})^\dagger \leq (X \cdot X^\dagger - I_{m\times m})^2.$$

Proof. The paper diagonalizes the positive operator $X\widehat{X}^\dagger$ using the SVD notation. The formal proof avoids storing that SVD explicitly: set $Y = X\widehat{X}^\dagger$. The primitive identities imply that Y is hermitian, nonnegative, and satisfies $Y^2 = XX^\dagger$. Hence $(X - \widehat{X})(X - \widehat{X})^\dagger = (Y - I)^2$, and positivity of Y gives $(Y - I)^2 \leq (Y^2 - I)^2 = (XX^\dagger - I)^2$. □

Lemma 4.32 (Projectivity of P)**.** $P = \{P_a\}$ *forms a projective sub-measurement.*

Proof. For outcomes a, b,

$$\begin{aligned} P_a \cdot P_b &= (\widehat{X}^\dagger \cdot T_a \cdot \widehat{X})(\widehat{X}^\dagger \cdot T_b \cdot \widehat{X}) \\ &= \widehat{X}^\dagger \cdot T_a \cdot I_{m\times m} \cdot T_b \cdot \widehat{X} \\ &= \widehat{X}^\dagger \cdot T_a \cdot \widehat{X} \cdot \mathbf{1}[[]\, a = b] \\ &= P_a \cdot \mathbf{1}[[]\, a = b], \end{aligned}$$

where the second line uses 4.29, the third uses that T is projective, and the last uses 4.28. Summing over a shows $\sum_a P_a \leq I$, so P is a projective sub-measurement. The formalization also records the sharper identity

$$\sum_a P_a = \widehat{X}^\dagger \widehat{X},$$

and the resulting total-domination estimates used later in the monotone-total self-improvement route. □

Lemma 4.33 (P is close to Q)**.**

$$Q_a \otimes I \approx_{30\zeta^{1/4}} P_a \otimes I.$$

Proof. The formal proof follows this argument directly from the primitive X, $\widehat{X}$, and P identities; the $30\zeta^{1/4}$ estimate is derived here rather than stored as part of the reduced datum. The rectangular-SVD lemma combines the same estimate with the positive-square identification of the middle factor from 4.30. The positive-Gram sigma-space lemma also records the coisometry $XX^\dagger = I$ for the original sigma embedding when the projective Q-family is subnormalized; this is the construction-level hypothesis later used in the residual-domination argument for the fresh completed outcome. The quantity to bound is

$$\sum_a \langle\psi|(Q_a - P_a)^2 \otimes I|\psi\rangle = \langle\psi|Q \otimes I|\psi\rangle + \langle\psi|P \otimes I|\psi\rangle - \sum_a \langle\psi|Q_a P_a \otimes I|\psi\rangle - \sum_a \langle\psi|P_a Q_a \otimes I|\psi\rangle, \tag{4.11}$$

using the projectivity of Q and P. The first term is at most $1 + 2\sqrt{\zeta}$ by 4.16, and the second is at most 1 because P is a sub-measurement.

For the last two terms,

$$\sum_a \langle\psi|Q_a P_a \otimes I|\psi\rangle + \sum_a \langle\psi|P_a Q_a \otimes I|\psi\rangle = 2 \cdot \Re\Big(\sum_a \langle\psi|Q_a P_a \otimes I|\psi\rangle\Big). \tag{4.12}$$

By 4.25,

$$\sum_a \langle\psi|Q_a P_a \otimes I|\psi\rangle = \sum_a \langle\psi|(X_a^\dagger \cdot X \cdot P_a) \otimes I|\psi\rangle.$$

The main estimate is

$$\sum_a \langle\psi|(X_a^\dagger \cdot X \cdot P_a) \otimes I|\psi\rangle \approx_{2\zeta^{1/4}} \sum_a \langle\psi|(X_a^\dagger \cdot \widehat{X} \cdot P_a) \otimes I|\psi\rangle. \tag{4.13}$$

This follows by Cauchy–Schwarz:

$$\Big|\sum_a \langle\psi|((X_a^\dagger \cdot (X - \widehat{X})) \otimes I) \cdot (P_a \otimes I)|\psi\rangle\Big|$$

is bounded by the square root of

$$\sum_a \langle\psi|(X_a^\dagger \cdot (X - \widehat{X})(X - \widehat{X})^\dagger \cdot X_a) \otimes I|\psi\rangle,$$

times the square root of $\sum_a \langle\psi|(P_a)^2 \otimes I|\psi\rangle \le 1$. The first factor is at most $4\sqrt{\zeta}$ by 4.31, 4.27, and 4.19, so the total error is at most $2\zeta^{1/4}$.

The right-hand side of 4.13 is real, because

$$\sum_a \langle\psi|(X_a^\dagger \cdot \widehat{X} \cdot P_a) \otimes I|\psi\rangle = \langle\psi|(X^\dagger \cdot \widehat{X}) \otimes I|\psi\rangle = \langle\psi|\sqrt{Q} \otimes I|\psi\rangle,$$

where the first equality uses 4.24, 4.28, and 4.29 and that T is a measurement, and the second uses 4.30. Hence 4.18 and 4.13 imply

$$\Re\Big(\sum_a \langle\psi|Q_a P_a \otimes I|\psi\rangle\Big) \ge 1 - 14\zeta^{1/4}.$$

Substituting this into 4.12 and then into 4.11 gives

$$\sum_a \langle\psi|(Q_a - P_a)^2 \otimes I|\psi\rangle \le (1 + 2\sqrt{\zeta}) + 1 - 2(1 - 14\zeta^{1/4}) = 2\sqrt{\zeta} + 28\zeta^{1/4} \le 30\zeta^{1/4}.$$

$\square$

Finally, 4.16 gives

$$A_a \otimes I \approx_{12\sqrt{\zeta}} Q_a \otimes I,$$

and 4.33 gives

$$Q_a \otimes I \approx_{30\zeta^{1/4}} P_a \otimes I.$$

The triangle inequality for $\approx_\delta$ yields

$$A_a \otimes I \approx_{84\zeta^{1/4}} P_a \otimes I.$$

□

Chapter 5

Expansion in the hypercube graph

5.1 The graph and its spectrum

Definition 5.1 (Rerandomizing one coordinate)**.** For $u \in \mathbb{F}_q^m$, an index $i \in \{1, \dots, m\}$, and $x \in \mathbb{F}_q$, let

$$\mathrm{rerand}_i(u, x) = u + xe_i.$$

This is the move obtained by rerandomizing the i-th coordinate of u by a uniform additive shift.

Definition 5.2 (Hypercube graph)**.** The hypercube graph $C = (V, E)$ has vertex set $V = \mathbb{F}_q^m$, and an edge between u and v whenever they differ in at most one coordinate. A random edge $(u, v) \sim C$ is sampled by drawing $u \sim \mathbb{F}_q^m$, $i \in \{1, \dots, m\}$, and $x \in \mathbb{F}_q$ uniformly and then setting $v = \mathrm{rerand}_i(u, x)$.

Definition 5.3 (Adjacency matrix and Laplacian)**.** Let $M = q^m$. The normalized adjacency matrix of C is

$$K = \mathbb{E}_{(u,v)\sim C} |u\rangle\langle v|,$$

and the Laplacian is

$$L = \frac{1}{M} I - K.$$

Lemma 5.4 (Laplacian rewrite)**.** *The Laplacian can be written as*

$$L = \frac{1}{2} \mathbb{E}_{(u,v)\sim C} (|u\rangle - |v\rangle)(\langle u| - \langle v|).$$

Proof. The two vertex marginals of a random edge are uniform on $\mathbb{F}_q^m$. Expanding the right-hand side yields $(1/M)I - K$. □

Lemma 5.5 (Average of a nontrivial additive character)**.** *For $\beta \in \mathbb{F}_q$,*

$$\mathbb{E}_{x\sim \mathbb{F}_q} \omega^{\mathrm{tr}(\beta x)} = \begin{cases} 1 & \text{if } \beta = 0, \\ 0 & \text{if } \beta \neq 0. \end{cases}$$

Proof. If $\beta = 0$ the character is constant. If $\beta \neq 0$, the map $x \mapsto \beta x$ permutes $\mathbb{F}_q$, so the average is the average of a nontrivial additive character, which vanishes. □

Lemma 5.6 (Orthogonality of finite-field characters)**.** *For $\alpha, \beta \in \mathbb{F}_q^m$,*

$$\frac{1}{q^m} \sum_{u \in \mathbb{F}_q^m} \omega^{\operatorname{tr}(u \cdot (\beta - \alpha))} = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta. \end{cases}$$

Proof. This is Lemma 3.3 applied to $v = \beta - \alpha$. □

Lemma 5.7 (Eigenvectors of the hypercube graph)**.** *For each $\alpha \in \mathbb{F}_q^m$, define*

$$|\varphi_\alpha\rangle = \frac{1}{M^{1/2}} \cdot \sum_{u \in \mathbb{F}_q^m} \omega^{\operatorname{tr}(u \cdot \alpha)} |u\rangle.$$

Then the following two statements hold.

1. *The $|\varphi_\alpha\rangle$'s form an orthonormal basis of $\mathbb{C}^V$.*
2. *For each $\alpha \in \mathbb{F}_q^m$, $|\varphi_\alpha\rangle$ is an eigenvector for K with eigenvalue $\frac{1}{M} \cdot \frac{m - |\alpha|}{m}$, where $|\alpha|$ is the number of nonzero coordinates of α.*

Proof. First we prove 1. For $\alpha, \beta \in \mathbb{F}_q^m$,

$$\langle \varphi_\alpha \mid \varphi_\beta \rangle = \frac{1}{M} \sum_{u \in \mathbb{F}_q^m} \omega^{\operatorname{tr}(u \cdot (\beta - \alpha))} = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta, \end{cases}$$

by Lemma 5.6. Thus the $|\varphi_\alpha\rangle$ form an orthonormal basis of $\mathbb{C}^V$.

Next we prove 2. For $\alpha \in \mathbb{F}_q^m$,

$$\begin{aligned} K \cdot |\varphi_\alpha\rangle &= \left(\mathbb{E}_{(u,v) \sim C} |u\rangle\langle v| \right) \cdot \left(\frac{1}{M^{1/2}} \cdot \sum_{u \in \mathbb{F}_q^m} \omega^{\operatorname{tr}(u \cdot \alpha)} |u\rangle \right) \\ &= \frac{1}{M^{1/2}} \cdot \mathbb{E}_{(u,v) \sim C} \omega^{\operatorname{tr}(v \cdot \alpha)} |u\rangle. \end{aligned} \tag{5.1}$$

By the edge-sampling rule from Definition 5.2, we may write $v = u + x e_i$, where i is uniformly random in $\{1, \dots, m\}$ and x is uniformly random in $\mathbb{F}_q$. Therefore

$$(5.1) = \frac{1}{M^{1/2}} \cdot \mathbb{E}_{u,i,x} \omega^{\operatorname{tr}((u + x e_i) \cdot \alpha)} |u\rangle = \left(\mathbb{E}_{i,x} \omega^{\operatorname{tr}(x \alpha_i)} \right) \cdot \frac{1}{M^{1/2}} \cdot \mathbb{E}_u \omega^{\operatorname{tr}(u \cdot \alpha)} |u\rangle = \frac{1}{M} \left(\mathbb{E}_{i,x} \omega^{\operatorname{tr}(x \alpha_i)} \right) \cdot |\varphi_\alpha\rangle.$$

Hence $|\varphi_\alpha\rangle$ is an eigenvector of K with eigenvalue

$$\frac{1}{M} \cdot \mathbb{E}_i \left[\mathbb{E}_x \omega^{\operatorname{tr}(x \alpha_i)} \right] = \frac{1}{M} \cdot \mathbb{E}_i \left[\mathbf{1}[[] \, \alpha_i = 0] \right] = \frac{1}{M} \cdot \frac{m - |\alpha|}{m},$$

by Lemma 5.5. □

Lemma 5.8 (Spectral gap of the Laplacian)**.** *If $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{q^m}$ are the eigenvalues of L, then*

$$\lambda_1 = 0, \qquad \lambda_2 = \frac{1}{m q^m}.$$

Proof. Lemma 5.7 identifies the two largest eigenvalues of K, hence the two smallest eigenvalues of $L = (1/q^m) I - K$. □

5.2 Local and global variance

Definition 5.9 (Local and global variance)**.** Let $|\psi\rangle \in \mathscr{H}_{\mathrm{A}} \otimes \mathscr{H}_{\mathrm{B}}$ and let $0 \leq A^u \leq I$ be an operator for each $u \in \mathbb{F}_q^m$. The local variance is

$$\mathbf{Var}_{\mathrm{local}}(A, \psi) = \frac{1}{2}\mathbb{E}_{(u,v)\sim C}\langle\psi|(A^u - A^v)^2 \otimes I|\psi\rangle,$$

and the global variance is

$$\mathbf{Var}_{\mathrm{global}}(A, \psi) = \frac{1}{2}\mathbb{E}_{u,v\sim\mathbb{F}_q^m}\langle\psi|(A^u - A^v)^2 \otimes I|\psi\rangle.$$

Lemma 5.10 (Local variance as a Laplacian form)**.** *Define*

$$A_{\mathrm{comb}} = \sum_{u\in\mathbb{F}_q^m} |u\rangle \otimes A^u \otimes I.$$

Then

$$\mathrm{Tr}(A_{\mathrm{comb}}^\dagger(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}}) = \mathbf{Var}_{\mathrm{local}}(A, \psi).$$

Proof. For $u, v \in \mathbb{F}_q^m$,

$$\begin{aligned}(((\langle u| - \langle v|) \otimes \langle\psi|) \cdot A_{\mathrm{comb}} &= (((\langle u| - \langle v|) \otimes \langle\psi|) \cdot \sum_{w\in\mathbb{F}_q^m} |w\rangle \otimes A^w \otimes I \\ &= \langle\psi| \cdot ((A^u - A^v) \otimes I).\end{aligned} \tag{5.2}$$

Therefore

$$A_{\mathrm{comb}}^\dagger(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}} = \frac{1}{2}\mathbb{E}_{(u,v)\sim C}A_{\mathrm{comb}}^\dagger((|u\rangle - |v\rangle)(\langle u| - \langle v|) \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}}$$

by Lemma 5.4, and hence

$$A_{\mathrm{comb}}^\dagger(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}} = \frac{1}{2}\mathbb{E}_{(u,v)\sim C}((A^u - A^v) \otimes I)|\psi\rangle\langle\psi|((A^u - A^v) \otimes I)$$

by (5.2). Taking the trace gives

$$\mathrm{Tr}(A_{\mathrm{comb}}^\dagger(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}}) = \frac{1}{2}\mathbb{E}_{(u,v)\sim C}\langle\psi|(A^u - A^v)^2 \otimes I|\psi\rangle = \mathbf{Var}_{\mathrm{local}}(A, \psi).$$

$\square$

Lemma 5.11 (Global variance as the orthogonal Fourier mass)**.** *Write*

$$A_{\mathrm{comb}} = |\varphi_0\rangle \otimes A_0 + |\varphi_\perp\rangle \otimes A_\perp,$$

where $|\varphi_\perp\rangle$ is orthogonal to $|\varphi_0\rangle$. Then

$$\frac{1}{q^m}\mathrm{Tr}((\langle\varphi_\perp| \otimes A_\perp)(I \otimes |\psi\rangle\langle\psi|)(|\varphi_\perp\rangle \otimes A_\perp)) = \mathbf{Var}_{\mathrm{global}}(A, \psi).$$

Proof. We first compute

$$A_0 = (\langle\varphi_0| \otimes I) \cdot A_{\mathrm{comb}} = \left(\frac{1}{M^{1/2}} \cdot \sum_{u\in\mathbb{F}_q^m} \langle u|\right) \otimes I \cdot \left(\sum_{u\in\mathbb{F}_q^m} |u\rangle \otimes A^u \otimes I\right) = \frac{1}{M^{1/2}} \cdot \sum_{u\in\mathbb{F}_q^m} A^u \otimes I.$$

Writing $A_{\mathrm{avg}} = \mathbb{E}_u A^u$, it follows that

$$|\varphi_0\rangle \otimes A_0 = \sum_{u\in\mathbb{F}_q^m} |u\rangle \otimes A_{\mathrm{avg}} \otimes I, \qquad |\varphi_\perp\rangle \otimes A_\perp = \sum_{u\in\mathbb{F}_q^m} |u\rangle \otimes (A^u - A_{\mathrm{avg}}) \otimes I.$$

Hence

$$\frac{1}{M} \cdot \mathrm{Tr}((\langle\varphi_\perp| \otimes A_\perp)(I \otimes |\psi\rangle\langle\psi|)(|\varphi_\perp\rangle \otimes A_\perp)) = \mathbb{E}_{u\sim\mathbb{F}_q^m} \langle\psi|(A^u - A_{\mathrm{avg}})^2 \otimes I|\psi\rangle. \tag{5.3}$$

Moreover,

$$\mathbb{E}_{u\sim\mathbb{F}_q^m}(A^u - A_{\mathrm{avg}})^2 = \mathbb{E}_{u\sim\mathbb{F}_q^m}((A^u)^2 - (A_{\mathrm{avg}})^2) = \frac{1}{2}\mathbb{E}_{u,v\sim\mathbb{F}_q^m}(A^u - A^v)^2.$$

Substituting this identity into (5.3) yields

$$\frac{1}{M} \cdot \mathrm{Tr}((\langle\varphi_\perp| \otimes A_\perp)(I \otimes |\psi\rangle\langle\psi|)(|\varphi_\perp\rangle \otimes A_\perp)) = \frac{1}{2}\mathbb{E}_{u,v\sim\mathbb{F}_q^m} \langle\psi|(A^u - A^v)^2 \otimes I|\psi\rangle = \mathbf{Var}_{\mathrm{global}}(A, \psi).$$

$\square$

Lemma 5.12 (Local-to-global inequality)**.** *For every family* $\{A^u\}$*,*

$$\mathbf{Var}_{\mathrm{global}}(A, \psi) \leq m\, \mathbf{Var}_{\mathrm{local}}(A, \psi).$$

Proof.

$$\begin{aligned} A^\dagger_{\mathrm{comb}}(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}} &= ((\langle\varphi_0| \otimes A_0 + \langle\varphi_\perp| \otimes A_\perp)(L \otimes |\psi\rangle\langle\psi|)(|\varphi_0\rangle \otimes A_0 + |\varphi_\perp\rangle \otimes A_\perp) \\ &= ((\langle\varphi_\perp| \otimes A_\perp)(L \otimes |\psi\rangle\langle\psi|)(|\varphi_\perp\rangle \otimes A_\perp) \\ &= \langle\varphi_\perp|L|\varphi_\perp\rangle \cdot A_\perp|\psi\rangle\langle\psi|A_\perp. \end{aligned} \tag{5.4}$$

The second step uses that $|\varphi_0\rangle$ is a 0-eigenvector of L. Since $|\varphi_\perp\rangle$ is orthogonal to $|\varphi_0\rangle$, Lemma 5.8 gives

$$\langle\varphi_\perp|L|\varphi_\perp\rangle \geq \frac{1}{mM} \cdot \langle\varphi_\perp \mid \varphi_\perp\rangle.$$

Therefore

$$\mathbf{Var}_{\mathrm{local}}(A, \psi) = \mathrm{Tr}(A^\dagger_{\mathrm{comb}}(L \otimes |\psi\rangle\langle\psi|)A_{\mathrm{comb}}) = \langle\varphi_\perp|L|\varphi_\perp\rangle \cdot \mathrm{Tr}(A_\perp|\psi\rangle\langle\psi|A_\perp)$$

by Lemma 5.10 and (5.4), so

$$\mathbf{Var}_{\mathrm{local}}(A, \psi) \geq \frac{1}{mM} \cdot \mathrm{Tr}((\langle\varphi_\perp| \otimes A_\perp)(I \otimes |\psi\rangle\langle\psi|)(|\varphi_\perp\rangle \otimes A_\perp)) = \frac{1}{m}\mathbf{Var}_{\mathrm{global}}(A, \psi),$$

where the last step is Lemma 5.11. Rearranging gives the claim. $\square$

Chapter 6

Global variance of the points measurements

Throughout this chapter, (ψ, A, B, L) is an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the (m, q, d) low individual degree test.

Lemma 6.1 (Replacing a line answer by a fixed polynomial)**.** *Let $G \in \mathrm{PolySub}(m, q, d)$. On the axis-parallel line-test distribution,*

$$B^{\ell}_{[f(u)=g(u)]} \otimes (G_g)^{1/2} \approx_{md/q} B^{\ell}_{g|_{\ell}} \otimes (G_g)^{1/2}.$$

Proof. Expand the $\approx_{md/q}$ norm on the weighted vector $(I \otimes (G_g)^{1/2})|\psi\rangle$. Because B^{ℓ} is a projective measurement, only those line polynomials f with $f \neq g|_{\ell}$ and $f(u) = g(u)$ contribute. Lemma 3.9 is the univariate Schwartz–Zippel wrapper used in Lean for this line-restriction event; it supplies the paper's md/q loss, and the remaining operator sum is bounded by the fact that G is a submeasurement. □

Remark 6.2 (Conditional variants for the collision-residual estimate)*.* The paper proof of Lemma 6.1 bounds the collision event directly by the Schwartz–Zippel estimate. The following conditional variants instead assume this collision estimate, or one of the reindexing identities that implies it, as an explicit hypothesis. These conditional statements record proof obligations to be discharged; they are not additional hypotheses in the paper statement of Lemma 6.1.

Lemma 6.3 (Local variance of the points measurements)**.** *Let $G \in \mathrm{PolySub}(m, q, d)$. On the hypercube edge distribution $(u, v) \sim C$,*

$$A^{u}_{g(u)} \otimes (G_g)^{1/2} \approx_{24\cdot\left(\varepsilon+\delta+\frac{md}{q}\right)} A^{v}_{g(v)} \otimes (G_g)^{1/2}. \tag{6.1}$$

Proof. Fix g and keep the weight $(G_g)^{1/2}$ on the second tensor factor throughout. On the vector $(I \otimes (G_g)^{1/2})|\psi\rangle$ one has the six-step chain

$$A^{u}_{g(u)} \otimes (G_g)^{1/2} \approx_{2\delta} I \otimes (G_g)^{1/2} A^{u}_{g(u)} \approx_{2\varepsilon} B^{\ell}_{[f(u)=g(u)]} \otimes (G_g)^{1/2} \approx_{md/q} B^{\ell}_{g|_{\ell}} \otimes (G_g)^{1/2} \approx_{md/q} B^{\ell}_{[f(v)=g(v)]} \otimes (G_g)^{1/2} \approx_{2\varepsilon} I \otimes (G$$

The first, second, fifth, and sixth comparisons use Lemma 3.21 together with Theorem 3.25 and the good-strategy hypotheses, and the middle pair is Lemma 6.1. Lemma 3.27 sums these six weighted transports. □

Lemma 6.4 (Equivalent local-variance bound)**.** *The local-variance bound for the point measurements is*

$$\sum_{g \in \mathcal{P}(m,q,d)} \mathbb{E}_{(u,v)\sim C} \langle \psi | (A^u_{g(u)} - A^v_{g(v)})^2 \otimes G_g | \psi \rangle \le 24 \left(\varepsilon + \delta + \frac{md}{q} \right). \tag{6.2}$$

Remark 6.5 (Polynomial-sum formalization for 6.2). The three referenced declarations and Lemma 6.4 together provide the full polynomial-sum (cardinality-free) chain proving the equation. The chain assembly lemma `localVarianceDeviation_sum_le_localVarianceOfPointsError` telescopes the six operator differences via `ev_sum_conjTranspose_mul_sum_le`, sums over g, and applies the six individual `_sum` bounds (including the reverse `generalizeB` sum) to avoid the per-polynomial cardinality blow-up.

Lemma 6.6 (Global variance of the points measurements)**.** *Let* $G \in \mathrm{PolySub}(m,q,d)$*. On the uniform distribution over independent* $u, v \in \mathbb{F}_q^m$*,*

$$A^u_{g(u)} \otimes (G_g)^{1/2} \approx_{24m\left(\varepsilon+\delta+\frac{md}{q}\right)} A^v_{g(v)} \otimes (G_g)^{1/2}. \tag{6.3}$$

Proof. We want to bound

$$\mathbb{E}_{u,v\sim\mathbb{F}_q^m} \sum_{g \in \mathcal{P}(m,q,d)} \|(A^u_{g(u)} - A^v_{g(v)}) \otimes (G_g)^{1/2} |\psi\rangle\|^2 = \mathbb{E}_{u,v\sim\mathbb{F}_q^m} \sum_{g \in \mathcal{P}(m,q,d)} \langle \psi | (A^u_{g(u)} - A^v_{g(v)})^2 \otimes G_g | \psi \rangle. \tag{6.4}$$

For each $g \in \mathcal{P}(m,q,d)$, define

$$A(g)^u := A^u_{g(u)}, \qquad |\psi_g\rangle := (I \otimes (G_g)^{1/2})|\psi\rangle.$$

Then

$$(6.4) = \sum_{g \in \mathcal{P}(m,q,d)} \mathbb{E}_{u,v\sim\mathbb{F}_q^m} \langle \psi_g | (A(g)^u - A(g)^v)^2 \otimes I | \psi_g \rangle = \sum_{g \in \mathcal{P}(m,q,d)} 2\,\mathbf{Var}_{\mathrm{global}}(A(g), \psi_g).$$

Applying Lemma 5.12 gives

$$(6.4) \le \sum_{g \in \mathcal{P}(m,q,d)} 2m\,\mathbf{Var}_{\mathrm{local}}(A(g), \psi_g) = m \cdot \sum_{g \in \mathcal{P}(m,q,d)} \mathbb{E}_{(u,v)\sim C} \langle \psi | (A^u_{g(u)} - A^v_{g(v)})^2 \otimes G_g | \psi \rangle.$$

The last quantity is at most $m \cdot 24 \left(\varepsilon + \delta + \frac{md}{q}\right)$ by Lemma 6.4, which is exactly 6.3. □

Chapter 7

Self-improvement

Throughout this chapter, (ψ, A, B, L) is an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the (m, q, d) low individual degree test.

7.1 Non-projective Output

A large part of this chapter is devoted to proving Lemma 7.1, which is a slightly weaker form of the projective self-improvement theorem proved in Section 7.2. The key difference is that the output family H is only required to be a submeasurement, rather than a projective submeasurement. Once this non-projective statement is available, Theorem 4.2 upgrades H to a projective family without losing control of the four quantitative conclusions.

Compared with the projective theorem, the helper lemma packages strong self-consistency using the explicit inequality from Definition 3.34, keeps the boundedness conclusion in terms of an auxiliary operator Z, and produces a much smaller error parameter.

Lemma 7.1 (Self-improvement with non-projective output)**.** *Let $G \in \mathrm{PolyMeas}(m, q, d)$ be a measurement with the following property:*

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\nu I \otimes G_{[g(u)=a]}.$$

Let

$$\zeta = 100m \cdot \left(\varepsilon^{1/2} + \delta^{1/2} + (d/q)^{1/2}\right).$$

Then there exists $H \in \mathrm{PolySub}(m, q, d)$ with the following properties:

1. *(Completeness): If $H = \sum_h H_h$, then*

$$\langle \psi | H \otimes I | \psi \rangle \geq (1 - \nu) - \zeta.$$

2. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\zeta I \otimes H_{[h(u)=a]}.$$

3. *(Strong self-consistency):*

$$\sum_h \langle \psi | H_h \otimes H_h | \psi \rangle \geq \langle \psi | H \otimes I | \psi \rangle - \zeta.$$

4. (Boundedness): There exists a positive-semidefinite matrix Z such that

$$\langle\psi|Z \otimes I|\psi\rangle - \mathbb{E}_u \sum_a \langle\psi|A^u_a \otimes H_{[h(u)=a]}|\psi\rangle \leq \zeta$$

and for each $h \in \mathcal{P}(m,q,d)$,

$$Z \geq \left(\mathbb{E}_u A^u_{h(u)}\right).$$

Proof. Let $T = \{T_g\}$ and Z be the optimal solutions to the SDPs (7.23) and (7.24) given by Lemma 7.7. Then T is a measurement, and

$$\forall g \in \mathcal{P}(m,q,d): \qquad Z \geq \left(\mathbb{E}_u A^u_{g(u)}\right), \tag{7.1}$$

$$T_g \cdot Z = T_g \cdot \left(\mathbb{E}_u A^u_{g(u)}\right). \tag{7.2}$$

For each $u \in \mathbb{F}_q^m$, define $H^u = \{H^u_h\}_{h \in \mathcal{P}(m,q,d)}$ by

$$H^u_h := A^u_{h(u)} \cdot T_h \cdot A^u_{h(u)},$$

and let

$$H_h := \mathbb{E}_u H^u_h.$$

The pointwise family H^u is a submeasurement because

$$\sum_h H^u_h = \sum_a A^u_a \cdot \Big(\sum_{h:h(u)=a} T_h\Big) \cdot A^u_a \leq \sum_a (A^u_a)^2 = I,$$

where the inequality uses that T is a measurement and the last identity uses that A is projective. Averaging over u gives

$$\sum_h H_h = \mathbb{E}_u \sum_h H^u_h \leq I,$$

so H is also a submeasurement and therefore lies in $\mathrm{PolySub}(m,q,d)$.

Set

$$\zeta_{\text{variance}} = 24m \cdot \Big(\varepsilon + \delta + \frac{md}{q}\Big).$$

Lemma 7.8 below is the technical transfer from the averaged family H back to the sandwiched operators $A^u_{h(u)} T_h A^u_{h(u)}$. We use it in the completeness, consistency, strong self-consistency, and boundedness estimates that follow.

We now verify the four conclusions of Lemma 7.1.

Proof of 1. The completeness of H is

$$\begin{aligned}\sum_h \langle\psi|H_h \otimes I|\psi\rangle &= \mathbb{E}_u \sum_h \langle\psi|H^u_h \otimes I|\psi\rangle \\ &= \mathbb{E}_u \sum_h \langle\psi|(A^u_{h(u)} \cdot T_h \cdot A^u_{h(u)}) \otimes I|\psi\rangle. \end{aligned} \tag{7.3}$$

Grouping the polynomials according to their value at u gives

$$\mathbb{E}_u \sum_h \langle\psi|(A^u_{h(u)} \cdot T_h \cdot A^u_{h(u)}) \otimes I|\psi\rangle = \mathbb{E}_u \sum_a \langle\psi|(A^u_a \cdot T_{[h(u)=a]} \cdot A^u_a) \otimes I|\psi\rangle. \tag{7.4}$$

We first move the leftmost copy of A^u_a across the bipartition:

$$(7.4) \approx_{2\sqrt{\delta}} \mathbb{E}_u \sum_a \langle\psi|(T_{[h(u)=a]} \cdot A^u_a) \otimes A^u_a|\psi\rangle. \tag{7.5}$$

The difference is bounded by Cauchy–Schwarz as in (7.33); the first factor is at most $\sqrt{2\delta}$ by self-consistency of A, and the second is at most 1 because $T_{[h(u)=a]} \leq I$. We next remove the remaining copy of A^u_a on Bob's side:

$$(7.5) \approx_{\sqrt{\delta}} \mathbb{E}_u \sum_a \langle\psi|(T_{[h(u)=a]} \cdot A^u_a) \otimes I|\psi\rangle. \tag{7.6}$$

The first Cauchy–Schwarz factor is at most 1 because the fiber operators form a submeasurement after grouping T by the value of $h(u)$. Here the second Cauchy–Schwarz factor is

$$\mathbb{E}_u \sum_{a\neq b} \langle\psi|A^u_a \otimes A^u_b|\psi\rangle,$$

which is at most δ because A is self-consistent and projective. Reindexing by h and using (7.2),

$$(7.6) = \sum_h \langle\psi|(T_h \cdot \mathbb{E}_u A^u_{h(u)}) \otimes I|\psi\rangle = \sum_h \langle\psi|(T_h \cdot Z) \otimes I|\psi\rangle = \langle\psi|Z \otimes I|\psi\rangle.$$

Thus

$$\sum_h \langle\psi|H_h \otimes I|\psi\rangle \geq \langle\psi|Z \otimes I|\psi\rangle - 3\sqrt{\delta}. \tag{7.7}$$

Since G is a measurement, (7.1) and Lemma 3.18 give

$$\langle\psi|Z \otimes I|\psi\rangle \geq \mathbb{E}_u \sum_a \langle\psi|A^u_a \otimes G_{[g(u)=a]}|\psi\rangle \geq 1 - \nu.$$

Therefore the completeness is at least $1 - \nu - 3\sqrt{\delta}$, which is bounded below by $(1-\nu) - \zeta$.

Proof of 2. The inconsistency of H with A is

$$\mathbb{E}_u \sum_{a\neq b} \langle\psi|A^u_a \otimes H_{[h(u)=b]}|\psi\rangle = \mathbb{E}_u \sum_{a,h:h(u)\neq a} \langle\psi|A^u_a \otimes H_h|\psi\rangle. \tag{7.8}$$

The theorem-side off-diagonal selection used for this 'add-in-u' application is , and the corresponding left/right quantity wrappers are and . The selected scalar chain for this specialization is recorded by . Apply Lemma 7.8 with $\mathcal{O} = \mathbb{F}_q$, $M = A$, and $S_u = \{(a,h) : h(u) \neq a\}$. Then

$$(7.8) \approx_{4\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_u \sum_{a,h:h(u)\neq a} \langle\psi|(A^u_{h(u)} \cdot A^u_a \cdot A^u_{h(u)}) \otimes T_h|\psi\rangle,$$

and the right-hand side is 0 because A is projective. Hence

$$\mathbb{E}_u \sum_{a\neq b} \langle\psi|A^u_a \otimes H_{[h(u)=b]}|\psi\rangle \leq 4\sqrt{\zeta_{\text{variance}}}. \tag{7.9}$$

Since

$$4\sqrt{\zeta_{\text{variance}}} = 4\sqrt{24m \cdot \left(\varepsilon + \delta + \frac{md}{q}\right)} \leq 20m \cdot \left(\varepsilon^{1/2} + \delta^{1/2} + (d/q)^{1/2}\right) \leq \zeta,$$

this proves 2.

Proof of 3. From the definition of H^u_h and the projectivity of A,

$$H^u_h = A^u_{h(u)} \cdot T_h \cdot A^u_{h(u)} = A^u_{h(u)} \cdot H^u_h \cdot A^u_{h(u)}, \tag{7.10}$$

$$A^u_{h(u)} \cdot H^u_{h'} \cdot A^u_{h(u)} = H^u_{h'} \cdot A^u_{h(u)} = (A^u_{h(u)} \cdot T_{h'} \cdot A^u_{h(u)}) \cdot \mathbf{1}[h(u) = h'(u)]\,. \tag{7.11}$$

The strong self-consistency expression is

$$\sum_{h\in\mathcal{P}(m,q,d)} \langle\psi|H_h\otimes H_h|\psi\rangle = \mathbb{E}_u \sum_{h\in\mathcal{P}(m,q,d)} \langle\psi|H_h^u\otimes H_h|\psi\rangle. \tag{7.12}$$

Apply Lemma 7.8 with $\mathcal{O}=\mathcal{P}(m,q,d)$, $M=H$, and $S_u=\{(h,h):h\in\mathcal{P}(m,q,d)\}$. This gives

$$(7.12) \approx_{4\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_u \sum_{h\in\mathcal{P}(m,q,d)} \langle\psi|(A^u_{h(u)}\cdot H^u_h\cdot A^u_{h(u)})\otimes T_h|\psi\rangle. \tag{7.13}$$

We next enlarge the sum from $h'=h$ to all pairs h,h':

$$(7.13) \approx_{2\sqrt{\zeta_{\text{variance}}}+\frac{md}{q}} \mathbb{E}_u \sum_{h,h'\in\mathcal{P}(m,q,d)} \langle\psi|(A^u_{h(u)}\cdot H^u_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle. \tag{7.14}$$

The added off-diagonal contribution is

$$\begin{aligned}(7.14)-(7.13) &= \mathbb{E}_u \sum_{h\neq h'} \langle\psi|(A^u_{h(u)}\cdot H^u_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle \\ &= \mathbb{E}_u \sum_{h\neq h'} \langle\psi|(A^u_{h(u)}\cdot T_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle\cdot \mathbf{1}[h(u)=h'(u)]\,,\end{aligned} \tag{7.15}$$

where the second equality is (7.11). To estimate (7.15), first replace the left copy of $A^u_{h(u)}$ by $A^v_{h(v)}$:

$$(7.15) \approx_{\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|(A^v_{h(v)}\cdot T_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle\cdot \mathbf{1}[h(u)=h'(u)]\,. \tag{7.16}$$

The corresponding Cauchy–Schwarz bound is

$$\begin{aligned}&\left|\mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|((A^u_{h(u)}-A^v_{h(v)})\cdot T_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle\cdot \mathbf{1}[h(u)=h'(u)]\right| \\ &\qquad\le \sqrt{\mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|((A^u_{h(u)}-A^v_{h(v)})\cdot T_{h'}\cdot (A^u_{h(u)}-A^v_{h(v)}))\otimes T_h|\psi\rangle} \\ &\qquad\qquad\cdot \sqrt{\mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|(A^u_{h(u)}\cdot T_{h'}\cdot A^u_{h(u)})\otimes T_h|\psi\rangle\cdot \mathbf{1}[h(u)=h'(u)]}.\end{aligned} \tag{7.17}$$

The first factor is bounded by $\sqrt{\zeta_{\text{variance}}}$ via Lemma 6.6, and the second by 1 because T and H^u are submeasurements.

Repeating the same argument for the remaining copy of A yields

$$(7.16) \approx_{\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|(A^v_{h(v)}\cdot T_{h'}\cdot A^v_{h(v)})\otimes T_h|\psi\rangle\cdot \mathbf{1}[h(u)=h'(u)]\,. \tag{7.18}$$

The same intermediate estimate gives

$$\mathbb{E}_{u,v} \sum_{h\neq h'} \langle\psi|(A^v_{h(v)}\cdot T_{h'}\cdot A^v_{h(v)})\otimes T_h|\psi\rangle \le 1, \tag{7.19}$$

so Schwartz–Zippel bounds the indicator average by md/q and proves (7.14).

Using (7.11), the right-hand side of (7.14) is

$$\mathbb{E}_u \sum_{h,h' \in \mathcal{P}(m,q,d)} \langle\psi|(H^u_{h'} \cdot A^u_{h(u)}) \otimes T_h|\psi\rangle. \tag{7.20}$$

Replace the remaining copy of $A^u_{h(u)}$ by $A^v_{h(v)}$:

$$(7.20) \approx_{\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_{u,v} \sum_{h,h' \in \mathcal{P}(m,q,d)} \langle\psi|(H^u_{h'} \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle. \tag{7.21}$$

Here the first Cauchy–Schwarz factor is at most 1 because T and H^u are submeasurements, and the second is again bounded by Lemma 6.6. Finally move this last copy of $A^v_{h(v)}$ to Bob's side:

$$(7.21) \approx_{\sqrt{2\delta}} \mathbb{E}_{u,v} \sum_{h,h' \in \mathcal{P}(m,q,d)} \langle\psi|H^u_{h'} \otimes (T_h \cdot A^v_{h(v)})|\psi\rangle. \tag{7.22}$$

Substituting (7.2), then (7.1), and finally the already proved bound (7.9), one obtains

$$(7.22) \geq \sum_h \langle\psi|H_h \otimes I|\psi\rangle - 4\sqrt{\zeta_{\text{variance}}}.$$

Combining (7.13), (7.14), (7.21), and (7.22) gives

$$\sum_h \langle\psi|H_h \otimes H_h|\psi\rangle \geq \sum_h \langle\psi|H_h \otimes I|\psi\rangle - 11\sqrt{\zeta_{\text{variance}}} - \sqrt{2\delta} - \frac{md}{q}.$$

The Lean producer-side packaging of this step starts with the input structure and the exact residual-side expansion and then uses the proved wrappers to recover the final helper-stage self-consistency input. The corresponding bridge-package constructor is . The paper's arithmetic shows that the right-hand error is at most

$$57m \cdot \left(\varepsilon^{1/2} + \delta^{1/2} + (d/q)^{1/2}\right) \leq \zeta,$$

which proves 3.

Proof of 4. The helper-stage boundedness quantity is

$$\langle\psi|Z \otimes I|\psi\rangle - \mathbb{E}_u \sum_a \langle\psi|A^u_a \otimes H_{[h(u)=a]}|\psi\rangle.$$

The reindexing of the sum by h is the algebraic identity

$$\sum_a A^u_a \otimes H_{[h(u)=a]} = \sum_h A^u_{h(u)} \otimes H_h,$$

whose averaged scalar form reads

$$\mathbb{E}_u \sum_a \langle\psi|\, A^u_a \otimes H_{[h(u)=a]}\,|\psi\rangle = \mathbb{E}_u \sum_h \langle\psi|\, A^u_{h(u)} \otimes H_h\,|\psi\rangle.$$

Using $\sum_a A^u_a = I$, the pointwise difference between the right-placed total $I \otimes H.\text{total} = \sum_h I \otimes H_h$ and the helper-agreement operator at u reindexes as the off-diagonal sum

$$\sum_h I \otimes H_h - \sum_a A^u_a \otimes H_{[h(u)=a]} = \sum_h \sum_{a \neq h(u)} A^u_a \otimes H_h,$$

whose averaged scalar form reads

$$\sum_h \langle\psi|\, I\otimes H_h\,|\psi\rangle - \mathbb{E}_u \sum_a \langle\psi|\, A^u_a \otimes H_{[h(u)=a]}\,|\psi\rangle = \mathbb{E}_u \sum_h \sum_{a\neq h(u)} \langle\psi|\, A^u_a\otimes H_h\,|\psi\rangle.$$

Combined with (7.9) this gives

$$\langle\psi|Z\otimes I|\psi\rangle - \mathbb{E}_u \sum_a \langle\psi|A^u_a\otimes H_{[h(u)=a]}|\psi\rangle \le \langle\psi|Z\otimes I|\psi\rangle - \sum_h \langle\psi|I\otimes H_h|\psi\rangle + 4\sqrt{\zeta_{\text{variance}}}.$$

By (7.7), the right-hand side is at most $3\sqrt{\delta}+4\sqrt{\zeta_{\text{variance}}}$. Substituting the definition of ζ_{variance} and collecting terms gives

$$23m\cdot\left(\varepsilon^{1/2}+\delta^{1/2}+(d/q)^{1/2}\right)\le\zeta.$$

Together with (7.1), this proves 4. □

Remark 7.2 (Slackness-carrying self-improvement helper). The formalization also records a strengthened helper theorem whose output carries the complementary-slackness equations used in the completeness chain, together with the residual-domination bridge that assembles the full self-improvement conclusion from the slackness-carrying helper. An intermediate assembly route routes a QXP repair producer plus the construction-level coisometry identity through the same bridge. These theorems use the Section 9 statement above. Their proofs are not yet complete, because the strong-duality proof for the SDP remains to be formalized.

7.1.1 A Semidefinite Program

Lemma 7.3 (Uniform SDP feasible witness)**.** *The uniform primal family $T_g = (2|\mathcal{P}(m,q,d)|)^{-1}I$ has total mass $(1/2)I$, and the dual operator $Z = 2I$ is positive semidefinite and dominates the identity.*

Proof. This is the elementary Slater-type feasible witness recorded in the basic self-improvement definitions. □

Lemma 7.4 (Explicit matrix feasible bounds)**.** *In every concrete matrix realization, the uniform family $T_g = (2|\mathcal{P}(m,q,d)|)^{-1}I$ has total mass $(1/2)I$. The dual operator $Z = 2I$ dominates the identity and satisfies $Z - A_g \ge I$, hence $Z \ge A_g$, for every polynomial g. In the canonical block SDP, the corresponding block matrix is feasible, its slack block is $(1/2)I$, and the canonical dual slack for the same dual witness also dominates the block identity. More generally, dual feasibility implies positivity of Z, since the averaged point operators A_g are positive semidefinite.*

Proof. The formal proof is the matrix specialization of the uniform SDP witness recorded in the basic definitions. The inequality $A_g \le I$ follows by averaging the point-measurement effects and using the submeasurement bound at each point. □

Lemma 7.5 (Matrix slackness output)**.** *From a matrix-level strong-duality witness with complementary slackness of the shape asserted by Lemma 7.7, one obtains a complete measurement $\{T_g\}$, a feasible dual operator Z, equality of the primal and dual objectives, and the equations $T_gZ = T_gA_g$ for all g. The formal bridge to the current abstract helper interface is conditional on the additional reduced-interface bound $I \le Z$ for this selected dual witness. The same optimal witness can also be considered together with this dominance condition. With the usual helper hypotheses that the strategy is good, the error parameter ν, and the comparison measurement G, this datum feeds the slackness-carrying self-improvement helper. Lean also isolates the saturated* *canonical optimal-pair output still required from strong duality: a feasible canonical primal matrix, a dual-feasible operator with equal objective value, canonical complementary slackness, and vanishing of the extra canonical slack block. This datum gives the displayed measurement witness without assuming $I \le Z$. Lean also proves the converse saturation step for the dominance-carrying optimal pair: the canonical complementary-slackness equation on the extra block gives $X_{\mathrm{none,none}} Z = 0$, and the lower bound $I \le Z$ forces $X_{\mathrm{none,none}} = 0$. Thus the dominance-carrying optimal pair canonically determines the saturated optimal pair, and hence also the matrix slackness statement without the dominance field. The remaining bridge is first stated as an abstract SDP statement, and then as a direct helper input, for a canonical feasible primal matrix satisfying objective equality and canonical complementary slackness together with the auxiliary dominance hypothesis.*

Proof. The identity $\sum_g T_g = I$ turns the optimal primal submeasurement into a measurement. The zero-defect form $T_g(Z - A_g) = 0$ is equivalent, by distributivity, to the displayed equation $T_g Z = T_g A_g$. □

Remark 7.6 (Residual-dominating self-improvement assembly). These auxiliary results combine the matrix slackness output with a residual-dominating orthonormalization input. The residual-domination hypothesis can be supplied either as an intrinsic self-improvement orthonormalization input or by a QXP repair construction carrying the fresh-outcome domination invariant. These are intermediate proof obligations for the projective-output step, not part of the paper statement of Lemma 7.5.

Lemma 7.7 (Dual SDP and complementary slackness). *Let*

$$A_g = \mathbb{E}_{u \sim \mathbb{F}_q^m} A^u_{g(u)}.$$

Then the primal is

$$\begin{aligned} \sup \quad & \sum_g \mathrm{Tr}(T_g \cdot A_g) && (7.23) \\ s.t. \quad & T_g \ge 0 \qquad \forall g \in \mathcal{P}(m,q,d), \\ & \sum_g T_g \le I, \end{aligned}$$

and the dual is

$$\begin{aligned} \inf \quad & \mathrm{Tr}(Z) && (7.24) \\ s.t. \quad & Z \ge A_g. && (7.25) \end{aligned}$$

These semidefinite programs are dual to each other. Moreover there is an optimal pair of solutions $\{T_g\}$ to (7.23) and Z to (7.24) such that $\sum_g T_g = I$ and

$$T_g Z \;=\; T_g A_g, \qquad \forall g \in \mathcal{P}(m,q,d). \tag{7.26}$$

Proof. Rewrite (7.23) in canonical block form. Let r be the dimension of the space on which A acts, let $M = |\mathcal{P}(m,q,d)|$, and fix an ordering $g_1, \dots, g_M$ of the polynomials. Consider

$$\begin{aligned} \sup \quad & \mathrm{Tr}(C^\dagger X) && (7.27) \\ \text{s.t.} \quad & \mathrm{Tr}(D_{ij}^\dagger X) = b_{ij} \qquad \forall i,j \in \{1,\dots,r\}, \\ & X \ge 0, \end{aligned}$$

where

$$C = \sum_{i=1}^{M} |i\rangle\langle i| \otimes A_{g_i},$$

and, for $i, j \in \{1, \dots, r\}$,

$$D_{ij} = \sum_{k=1}^{M+1} |k\rangle\langle k| \otimes |i\rangle\langle j|, \qquad b_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

Writing $X = \sum_{i,j=1}^{M+1} |i\rangle\langle j| \otimes X_{ij}$, the constraints say precisely that $\sum_{i=1}^{M+1} X_{ii} = I$, hence $\sum_{i=1}^{M} X_{ii} \leq I$, and the objective is

$$\operatorname{Tr}(C^\dagger X) = \sum_{i=1}^{M} \operatorname{Tr}(A_{g_i} \cdot X_{ii}).$$

Thus (7.27) is equivalent to (7.23) under the identification $T_{g_i} = X_{ii}$.

The canonical dual is

$$\inf \quad \sum_{i} z_{ij} b_{ij} \tag{7.28}$$

$$\text{s.t.} \quad \sum_{i,j} z_{ij} D_{ij} \geq C. \tag{7.29}$$

Since

$$\sum_{i,j=1}^{r} z_{ij} D_{ij} = \sum_{k=1}^{M+1} |k\rangle\langle k| \otimes \left(\sum_{i,j=1}^{r} z_{ij} |i\rangle\langle j| \right) =: \sum_{k=1}^{M+1} |k\rangle\langle k| \otimes Z,$$

the constraint (7.29) is exactly $Z \geq A_{g_i}$ for every i, and the objective is $\operatorname{Tr}(Z)$. In Lean, the latter statement is recorded as the trace-pairing identity

$$\Re \operatorname{Tr}(D(Z)X) = \Re \operatorname{Tr}(Z)$$

for every feasible canonical primal matrix X, where $D(Z)$ denotes the block-diagonal canonical dual operator. Consequently the canonical primal-dual gap is the real trace pairing of X with the canonical dual slack $D(Z) - C$. Since the real trace pairing of positive semidefinite operators is nonnegative, Lean also records the resulting weak-duality inequality for any feasible canonical primal and dual pair. Hence (7.28) is equivalent to (7.24).

Slater's condition holds because

$$T_g = \frac{1}{2M} \cdot I \qquad \forall g \in \mathcal{P}(m, q, d), \qquad Z = 2I$$

are strictly feasible for the primal and dual respectively. Therefore strong duality holds. For an optimal pair $(X, (z_{ij}))$ in canonical form, complementary slackness gives

$$X\Big(\sum_{i,j} z_{ij} D_{ij} - C \Big) \;=\; 0. \tag{7.30}$$

We may assume that X is block diagonal. Translating (7.30) back to the variables $\{T_g\}$ and Z yields $T_g(Z - A_g) = 0$ for every g and $SZ = 0$, where $S = I - \sum_g T_g$ is the slack block. The paper then treats this as forcing $S = 0$. The current Lean development separates this point from the unconditional SDP statement: it proves the displayed saturation from the auxiliary dominance hypothesis $I \le Z$. Thus, if the canonical complementary-slackness pair is supplied together with $I \le Z$, Lean packages these conditional conclusions as a `MatrixSdpOptimalWitnessWithDominance` once dual feasibility and equality of the paper primal and dual objectives have also been supplied. It also records the corresponding statement-level package, and an elimination theorem which exposes the complete measurement, the dual witness Z, the retained dominance $I \le Z$, dual feasibility, strong duality, and the equations (7.26). The additional dominance requirement is not a consequence of the dual constraints $Z \ge A_g$ alone; the discrepancy and the intended saturated-witness replacement are recorded in `docs/paper-gaps/issue-1230-self-improvement-sdp-usage.tex`. In the concrete matrix realization, Lean records the defect-zero condition as `matrixSdpComplementarySlacknessDefe` the theorem `MatrixSdpOptimalWitness.complementarySlacknessEquation` translates it to the displayed equation $T_g Z = T_g A_g$, while `MatrixSdpOptimalWitness.primalMeasurement` packages $\sum_g T_g = I$ as a complete measurement. The statement `MatrixSdpStatementWithSlackness` records this matrix-level strong-duality output without identifying it with the reduced feasible witness supplied by `sdp`. The explicit Slater witnesses used above are separately recorded by `matrixSdpStrictPrimalSubmeasurement`, `matrixSdpStrictDualWitness`, and `matrixSdpFeasibleBounds_canoni` the dual feasibility of $2I$ follows from `matrixAveragedPointOperator_le_one`. The block-diagonal reduction is also recorded for arbitrary feasible canonical matrices: extracting the diagonal blocks preserves the objective and transfers canonical complementary slackness to the extracted paper primal submeasurement. $\square$

Lemma 7.8 (Averaging over the evaluation point)**.** *Let $T = \{T_h\}$ be an optimal solution of the primal SDP, let*

$$H_h = \mathbb{E}_{u \sim \mathbb{F}_q^m} A^u_{h(u)} T_h A^u_{h(u)},$$

and set

$$\zeta_{\text{variance}} = 24m \cdot \left(\varepsilon + \delta + \frac{md}{q}\right).$$

Suppose $M = \{M^u_o\}$ is a submeasurement with outcomes in some set $\mathcal{O}$. For each $u \in \mathbb{F}_q^m$, let S_u be a subset of $\mathcal{O} \times \mathcal{P}(m, q, d)$. Then

$$\mathbb{E}_u \sum_{(o,h) \in S_u} \langle \psi | M^u_o \otimes H_h | \psi \rangle \approx_{4\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_u \sum_{(o,h) \in S_u} \langle \psi | (A^u_{h(u)} \cdot M^u_o \cdot A^u_{h(u)}) \otimes T_h | \psi \rangle.$$

Proof. We begin by expanding

$$\begin{aligned}
\mathbb{E}_u \sum_{(o,h) \in S_u} \langle \psi | M^u_o \otimes H_h | \psi \rangle &= \mathbb{E}_{u,v} \sum_{(o,h) \in S_u} \langle \psi | M^u_o \otimes H^v_h | \psi \rangle \\
&= \mathbb{E}_{u,v} \sum_{(o,h) \in S_u} \langle \psi | M^u_o \otimes (A^v_{h(v)} \cdot T_h \cdot A^v_{h(v)}) | \psi \rangle. \qquad (7.31)
\end{aligned}$$

We claim that

$$(7.31) \approx_{\sqrt{2\delta}} \mathbb{E}_{u,v} \sum_{(o,h) \in S_u} \langle \psi | (A^v_{h(v)} \cdot M^u_o) \otimes (T_h \cdot A^v_{h(v)}) | \psi \rangle. \qquad (7.32)$$

To show this, we bound the magnitude of the difference:

$$\begin{aligned}
&\Big|\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)}) \cdot (M^u_o \otimes (T_h \cdot A^v_{h(v)}))|\psi\rangle\Big| \\
&\quad\le \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)}) \cdot (M^u_o \otimes T_h) \cdot (A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)})|\psi\rangle} \\
&\qquad\qquad \cdot \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|M^u_o \otimes (A^v_{h(v)} \cdot T_h \cdot A^v_{h(v)})|\psi\rangle}. \qquad (7.33)
\end{aligned}$$

The first factor is controlled by summing over the value $a = h(v)$:

$$\begin{aligned}
&\mathbb{E}_{u,v} \sum_{a\in\mathbb{F}_q} \langle\psi|(A^v_a \otimes I - I \otimes A^v_a) \cdot \Big(\sum_{(o,h)\in S_u,\, h(v)=a} M^u_o \otimes T_h\Big) \cdot (A^v_a \otimes I - I \otimes A^v_a)|\psi\rangle \\
&\le \mathbb{E}_{u,v} \sum_{a\in\mathbb{F}_q} \langle\psi|(A^v_a \otimes I - I \otimes A^v_a)^2|\psi\rangle,
\end{aligned}$$

where the inequality uses that M^u and T are submeasurements. By Lemma 3.21 and the self-consistency of A, this is at most 2δ. The second factor is

$$\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|M^u_o \otimes H^v_h|\psi\rangle,$$

which is at most 1 because M^u and H^v are submeasurements.

Next, we claim that

$$(7.32) \approx_{\sqrt{2\delta}} \mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \cdot M^u_o \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle. \qquad (7.34)$$

Again, we bound the magnitude of the difference:

$$\begin{aligned}
&\Big|\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|((A^v_{h(v)} \cdot M^u_o) \otimes T_h) \cdot (A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)})|\psi\rangle\Big| \\
&\le \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \cdot M^u_o \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle} \qquad (7.35) \\
&\quad \cdot \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)}) \cdot (M^u_o \otimes T_h) \cdot (A^v_{h(v)} \otimes I - I \otimes A^v_{h(v)})|\psi\rangle}.
\end{aligned}$$

The first factor is bounded by

$$\begin{aligned}
&\mathbb{E}_{u,v} \sum_h \langle\psi|\Big(A^v_{h(v)} \cdot \sum_{o:(o,h)\in S_u} M^u_o \cdot A^v_{h(v)}\Big) \otimes T_h|\psi\rangle \\
&\le \mathbb{E}_{u,v} \sum_h \langle\psi|(A^v_{h(v)})^2 \otimes T_h|\psi\rangle \le \mathbb{E}_{u,v} \sum_h \langle\psi|I \otimes T_h|\psi\rangle \le 1,
\end{aligned}$$

using first that M is a submeasurement, then that $A^v_{h(v)} \le I$, and finally that T is a submeasurement ($\sum_g T_g \le I$). The second factor is exactly the same expression as in (7.33), hence at most $\sqrt{2\delta}$.

Having moved both copies of A to the left-hand side, we next show that

$$(7.34) \approx_{\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^u_{h(u)} \cdot M^u_o \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle. \tag{7.36}$$

The difference is bounded by

$$\begin{aligned}
&\Big|\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|((A^v_{h(v)} - A^u_{h(u)}) \cdot M^u_o \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle\Big| \\
&\qquad \le \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|((A^v_{h(v)} - A^u_{h(u)}) \cdot M^u_o \cdot (A^v_{h(v)} - A^u_{h(u)})) \otimes T_h|\psi\rangle} \\
&\qquad\qquad \cdot \sqrt{\mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^v_{h(v)} \cdot M^u_o \cdot A^v_{h(v)}) \otimes T_h|\psi\rangle}.
\end{aligned} \tag{7.37}$$

The first factor can be rewritten as

$$\begin{aligned}
&\mathbb{E}_{u,v} \sum_h \langle\psi|\Big((A^v_{h(v)} - A^u_{h(u)}) \cdot \sum_{o:(o,h)\in S_u} M^u_o \cdot (A^v_{h(v)} - A^u_{h(u)})\Big) \otimes T_h|\psi\rangle \\
&\le \mathbb{E}_{u,v} \sum_h \langle\psi|(A^v_{h(v)} - A^u_{h(u)})^2 \otimes T_h|\psi\rangle,
\end{aligned}$$

because M^u is a submeasurement. Lemma 6.6 bounds this by ζ_{variance} since $T \in \text{PolySub}(m, q, d)$. The second factor is exactly the first factor from (7.35), so it is at most 1.

Finally, we show that

$$(7.36) \approx_{\sqrt{\zeta_{\text{variance}}}} \mathbb{E}_{u,v} \sum_{(o,h)\in S_u} \langle\psi|(A^u_{h(u)} \cdot M^u_o \cdot A^u_{h(u)}) \otimes T_h|\psi\rangle. \tag{7.38}$$

The same Cauchy–Schwarz estimate as in (7.37), with the final copy of $A^v_{h(v)}$ replaced by $A^u_{h(u)}$, gives the same bound $\sqrt{\zeta_{\text{variance}}}$. Indeed, the factor containing $A^u_{h(u)} \cdot M^u_o \cdot A^u_{h(u)}$ is bounded by 1 by the same submeasurement argument as above, while the factor containing $(A^v_{h(v)} - A^u_{h(u)})^2$ is unchanged. Summing the four transports gives an error of $2\sqrt{2\delta} + 2\sqrt{\zeta_{\text{variance}}}$, and the lemma follows from $2\delta \le \zeta_{\text{variance}}$. □

7.2 Projective Output

This final step applies orthonormalization to the non-projective family produced by Lemma 7.1. The bookkeeping here is entirely about showing that completeness, consistency, self-consistency, and boundedness survive the passage from $\widehat{H}$ to a projective submeasurement H.

Remark 7.9 (Auxiliary self-improvement lemmas). Several auxiliary results are used to prove Theorem 7.10 from the helper output and the orthonormalization theorem. They record intermediate proof obligations that do not appear as hypotheses in the paper statement. The current Lean theorem still takes these obligations as explicit assumptions, so the theorem below is linked to the current Lean statement but is not marked as completely formalized.

Theorem 7.10 (Self-improvement)**.** *Let $G \in \text{PolyMeas}(m, q, d)$ be a measurement with the following properties:*

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\nu I \otimes G_{[g(u)=a]}.$$

Let

$$\zeta = 3000m \cdot \left(\varepsilon^{1/32} + \delta^{1/32} + (d/q)^{1/32}\right).$$

Then there exists a projective submeasurement $H \in \mathrm{PolySub}(m, q, d)$ with the following properties:

1. *(Completeness): If $H = \sum_h H_h$, then*

$$\langle \psi | H \otimes I | \psi \rangle \geq (1 - \nu) - \zeta.$$

2. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\zeta I \otimes H_{[h(u)=a]}.$$

3. *(Strong self-consistency):*

$$H_h \otimes I \approx_\zeta I \otimes H_h.$$

4. *(Boundedness): There exists a positive-semidefinite matrix Z such that*

$$\langle \psi | Z \otimes (I - H) | \psi \rangle \leq \zeta$$

and for each $h \in \mathcal{P}(m, q, d)$,

$$Z \geq \left(\mathbb{E}_u A_{h(u)}^u\right).$$

Proof. The bound is trivial if one of ε, δ, or d/q is at least 1, so assume all three are at most 1. In Lean, the third assumption is recorded as $d \leq q$, which is equivalent to $d/q \leq 1$ since $q > 0$. Apply Lemma 7.1 to G and let $\widehat{H} \in \mathrm{PolySub}(m, q, d)$ be the output, with

$$\hat{\zeta} = 100m \cdot \left(\varepsilon^{1/2} + \delta^{1/2} + (d/q)^{1/2}\right).$$

By 3 of Lemma 7.1,

$$\sum_h \langle \psi | \widehat{H}_h \otimes \widehat{H}_h | \psi \rangle \geq \langle \psi | \widehat{H} \otimes I | \psi \rangle - \hat{\zeta}.$$

Therefore Theorem 4.2 supplies a projective submeasurement $H \in \mathrm{PolySub}(m, q, d)$ such that

$$\widehat{H}_h \otimes I \approx_{\hat{\zeta}_{\mathrm{ortho}}} H_h \otimes I, \tag{7.39}$$

where

$$\hat{\zeta}_{\mathrm{ortho}} = 100\hat{\zeta}^{1/4}.$$

In addition, Lemma 3.39 implies that

$$\widehat{H}_{[h(u)=a]} \otimes I \approx_{\hat{\zeta}_{\mathrm{dataprocess}}} H_{[h(u)=a]} \otimes I, \tag{7.40}$$

where

$$\hat{\zeta}_{\mathrm{dataprocess}} = 8\hat{\zeta} + 8\sqrt{\hat{\zeta}_{\mathrm{ortho}}}.$$

The four conclusions are transferred one by one.

Proof of 1. Lemma 3.38 implies

$$\langle\psi|H\otimes I|\psi\rangle \geq \langle\psi|\widehat{H}\otimes I|\psi\rangle - \hat{\zeta} - 2\sqrt{\hat{\zeta}_{\mathrm{ortho}}} \geq (1-\nu) - 2\hat{\zeta} - 2\sqrt{\hat{\zeta}_{\mathrm{ortho}}},$$

where the second inequality is 1 of Lemma 7.1.

Proof of 2. Lemma 3.20 applied to 2 of Lemma 7.1 and the post-processed comparison (7.40) gives

$$A^u_a \otimes I \simeq_{\hat{\zeta}+\sqrt{\hat{\zeta}_{\mathrm{dataprocess}}}} I \otimes H_{[h(u)=a]}.$$

In Lean the corresponding transport is stated for submeasurements with the total-overlap displacement recorded explicitly; this is the term which vanishes in the measurement-valued form of Lemma 3.20. The monotone-total route records a sufficient structural replacement: if the orthonormalization repair preserves the completed residual outcome, then the projective total is bounded by the helper total as an operator, hence also has no larger right-register expectation in the strategy state. Under this structural hypothesis the alphabet-size displacement term is not introduced. The QXP-layer interface used by the formalization records both the total domination of the canonical projective family and the sharper residual-domination field for the fresh option-completion outcome. Under the identification of the Q-layer with the option-completed source submeasurement, this residual-domination field is equivalent to the fresh-outcome comparison $Q_{\mathrm{none}} \leq P_{\mathrm{none}}$. Algebraically, this fresh-outcome comparison follows as soon as the construction of $\widehat{X}$ preserves the corresponding fresh row block, namely $\widehat{X}_{\mathrm{none}} = X_{\mathrm{none}}$. Since $A^u_{\mathrm{tot}} = I$ and postprocessing does not change the total operator, Lean also records the equivalent form in which this displacement is the single scalar difference

$$\left|\langle\psi, I\otimes H_{\mathrm{tot}}\psi\rangle - \langle\psi, I\otimes \widehat{H}_{\mathrm{tot}}\psi\rangle\right|$$

between the right-register totals of H and $\widehat{H}$. The outcomewise data-processing bound controls this scalar by applying the vector triangle inequality to the sum over $a \in \mathbb{F}_q$; the formal estimate gives the additional term

$$\sqrt{q\,\hat{\zeta}_{\mathrm{dataprocess}}}.$$

Thus the total-overlap term is no longer an independent hypothesis, although the numerical absorption of this alphabet-size loss is recorded separately. There is also a sharper formal route: if the right-register total of the projective family has no larger expectation than that of $\widehat{H}$, then the total-overlap term can only improve the consistency defect. Under this monotonicity hypothesis the transported error is again $\hat{\zeta} + \sqrt{\hat{\zeta}_{\mathrm{dataprocess}}}$, and the standard final-stage numerical estimate absorbs it into ζ.

Proof of 3. Lemma 3.36 implies

$$\widehat{H}_h \otimes I \approx_{2\hat{\zeta}} I \otimes \widehat{H}_h.$$

Combining this with (7.39) on both sides and summing the three transports with Lemma 3.27 yields

$$H_h \otimes I \approx_{6\hat{\zeta}+6\hat{\zeta}_{\mathrm{ortho}}} I \otimes H_h.$$

Proof of 4. The boundedness quantity is

$$\begin{aligned}
&\langle\psi|Z\otimes(I-H)|\psi\rangle \\
&=\langle\psi|Z\otimes I|\psi\rangle-\sum_h\langle\psi|Z\otimes H_h|\psi\rangle \\
&\leq\langle\psi|Z\otimes I|\psi\rangle-\sum_h\langle\psi|\mathbb{E}_u A^u_{h(u)}\otimes H_h|\psi\rangle \\
&=\langle\psi|Z\otimes I|\psi\rangle-\mathbb{E}_u\sum_h\langle\psi|A^u_{h(u)}\otimes H_h|\psi\rangle \\
&=\langle\psi|Z\otimes I|\psi\rangle-\mathbb{E}_u\sum_a\langle\psi|A^u_a\otimes H_{[h(u)=a]}|\psi\rangle.
\end{aligned}\tag{7.41}$$

Proposition 3.24 upgrades (7.40) to the scalar estimate

$$(7.41)\leq\langle\psi|Z\otimes I|\psi\rangle-\mathbb{E}_u\sum_a\langle\psi|A^u_a\otimes\widehat{H}_{[h(u)=a]}|\psi\rangle+\sqrt{\hat{\zeta}_{\text{dataprocess}}},$$

and now 4 of Lemma 7.1 bounds this by $\hat{\zeta}+\sqrt{\hat{\zeta}_{\text{dataprocess}}}$. The formal boundedness field is the corresponding operator-monotonicity step: once the SDP dual witness satisfies $I\leq Z$, the total mass of any left-placed submeasurement, and in particular of the final projective submeasurement, is bounded by $Z\otimes I$ with zero boundedness error.

Thus the four conclusions hold with errors

$$2\hat{\zeta}+2\sqrt{\hat{\zeta}_{\text{ortho}}},\qquad\hat{\zeta}+\sqrt{\hat{\zeta}_{\text{dataprocess}}},\qquad 6\hat{\zeta}+6\hat{\zeta}_{\text{ortho}},\qquad\hat{\zeta}+\sqrt{\hat{\zeta}_{\text{dataprocess}}},$$

respectively. The paper's exponent bookkeeping then shows the following bounds. The formal bookkeeping uses the three-term power sum $\varepsilon^p+\delta^p+(d/q)^p$, its monotonicity on the unit interval, and the corresponding square-root estimate. The displayed Lean theorems record these arithmetic identities together with the completeness, self-closeness, and projective-residual absorptions $2\hat{\zeta}+2\sqrt{\hat{\zeta}_{\text{ortho}}}\leq\zeta$, $6\hat{\zeta}+6\hat{\zeta}_{\text{ortho}}\leq\zeta$ and $\hat{\zeta}+\sqrt{\hat{\zeta}_{\text{dataprocess}}}\leq\zeta$. The two threshold lemmas involving $|\mathbb{F}_q|$ record the scalar part of the alphabet-size obstruction in the current final-fields point-consistency transport: the data-processing threshold contains $8\hat{\zeta}$, but the transported estimate still carries the factor $\sqrt{|\mathbb{F}_q|}$.

$$\hat{\zeta}_{\text{ortho}}\leq 400m\cdot\left(\varepsilon^{1/8}+\delta^{1/8}+(d/q)^{1/8}\right),$$

$$\hat{\zeta}_{\text{dataprocess}}\leq 960m\cdot\left(\varepsilon^{1/16}+\delta^{1/16}+(d/q)^{1/16}\right),$$

and hence

$$6\hat{\zeta}+6\hat{\zeta}_{\text{ortho}}\leq\zeta,\qquad\hat{\zeta}+\sqrt{\hat{\zeta}_{\text{dataprocess}}}\leq\zeta.$$

Since $2\hat{\zeta}+2\sqrt{\hat{\zeta}_{\text{ortho}}}\leq\hat{\zeta}_{\text{dataprocess}}\leq\zeta$, all four bounds are at most ζ. □

Chapter 8

Commutativity

8.1 Commutativity of the point measurements

Definition 8.1 (Shared diagonal-line sample)**.** Sample a uniformly random ordered point pair (u, v) and a uniformly random parameter $t \in \mathbb{F}_q$. Package this data as the diagonal line whose direction is $v - u$ and whose parametrization visits u at t and v at $t + 1$.

Theorem 8.2 (Commutativity of the point measurements)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the (m, q, d) low individual degree test. On average over independent and uniformly random $u, v \sim \mathbb{F}_q^m$,*

$$(A_a^u A_b^v) \otimes I \approx_{32\gamma m} (A_b^v A_a^u) \otimes I.$$

Proof. By Definition 2.11, the strategy passes the m-restricted diagonal lines test with probability $1 - \gamma m$. Hence

$$A_a^u \otimes I \simeq_{\gamma m} I \otimes L_{[f(u)=a]}^{\ell}$$

on average over uniformly random $u \sim \mathbb{F}_q^m$ and a uniformly random line ℓ in $\mathbb{F}_q^m$ containing u. By 3.21,

$$A_a^u \otimes I \approx_{2\gamma m} I \otimes L_{[f(u)=a]}^{\ell}. \tag{8.1}$$

Let u and v be independent uniformly random points in $\mathbb{F}_q^m$, and let ℓ be a uniformly random line containing both points. The marginal distribution on (u, ℓ) and on (v, ℓ) is the same as above, so

$$\begin{aligned}
(A_a^u A_b^v) \otimes I &\approx_{2\gamma m} A_a^u \otimes L_{[f(v)=b]}^{\ell} && \text{(by 8.1)} \\
&\approx_{2\gamma m} I \otimes (L_{[f(v)=b]}^{\ell} L_{[f(u)=a]}^{\ell}) && \text{(by 8.1)} \\
&= I \otimes (L_{[f(u)=a]}^{\ell} L_{[f(v)=b]}^{\ell}) && \text{(because } L \text{ is projective)} \\
&\approx_{2\gamma m} A_b^v \otimes L_{[f(u)=a]}^{\ell} && \text{(by 8.1)} \\
&\approx_{2\gamma m} (A_b^v A_a^u) \otimes I. && \text{(by 8.1)}
\end{aligned}$$

The theorem follows from 3.27. $\square$

8.2 Commutativity of G after evaluation

Lemma 8.3 (Commutativity of G after evaluation)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m+1, q, d)$ low individual degree test. Let $\{G^x\} \in \mathrm{PolySub}(m, q, d)$ be a collection of projective sub-measurements indexed by $x \in \mathbb{F}_q$ with the following properties:*

1. *(Consistency with A) On average over $(u, x) \sim \mathbb{F}_q^{m+1}$,*
$$A_a^{u,x} \otimes I \simeq_\zeta I \otimes G^x_{[g(u)=a]}.$$

2. *(Strong self-consistency) On average over $x \sim \mathbb{F}_q$,*
$$G_g^x \otimes I \approx_\zeta I \otimes G_g^x.$$

3. *(Boundedness) There exists a positive-semidefinite matrix Z^x for each $x \in \mathbb{F}_q$ such that*
$$\mathbb{E}_x \langle \psi | (I - G^x) \otimes Z^x | \psi \rangle \leq \zeta$$
and for each $x \in \mathbb{F}_q$ and $g \in \mathcal{P}(m, q, d)$,
$$Z^x \geq \left(\mathbb{E}_u A^{u,x}_{g(u)} \right).$$

Let
$$\nu = 48m(\gamma^{1/2} + \zeta^{1/2}).$$
Then on average over independent and uniformly random $(u, x), (v, y) \sim \mathbb{F}_q^{m+1}$,
$$G^x_{[g(u)=a]} G^y_{[h(v)=b]} \otimes I \approx_\nu G^y_{[h(v)=b]} G^x_{[g(u)=a]} \otimes I.$$

Proof. Write $G_a^{u,x} = G^x_{[g(u)=a]}$. Expanding the commutator square gives

$$\begin{aligned}
&\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | (G_a^{u,x} G_b^{v,y} - G_b^{v,y} G_a^{u,x})^\dagger (G_a^{u,x} G_b^{v,y} - G_b^{v,y} G_a^{u,x}) \otimes I | \psi \rangle \\
&= 2\mathbb{E}_{u,v,x,y} \sum_{a,b} \Big(\langle \psi | G_b^{v,y} G_a^{u,x} G_b^{v,y} \otimes I | \psi \rangle - \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} G_b^{v,y} \otimes I | \psi \rangle \Big),
\end{aligned} \tag{8.2}$$

using the projectivity of G. To compare the two quartic terms, first note that

$$G^{u,x} = \sum_a G_a^{u,x} = \sum_a G^x_{[g(u)=a]} = G^x. \tag{8.3}$$

The identity (8.3) is exactly Proposition 3.13 for the evaluation map $g \mapsto g(u)$. By 1 and 3.31,

$$\begin{aligned}
G_a^{u,x} \otimes I &\approx_{4\zeta} G^{u,x} \otimes A_a^{u,x} \\
&= G^x \otimes A_a^{u,x},
\end{aligned} \tag{8.4}$$

where the last equality is 8.3. Applying 3.23 once to the second term in 8.2 gives

$$\begin{aligned}
&\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} G_b^{v,y} \otimes I | \psi \rangle \\
\approx_{2\sqrt{\zeta}} &\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} G^y \otimes A_b^{v,y} | \psi \rangle.
\end{aligned} \tag{8.5}$$

The first boundedness-driven stability step is

$$(8.5) \approx_{\sqrt{\zeta}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} \otimes A_b^{v,y} | \psi \rangle, \tag{8.6}$$

which is 8.4. Applying 8.4 once more and then 8.2, we obtain

$$\begin{aligned}(8.6) &\approx_{2\sqrt{\zeta}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G^x \otimes A_a^{u,x} A_b^{v,y} | \psi \rangle \\ &\approx_{6\sqrt{\gamma(m+1)}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G^x \otimes A_b^{v,y} A_a^{u,x} | \psi \rangle. \end{aligned} \tag{8.7}$$

The second boundedness-driven stability step is

$$(8.7) \approx_{\sqrt{\zeta}+6\sqrt{\gamma(m+1)}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} \otimes A_a^{u,x} A_b^{v,y} | \psi \rangle, \tag{8.8}$$

which is 8.5. Using 8.4 twice more and 3.23,

$$\begin{aligned}(8.8) &\approx_{2\sqrt{\zeta}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} \otimes A_b^{v,y} | \psi \rangle \\ &\approx_{2\sqrt{\zeta}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_b^{v,y} \otimes I | \psi \rangle. \end{aligned} \tag{8.9}$$

Next, 2 and the projectivity of G imply via 3.36 and 3.37 that

$$G_a^{u,x} \otimes I \approx_{\zeta} I \otimes G_a^{u,x}. \tag{8.10}$$

Applying 8.10 twice with 3.23 turns 8.9 into

$$\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_b^{v,y} G_a^{u,x} G_b^{v,y} \otimes I | \psi \rangle,$$

which is the first quartic term from 8.2. Collecting the ten approximation steps appearing between 8.2 and 8.10 yields the bound

$$48m(\gamma^{1/2} + \zeta^{1/2}),$$

so the commutator norm is at most ν. □

Lemma 8.4.

$$\begin{aligned} &\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} G^y \otimes A_b^{v,y} | \psi \rangle \\ &\approx_{\sqrt{\zeta}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G_a^{u,x} G_b^{v,y} G_a^{u,x} \otimes A_b^{v,y} | \psi \rangle. \end{aligned} \tag{8.11}$$

Proof. For each $y \in \mathbb{F}_q$ and $g \in \mathcal{P}(m,q,d)$, define

$$R_g^y = \mathbb{E}_{u,x} \sum_a G_a^{u,x} G_g^y G_a^{u,x}.$$

Since G is a sub-measurement, $R^y = \{R^y_g\}$ is also a sub-measurement. The difference between the two sides of the lemma is

$$\begin{aligned}
&\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^{u,x}_a G^{v,y}_b G^{u,x}_a (I - G^y) \otimes A^{v,y}_b | \psi \rangle \\
&= \mathbb{E}_{v,y} \sum_{g} \langle \psi | R^y_g (I - G^y) \otimes A^{v,y}_{g(v)} | \psi \rangle. \qquad (8.12)
\end{aligned}$$

Apply Cauchy–Schwarz to 8.12. The first factor is at most 1 because R^y is a sub-measurement. For the second factor, average over v and use 3 to replace $\mathbb{E}_v A^{v,y}_{g(v)}$ by Z^y; the resulting quantity is at most

$$\mathbb{E}_y \langle \psi | (I - G^y) \otimes Z^y | \psi \rangle \le \zeta.$$

Hence the difference has magnitude at most $\sqrt{\zeta}$. □

Lemma 8.5.

$$\begin{aligned}
&\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^{u,x}_a G^{v,y}_b G^x \otimes A^{u,x}_a A^{v,y}_b | \psi \rangle \\
&\approx_{\sqrt{\zeta} + 6\sqrt{\gamma(m+1)}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^{u,x}_a G^{v,y}_b \otimes A^{u,x}_a A^{v,y}_b | \psi \rangle.
\end{aligned}$$

Proof. The difference between the two sides is

$$\begin{aligned}
&\mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^{u,x}_a G^{v,y}_b (I - G^x) \otimes A^{u,x}_a A^{v,y}_b | \psi \rangle \\
&\approx_{6\sqrt{\gamma(m+1)}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^{u,x}_a G^{v,y}_b (I - G^x) \otimes A^{v,y}_b A^{u,x}_a | \psi \rangle, \qquad (8.13)
\end{aligned}$$

where 8.2 commutes the point measurements on the right register. Expanding $G^{u,x}_a = \sum_{g:g(u)=a} G^x_g$, we rewrite

$$(8.13) = \mathbb{E}_{u,v,x,y} \sum_{g,b} \langle \psi | G^x_g G^{v,y}_b (I - G^x) \otimes A^{v,y}_b A^{u,x}_{g(u)} | \psi \rangle. \qquad (8.14)$$

Apply Cauchy–Schwarz to 8.14. The first factor is at most 1 because both G and A are sub-measurements. In the second factor, average over u and use 3 to replace $\mathbb{E}_u A^{u,x}_{g(u)}$ by Z^x; then sum over g and b using the sub-measurement property. This leaves

$$\mathbb{E}_x \langle \psi | (I - G^x) \otimes Z^x | \psi \rangle \le \zeta.$$

Hence 8.14 has magnitude at most $\sqrt{\zeta}$, and the total loss is $\sqrt{\zeta} + 6\sqrt{\gamma(m+1)}$. □

8.3 Commutativity of G

Lemma 8.6 (Normalization condition for a sandwiched family)**.** *Let $P = \{P_a\}$ be a sub-measurement and $Q = \{Q_b\}$ be a projective sub-measurement. Define $C_{a,b} = Q_b P_a Q_b$. Then*

$$\sum_a \left(\sum_b C_{a,b} \right)^\dagger \left(\sum_b C_{a,b} \right) = \sum_a \left(\sum_b C_{a,b} \right) \left(\sum_b C_{a,b} \right)^\dagger \le I.$$

Proof. Expand the square, use the orthogonality $Q_b Q_{b'} = 0$ for $b \neq b'$, and then sum over the sub-measurement P. $\square$

Theorem 8.7 (Commutativity of G)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m+1, q, d)$ low individual degree test. Let $\{G^x\}_{x \in \mathbb{F}_q}$ denote a set of projective sub-measurements in* $\mathrm{PolySub}(m, q, d)$ *with the following properties:*

1. *(Consistency with A) On average over $(u, x) \sim \mathbb{F}_q^{m+1}$,*
$$A_a^{u,x} \otimes I \simeq_\zeta I \otimes G^x_{[g(u)=a]}.$$
2. *(Strong self-consistency) On average over $x \sim \mathbb{F}_q$,*
$$G_g^x \otimes I \approx_\zeta I \otimes G_g^x.$$
3. *(Boundedness) There exists a positive-semidefinite matrix Z^x for each $x \in \mathbb{F}_q$ such that*
$$\mathbb{E}_x \langle \psi | (I - G^x) \otimes Z^x | \psi \rangle \leq \zeta$$
and for each $x \in \mathbb{F}_q$ and $g \in \mathcal{P}(m, q, d)$,
$$Z^x \geq \left(\mathbb{E}_u A^{u,x}_{g(u)} \right).$$

Let
$$\nu = 30m \left(\gamma^{1/4} + \zeta^{1/4} + (d/q)^{1/4} \right).$$
Then on average over independent and uniformly random $(u, x), (v, y) \sim \mathbb{F}_q^{m+1}$,
$$G_g^x G_h^y \otimes I \approx_\nu G_h^y G_g^x \otimes I.$$

Proof. If at least one of γ, ζ, or d/q is at least 1, then $\nu \geq 30$ and the estimate is trivial from 3.26. We therefore assume $\gamma, \zeta, d/q \leq 1$.

Expanding the commutator square gives

$$\begin{aligned} &\mathbb{E}_{x,y} \sum_{g,h} \langle \psi | (G_g^x G_h^y - G_h^y G_g^x)^\dagger (G_g^x G_h^y - G_h^y G_g^x) \otimes I | \psi \rangle \\ &= 2\mathbb{E}_{x,y} \sum_{g,h} \langle \psi | G_g^x G_h^y G_g^x \otimes I | \psi \rangle - 2\mathbb{E}_{x,y} \sum_{g,h} \langle \psi | G_g^x G_h^y G_g^x G_h^y \otimes I | \psi \rangle. \end{aligned} \tag{8.15}$$

The first term is close to $\langle \psi | G \otimes G | \psi \rangle$, where $G = \mathbb{E}_x G^x$, by 3.32 and 2. For the second term, 3.23 and 8.6 yield

$$\mathbb{E}_{x,y} \sum_{g,h} \langle \psi | G_g^x G_h^y G_g^x G_h^y \otimes I | \psi \rangle \approx_{\sqrt{\zeta}} \mathbb{E}_{x,y} \sum_{g,h} \langle \psi | G_h^y G_g^x G_h^y \otimes G_g^x | \psi \rangle. \tag{8.16}$$

The first Schwartz–Zippel reduction evaluates the left factor at a random point:

$$\begin{aligned} (8.16) &= \mathbb{E}_{x,y} \sum_{g,h} \langle \psi | G_h^y G_g^x G_h^y \otimes G_g^x | \psi \rangle \\ &\approx_{\frac{dm}{q}} \mathbb{E}_{u,x,y} \sum_{a,h} \langle \psi | G_h^y G^x_{[g(u)=a]} G_h^y \otimes G^x_{[g(u)=a]} | \psi \rangle. \end{aligned} \tag{8.17}$$

Indeed, the difference is

$$\mathbb{E}_{u,x,y} \sum_{g \neq g', h} \mathbf{1}[[]\, g(u) = g'(u)] \langle \psi | G^y_h G^x_g G^y_h \otimes G^x_{g'} | \psi \rangle, \tag{8.18}$$

and 3.7 bounds this by dm/q.

Next, apply 3.23 twice:

$$\begin{aligned}
(8.17) &= \mathbb{E}_{u,x,y} \sum_{a,h} \langle \psi | G^y_h G^x_{[g(u)=a]} G^y_h \otimes G^x_{[g(u)=a]} | \psi \rangle \\
&\approx_{\sqrt{\zeta}} \mathbb{E}_{u,x,y} \sum_{a,h} \langle \psi | G^x_{[g(u)=a]} G^y_h G^x_{[g(u)=a]} G^y_h \otimes I | \psi \rangle \\
&\approx_{\sqrt{\zeta}} \mathbb{E}_{u,x,y} \sum_{a,h} \langle \psi | G^x_{[g(u)=a]} G^y_h G^x_{[g(u)=a]} \otimes G^y_h | \psi \rangle. \qquad (8.19)
\end{aligned}$$

A second Schwartz–Zippel reduction evaluates the right factor at an independent point:

$$\begin{aligned}
(8.19) &= \mathbb{E}_{u,x,y} \sum_{a,h} \langle \psi | G^x_{[g(u)=a]} G^y_h G^x_{[g(u)=a]} \otimes G^y_h | \psi \rangle \\
&\approx_{\frac{dm}{q}} \mathbb{E}_{u,v,x,y} \sum_{a,b} \langle \psi | G^x_{[g(u)=a]} G^y_{[h(v)=b]} G^x_{[g(u)=a]} \otimes G^y_{[h(v)=b]} | \psi \rangle. \qquad (8.20)
\end{aligned}$$

The difference is

$$\mathbb{E}_{u,v,x,y} \sum_{a, h \neq h'} \mathbf{1}[[]\, h(v) = h'(v)] \langle \psi | G^x_{[g(u)=a]} G^y_h G^x_{[g(u)=a]} \otimes G^y_{h'} | \psi \rangle, \tag{8.21}$$

and the same Schwartz–Zippel estimate again gives the loss dm/q.

Now set

$$\nu_{\text{evaluation}} = 48m(\gamma^{1/2} + \zeta^{1/2}),$$

the error from 8.3. Apply 8.3 to commute the evaluated families in 8.20; after one more use of 3.23 with 2, the resulting quantity is $\langle \psi | G \otimes G | \psi \rangle$. Thus the second term in 8.15 is also close to $\langle \psi | G \otimes G | \psi \rangle$.

Combining the first-term estimate, 8.16, the two Schwartz–Zippel losses, the two intermediate $\sqrt{\zeta}$ losses, and the evaluated commutation error $\nu^{1/2}_{\text{evaluation}}$ gives

$$30m \left(\gamma^{1/4} + \zeta^{1/4} + (d/q)^{1/4} \right),$$

where the constant $30m$ accounts for the $10m$ Schwartz–Zippel evaluation losses plus $20m$ from the $\approx$-chain in the first and second terms. This proves 8.7. □

Chapter 9

Pasting

Theorem 9.1 (Pasting)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m+1, q, d)$ low individual degree test. Let $\{G^x\}_{x \in \mathbb{F}_q}$ be projective submeasurements in* $\mathrm{PolySub}(m, q, d)$ *with the following properties:*

1. *(Completeness): If $G = \mathbb{E}_x \sum_g G^x_g$, then*
$$\langle \psi | G \otimes I | \psi \rangle \geq 1 - \kappa.$$
2. *(Consistency with A): On average over $(u, x) \sim \mathbb{F}_q^{m+1}$,*
$$A^{u,x}_a \otimes I \simeq_\zeta I \otimes G^x_{[g(u)=a]}.$$
3. *(Strong self-consistency): On average over $x \sim \mathbb{F}_q$,*
$$G^x_g \otimes I \approx_\zeta I \otimes G^x_g.$$
4. *(Boundedness): There exists a positive-semidefinite matrix Z^x for each $x \in \mathbb{F}_q$ such that*
$$\mathbb{E}_x \langle \psi | (I - G^x) \otimes Z^x | \psi \rangle \leq \zeta$$
and for each $x \in \mathbb{F}_q$ and $g \in \mathcal{P}(m, q, d)$,
$$Z^x \geq \mathbb{E}_u A^{u,x}_{g(u)}.$$

Let $k \geq 400md$ be an integer, and set
$$\nu = 100k^2 m \left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right),$$
$$\sigma = \kappa \left(1 + \frac{1}{100m}\right) + 2\nu + e^{-k/(80000m^2)}.$$
Then there exists a pasted measurement $H \in$ $\mathrm{PolyMeas}(m+1, q, d)$ *with the following property.*

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^{m+1}$,*
$$A^u_a \otimes I \simeq_\sigma I \otimes H_{[h(u)=a]}.$$

Proof. Let H be the submeasurement from Lemma 9.4, and fix an arbitrary polynomial $h^* \in \mathcal{P}(m+1,q,d)$. Define a measurement $H_{\text{meas}} \in \text{PolyMeas}(m+1,q,d)$ by

$$(H_{\text{meas}})_h = \begin{cases} H_{h^*} + (I - H) & \text{if } h = h^*, \\ H_h & \text{otherwise.} \end{cases}$$

Then $\sum_h (H_{\text{meas}})_h = H + (I - H) = I$. Moreover,

$$\begin{aligned}
&\mathbb{E}_u \sum_{a \neq b} \langle \psi | A_a^u \otimes (H_{\text{meas}})_{[h(u)=b]} | \psi \rangle \\
&= \mathbb{E}_u \sum_a \sum_{h : h(u) \neq a} \langle \psi | A_a^u \otimes (H_{\text{meas}})_h | \psi \rangle \\
&= \mathbb{E}_u \sum_a \sum_{h : h(u) \neq a} \langle \psi | A_a^u \otimes H_h | \psi \rangle + \mathbb{E}_u \sum_{a : h^*(u) \neq a} \langle \psi | A_a^u \otimes (I - H) | \psi \rangle \\
&\leq \mathbb{E}_u \sum_a \sum_{h : h(u) \neq a} \langle \psi | A_a^u \otimes H_h | \psi \rangle + \mathbb{E}_u \langle \psi | I \otimes (I - H) | \psi \rangle \\
&\leq \nu + \left(\kappa \left(1 + \frac{1}{100m} \right) + \nu + e^{-k/(80000m^2)} \right) && \text{(by 1 and 2)} \\
&= \sigma.
\end{aligned}$$

Thus $A_a^u \otimes I \simeq_\sigma I \otimes (H_{\text{meas}})_{[h(u)=a]}$. □

The bound is trivial when at least one of ε, δ, γ, ζ, or d/q is at least 1, since then $\nu \geq 1$. In the remainder of the chapter we therefore assume

$$\varepsilon, \delta, \gamma, \zeta, d/q \leq 1.$$

Definition 9.2 (Standing pasting context)**.** The standing pasting context consists of the data and hypotheses from Theorem 9.1: an $(\varepsilon, \delta, \gamma)$-good symmetric strategy (ψ, A, B, L) for the $(m+1, q, d)$ low individual degree test, a family of projective submeasurements $\{G^x\}_{x \in \mathbb{F}_q}$ in $\text{PolySub}(m, q, d)$ satisfying 1–4, an integer $k \geq 400md$, and the quantities ν and σ defined there.

9.0.1 From Measurements to Submeasurements

In the remainder of the chapter we only use the axis-parallel lines test in the last coordinate direction. We therefore write B_f^u for the line measurement on the line $\{(u, x) \mid x \in \mathbb{F}_q\}$ and first record the consistency relations that follow formally from the standing pasting context of Definition 9.2.

Conditioning the low individual degree test on the last coordinate direction gives, on average over $(u, x) \sim \mathbb{F}_q^{m+1}$,

$$A_a^{u,x} \otimes I \simeq_{(m+1)\varepsilon} I \otimes B^u_{[f(x)=a]}.$$

Hence 3.21 and $(m+1) \leq 2m$ imply

$$A_a^{u,x} \otimes I \approx_{4m\varepsilon} I \otimes B^u_{[f(x)=a]}.$$

Using also the $(\varepsilon, \delta, \gamma)$-goodness assumption and 3.21,

$$I \otimes A_a^{u,x} \approx_{2\delta} A_a^{u,x} \otimes I \approx_{4m\varepsilon} I \otimes B^u_{[f(x)=a]}.$$

Therefore 3.27 yields

$$I \otimes A_a^{u,x} \approx_{8m\varepsilon+4\delta} I \otimes B^u_{[f(x)=a]}. \tag{9.1}$$

Applying 3.20 to 2 and 9.1 gives

$$G^x_{[g(u)=a]} \otimes I \simeq_{\nu_1} I \otimes B^u_{[f(x)=a]}, \tag{9.2}$$

where

$$\nu_1 = \zeta + \sqrt{8m\varepsilon + 4\delta}. \tag{9.3}$$

Finally, 8.7 gives

$$G_g^x G_h^y \otimes I \approx_{\nu_{\mathrm{com}}} G_h^y G_g^x \otimes I, \tag{9.4}$$

where

$$\nu_{\mathrm{com}} = 30m\left(\gamma^{1/4} + \zeta^{1/4} + (d/q)^{1/4}\right).$$

Lemma 9.3 (Slice measurements are consistent with line answers)**.** *In the standing pasting context of Definition 9.2, set*

$$\nu_1 = \zeta + \sqrt{8m\varepsilon + 4\delta}.$$

Then, on average over $(u,x) \sim \mathbb{F}_q^{m+1}$,

$$G^x_{[g(u)=a]} \otimes I \simeq_{\nu_1} I \otimes B^u_{[f(x)=a]}.$$

Proof. This is exactly 9.2, with 9.3. □

Lemma 9.4 (Pasted submeasurement)**.** *In the standing pasting context of Definition 9.2, there exists a submeasurement $H \in \mathrm{PolySub}(m+1,q,d)$ with the following properties.*

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^{m+1}$,*

$$A_a^u \otimes I \simeq_{\nu} I \otimes H_{[h(u)=a]}.$$

2. *(Completeness): If $H = \sum_h H_h$, then*

$$\langle \psi | H \otimes I | \psi \rangle \geq 1 - \kappa\left(1 + \frac{1}{100m}\right) - \nu - e^{-k/(80000m^2)}.$$

Proof. Let H be the pasted submeasurement constructed in Definition 9.10. Lemma 9.28 implies

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu_6} I \otimes B^u_{[f(x)=a]}. \tag{9.5}$$

Applying 3.20 to 9.5 and 9.1 gives

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu_6 + \sqrt{8m\varepsilon+4\delta}} I \otimes A_a^{u,x}.$$

Since $\sqrt{8m\varepsilon + 4\delta} \leq 3m(\varepsilon^{1/32} + \delta^{1/32})$ for $\varepsilon, \delta \leq 1$, the total error is at most

$$\sqrt{8m\varepsilon + 4\delta} + \nu_6 \leq 47k^2 m\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right) \leq \nu,$$

which proves 1. The completeness claim is exactly Corollary 9.39. □

9.0.2 The Pasted Submeasurement

Definition 9.5 (Distinct tuples)**.** For $k \geq 1$, let $\mathsf{Distinct}_k \subseteq \mathbb{F}_q^k$ be the set of tuples $(x_1, \dots, x_k)$ with pairwise distinct coordinates.

Proposition 9.6 (Distinct tuples versus independent tuples)**.** *If $x = (x_1, \dots, x_k)$ is uniformly random in $\mathbb{F}_q^k$ and $y = (y_1, \dots, y_k)$ is uniformly random in $\mathsf{Distinct}_k$, then*

$$d_{\mathrm{TV}}(x, y) \leq \frac{k^2}{q}.$$

Proof. The total variation distance is exactly the probability mass of the collision event for a uniformly random tuple, because the distinct-tuple distribution is the uniform distribution conditioned on landing in $\mathsf{Distinct}_k$. A uniformly random tuple lies outside $\mathsf{Distinct}_k$ only if two coordinates collide, so a union bound gives

$$\Pr[x \notin \mathsf{Distinct}_k] \leq \sum_{i=2}^{k} \frac{i-1}{q} = \frac{k(k-1)}{2q} \leq \frac{k^2}{q},$$

and this bounds the total variation distance. □

We first record the natural construction obtained by sampling $d+1$ slices and interpolating. It motivates the eventual argument but is not the construction used in the formal theorem.

The First Construction

The first construction samples $d+1$ slice outcomes $g_1, \dots, g_{d+1} \in \mathcal{P}(m, q, d)$, interpolates them to a global polynomial $h \in \mathcal{P}(m+1, q, d)$, and then averages over distinct interpolation points. Its consistency is governed by the commutation of the slice measurements, but its completeness is subtler. The heuristic comparison is

$$\langle \psi | H \otimes I | \psi \rangle \approx \langle \psi | G^{d+1} \otimes I | \psi \rangle.$$

Thus one would like to show

$$\langle \psi | G^{d+1} \otimes I | \psi \rangle \approx \langle \psi | G \otimes I | \psi \rangle. \tag{9.6}$$

The naive estimate would instead suggest

$$\langle \psi | G^{d+1} \otimes I | \psi \rangle \approx 1 - (d+1)\kappa, \tag{9.7}$$

which is too weak for the induction.

The key input is that one can first show

$$\langle \psi | G^{d+2} \otimes I | \psi \rangle \approx_{\Delta} \langle \psi | G^{d+1} \otimes I | \psi \rangle, \tag{9.8}$$

for a small error $\Delta = \mathrm{poly}(m) \cdot \mathrm{poly}(\varepsilon, \delta, \gamma, \zeta, d/q)$. Writing the eigendecomposition $G = \sum_i \lambda_i |v_i\rangle\langle v_i|$ and the induced spectral distribution $\mu(i) = \langle \psi | (|v_i\rangle\langle v_i| \otimes I) | \psi \rangle$, this is equivalent to

$$\Delta \geq \langle \psi | G^{d+1} \otimes I | \psi \rangle - \langle \psi | G^{d+2} \otimes I | \psi \rangle = \mathbb{E}_{i \sim \mu}[\lambda_i^{d+1}(1 - \lambda_i)]. \tag{9.9}$$

The remaining comparison between G and G^{d+1} is then reduced to a scalar inequality.

Lemma 9.7 (A scalar inequality for the first construction)**.** *For every real number* $0 \le \lambda \le 1$*,*

$$\lambda(1-\lambda^d) \le 2(\lambda^{d+1}(1-\lambda))^{1/(d+1)}.$$

Proof. Compare $1-\lambda^d$ with $(1-\lambda)(1+\lambda+\cdots+\lambda^{d-1})$ and use that $d^{1/(d+1)} \le 2$. $\square$

As in the paper, Lemma 9.7 now converts (9.9) into the estimate

$$\mathbb{E}_{i\sim\mu}[\lambda_i(1-\lambda_i^d)] \le 2\cdot \mathbb{E}_{i\sim\mu}\left[(\lambda_i^{d+1}(1-\lambda_i))^{1/(d+1)}\right] \le 2\cdot\Delta^{1/(d+1)},$$

where the middle step uses concavity of $a \mapsto a^{1/(d+1)}$. Equivalently,

$$\langle\psi|G\otimes I|\psi\rangle - \langle\psi|G^{d+1}\otimes I|\psi\rangle \le 2\cdot\Delta^{1/(d+1)}.$$

For constant d, this is exactly the heuristic completeness comparison for the first construction recorded in ‘references/ldt-paper/ld-pasting.tex‘.

The Second Construction

Definition 9.8 (The incomplete part and the completed measurement)**.** For each $x \in \mathbb{F}_q$, write $G^x = \sum_g G_g^x$ and define the incomplete outcome $G_\perp^x = I - G^x$. The completed measurement $\widehat{G}^x$ has outcome set $\mathcal{P}(m,q,d) \cup \{\perp\}$ and is given by

$$\widehat{G}_g^x = G_g^x \quad (g \in \mathcal{P}(m,q,d)), \qquad \widehat{G}_\perp^x = G_\perp^x.$$

Definition 9.9 (Types)**.** A type is a vector $\tau \in \{0,1\}^k$. Its weight $|\tau|$ records how many coordinates lie in $\mathcal{P}(m,q,d)$ rather than equal $\perp$.

Definition 9.10 (The pasted measurement)**.** Fix $k \ge d+1$. For $x_1,\dots,x_k \in \mathbb{F}_q$ and outcomes $g_1,\dots,g_k \in \mathcal{P}(m,q,d) \cup \{\perp\}$, define

$$\widehat{H}_{g_1,\dots,g_k}^{x_1,\dots,x_k} = \widehat{G}_{g_1}^{x_1}\widehat{G}_{g_2}^{x_2}\cdots\widehat{G}_{g_k}^{x_k}\cdots\widehat{G}_{g_2}^{x_2}\widehat{G}_{g_1}^{x_1}.$$

If $(x_1,\dots,x_k) \in \mathsf{Distinct}_k$ and $h \in \mathcal{P}(m+1,q,d)$, define

$$H_h^{x_1,\dots,x_k} = \sum_{\tau:\,|\tau|\ge d+1} \widehat{H}_{h_\tau}^{x_1,\dots,x_k},$$

where the tuple h_τ has i-th entry $h|_{x_i}$ when $\tau_i = 1$ and $\perp$ when $\tau_i = 0$. The pasted submeasurement is then

$$H_h = \mathbb{E}_{(x_1,\dots,x_k)\sim\mathsf{Distinct}_k} H_h^{x_1,\dots,x_k}.$$

Definition 9.11 (Completed-outcome reindexing)**.** The completed outcome tuples admit the reindexings obtained by separating the first coordinate, moving a third slice to the front, and identifying the one-coordinate split with the original slice question and outcome spaces.

Lemma 9.12 (Vertical restriction identities)**.** *Restricting an interpolated polynomial to a vertical line and then postprocessing at a point agrees with evaluating the corresponding slice. The same identities commute with averaging over indexed submeasurements.*

9.0.3 Strong Self-Consistency and Commutation of $\widehat{G}$

Lemma 9.13 (Strong self-consistency of the complete part)**.** *In the standing pasting context of Definition 9.2,*

$$G^x \otimes I \approx_\zeta I \otimes G^x.$$

Proof. Because G is projective, 3.36 identifies the assumed strong self-consistency of the outcome-level family with

$$\mathbb{E}_x \sum_g \langle\psi|G^x_g \otimes G^x_g|\psi\rangle \geq \mathbb{E}_x \sum_g \langle\psi|G^x_g \otimes I|\psi\rangle - \frac{\zeta}{2}. \tag{9.10}$$

Hence

$$\begin{aligned}
&\mathbb{E}_x\|(G^x \otimes I - I \otimes G^x)|\psi\rangle\|^2 \\
&= 2\mathbb{E}_x\langle\psi|G^x \otimes I|\psi\rangle - 2\mathbb{E}_x\langle\psi|G^x \otimes G^x|\psi\rangle \\
&= 2\mathbb{E}_x \sum_g \langle\psi|G^x_g \otimes I|\psi\rangle - 2\mathbb{E}_x \sum_{g,h} \langle\psi|G^x_g \otimes G^x_h|\psi\rangle \\
&\leq 2\mathbb{E}_x \sum_g \langle\psi|G^x_g \otimes I|\psi\rangle - 2\mathbb{E}_x \sum_g \langle\psi|G^x_g \otimes G^x_g|\psi\rangle \leq \zeta
\end{aligned}$$

by 9.10. □

Corollary 9.14 (Strong self-consistency of the incomplete part)**.** *In the standing pasting context of Definition 9.2,*

$$G^x_\perp \otimes I \approx_\zeta I \otimes G^x_\perp.$$

Proof. Since $G^x_\perp = I - G^x$,

$$G^x_\perp \otimes I - I \otimes G^x_\perp = I \otimes G^x - G^x \otimes I,$$

so the claim is immediate from Lemma 9.13. □

Commutativity of $G_\perp$

Lemma 9.15 (Switcheroo contraction estimates)**.** *After one commutation of a G-factor through an auxiliary projective family, the two fourth-term contractions are bounded by the same self-consistency and commutation errors used in the switcheroo argument.*

Lemma 9.16 (Commutativity with G^x_g implies commutativity with G^x)**.** *Let $M = \{M^x_o\}$ be a projective submeasurement with outcomes in a set $\mathcal{O}$. Suppose that*

$$M^x_o \otimes I \approx_\omega I \otimes M^x_o, \tag{9.11}$$

and

$$G^x_g M^y_o \otimes I \approx_\chi M^y_o G^x_g \otimes I \tag{9.12}$$

on average over independent uniformly random $x, y \sim \mathbb{F}_q$. Then

$$G^x M^y_o \otimes I \approx_{6\sqrt{\zeta}+6\sqrt{\omega}+4\sqrt{\chi}} M^y_o G^x \otimes I.$$

Proof. Let $M = \mathbb{E}_y \sum_o M_o^y$. The error to bound is

$$\begin{aligned}
&\mathbb{E}_{x,y} \sum_o \|(G^x M_o^y - M_o^y G^x) \otimes I|\psi\rangle\|^2 \\
&= \mathbb{E}_{x,y} \sum_o \langle\psi| M_o^y (G^x)^2 M_o^y \otimes I|\psi\rangle + \mathbb{E}_{x,y} \sum_o \langle\psi| G^x (M_o^y)^2 G^x \otimes I|\psi\rangle \\
&\qquad - \mathbb{E}_{x,y} \sum_o \langle\psi| G^x M_o^y G^x M_o^y \otimes I|\psi\rangle - \mathbb{E}_{x,y} \sum_o \langle\psi| M_o^y G^x M_o^y G^x \otimes I|\psi\rangle. \qquad (9.13)
\end{aligned}$$

We compare all four terms with $\langle\psi|G \otimes M|\psi\rangle$.

For the first term in 9.13,

$$\begin{aligned}
\mathbb{E}_{x,y} \sum_o \langle\psi| M_o^y (G^x)^2 M_o^y \otimes I|\psi\rangle &= \mathbb{E}_{x,y} \sum_o \langle\psi| M_o^y G^x M_o^y \otimes I|\psi\rangle \\
&= \mathbb{E}_y \sum_o \langle\psi| M_o^y G M_o^y \otimes I|\psi\rangle \\
&\approx_{2\sqrt{\omega}} \mathbb{E}_y \sum_o \langle\psi| G \otimes M_o^y|\psi\rangle \qquad \text{(by 3.32 and 9.11)} \\
&= \langle\psi|G \otimes M|\psi\rangle.
\end{aligned}$$

For the second term,

$$\begin{aligned}
\mathbb{E}_{x,y} \sum_o \langle\psi| G^x (M_o^y)^2 G^x \otimes I|\psi\rangle &= \mathbb{E}_{x,y} \sum_o \langle\psi| G^x M_o^y G^x \otimes I|\psi\rangle \\
&= \mathbb{E}_x \langle\psi| G^x M G^x \otimes I|\psi\rangle \\
&\approx_{2\sqrt{\zeta}} \mathbb{E}_x \langle\psi| M \otimes G^x|\psi\rangle \qquad \text{(by 3.32 and 9.13)} \\
&= \langle\psi|M \otimes G|\psi\rangle.
\end{aligned}$$

For the third term, first

$$\begin{aligned}
\mathbb{E}_{x,y} \sum_o \langle\psi| G^x M_o^y G^x M_o^y \otimes I|\psi\rangle &= \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| G^x M_o^y G_g^x G_g^x M_o^y \otimes I|\psi\rangle \\
&\approx_{\sqrt{\chi}} \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| G^x M_o^y G_g^x M_o^y G_g^x \otimes I|\psi\rangle. \qquad (9.14)
\end{aligned}$$

This is the Cauchy–Schwarz estimate coming from 9.12. Next,

$$(9.14) \approx_{\sqrt{\zeta}} \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| G^x M_o^y G_g^x M_o^y \otimes G_g^x|\psi\rangle. \qquad (9.15)$$

Here the error is bounded using 3. Then

$$(9.15) \approx_{\sqrt{\zeta}} \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| G_g^x G^x M_o^y G_g^x M_o^y \otimes I|\psi\rangle, \qquad (9.16)$$

again by strong self-consistency of G_g^x. Since G is projective,

$$\begin{aligned}
(9.16) &= \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| G_g^x M_o^y G_g^x M_o^y \otimes I|\psi\rangle \\
&\approx_{\sqrt{\chi}} \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| M_o^y G_g^x G_g^x M_o^y \otimes I|\psi\rangle. \qquad (9.17)
\end{aligned}$$

Finally,

$$\begin{aligned}
(9.17) &= \mathbb{E}_{x,y} \sum_{o,g} \langle\psi| M_o^y G_g^x M_o^y \otimes I |\psi\rangle \\
&= \mathbb{E}_{y} \sum_{o} \langle\psi| M_o^y G M_o^y \otimes I |\psi\rangle \\
&\approx_{2\sqrt{\omega}} \mathbb{E}_{y} \sum_{o} \langle\psi| G \otimes M_o^y |\psi\rangle && \text{(by 3.32 and 9.11)} \\
&= \langle\psi| G \otimes M |\psi\rangle.
\end{aligned}$$

Therefore

$$\mathbb{E}_{x,y} \sum_{o} \langle\psi| G^x M_o^y G^x M_o^y \otimes I |\psi\rangle \approx_{2\sqrt{\zeta}+2\sqrt{\omega}+2\sqrt{\chi}} \langle\psi| G \otimes M |\psi\rangle. \tag{9.18}$$

The fourth term in 9.13 is the Hermitian conjugate of the third term, so it satisfies the same bound. Summing the four errors gives

$$2\sqrt{\omega} + 2\sqrt{\zeta} + 2(2\sqrt{\zeta} + 2\sqrt{\omega} + 2\sqrt{\chi}) = 6\sqrt{\zeta} + 6\sqrt{\omega} + 4\sqrt{\chi}.$$

□

Remark 9.17. The Lean theorem corresponding to Lemma 9.16 takes one extra symmetry hypothesis, `hfix : swapDensity bi.density = bi.density`. This is the `density_swap` component of the paper's permutation-invariance assumption on the symmetric two-prover strategy; the stronger `swap_ev` field of `PermInvState` is not assumed by this theorem. The hypothesis is used to identify $\langle\psi|G \otimes M|\psi\rangle$ with $\langle\psi|M \otimes G|\psi\rangle$ at the center of the triangle-inequality chain.

Corollary 9.18 (Commutativity of the complete part). *In the standing pasting context of Definition 9.2, the following commutation relations hold:*

$$G_g^x G^y \otimes I \approx_{\nu_2} G^y G_g^x \otimes I, \qquad G^x G^y \otimes I \approx_{\nu_2} G^y G^x \otimes I,$$

where

$$\nu_2 = 36m\left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right).$$

Proof. By 9.4,

$$G_g^x G_h^y \otimes I \approx_{\nu_{\mathrm{com}}} G_h^y G_g^x \otimes I. \tag{9.19}$$

Applying Lemma 9.16 to 9.19, with $\{M_o^y\}$ equal to the outcome family $\{G_h^y\}$, gives

$$G_g^x G^y \otimes I \approx_{\theta_1} G^y G_g^x \otimes I, \tag{9.20}$$

where $\theta_1 = 12\sqrt{\zeta} + 4\sqrt{\nu_{\mathrm{com}}}$. Applying Lemma 9.16 again to 9.20, now with the one-outcome family $M_o^y = G^y$, gives

$$G^x G^y \otimes I \approx_{\theta_2} G^y G^x \otimes I,$$

where $\theta_2 = 12\sqrt{\zeta} + 4\sqrt{\theta_1}$.

Since $\nu_{\mathrm{com}} = 30m(\gamma^{1/4} + \zeta^{1/4} + (d/q)^{1/4})$, these errors satisfy

$$\theta_1 \le 36m\left(\gamma^{1/8} + \zeta^{1/8} + (d/q)^{1/8}\right) \le \nu_2,$$

$$\theta_2 \le 36m\left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right) = \nu_2.$$

□

Corollary 9.19 (Commutativity of the incomplete part)**.** *In the standing pasting context of Definition 9.2,*

$$G^x_g G^y_\perp \otimes I \approx_{\nu_2} G^y_\perp G^x_g \otimes I, \qquad G^x_\perp G^y_\perp \otimes I \approx_{\nu_2} G^y_\perp G^x_\perp \otimes I.$$

Proof. The identities

$$G^x_g G^y_\perp - G^y_\perp G^x_g = G^y G^x_g - G^x_g G^y$$

and

$$G^x_\perp G^y_\perp - G^y_\perp G^x_\perp = G^x G^y - G^y G^x$$

reduce both claims to Corollary 9.18. □

Putting Everything Together

Corollary 9.20 (Strong self-consistency and commutation of $\widehat{G}$)**.** *The completed measurements $\widehat{G}^x$ satisfy*

$$\widehat{G}^x_g \otimes I \approx_{2\zeta} I \otimes \widehat{G}^x_g, \tag{9.21}$$

$$\widehat{G}^x_g \widehat{G}^y_h \otimes I \approx_{\nu_3} \widehat{G}^y_h \widehat{G}^x_g \otimes I, \tag{9.22}$$

where

$$\nu_3 = 138m\left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right).$$

Proof. For 9.21, split the outcome set into the genuine outcomes and $\perp$:

$$\mathbb{E}_x \sum_g \|(\widehat{G}^x_g \otimes I - I \otimes \widehat{G}^x_g)|\psi\rangle\|^2 \le \zeta + \zeta = 2\zeta$$

by 3 and 9.14. For 9.22, the same decomposition yields four cases: polynomial–polynomial, $\perp$–$\perp$, polynomial–$\perp$, and $\perp$–polynomial. These contribute ν_{com} and three copies of ν_2, so

$$\nu_{\text{com}} + 3\nu_2 \le 138m\left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right) = \nu_3.$$

□

Lemma 9.21 (Completed part bounded by a slice)**.** *The contribution of the completed part in the squared-distance defect is bounded by the corresponding slice contribution.*

9.0.4 Sandwiching Lemmas

Lemma 9.22 (Completed sandwich normalization)**.** *The left and right completed half-sandwiches split under the head–tail reindexing, and the corresponding sums of adjoint-products are bounded above by the identity.*

Lemma 9.23 (One-step completed sandwich commutation)**.** *The elementary head–tail moves, split equivalences, and two-factor case provide the one-step estimates used to move a completed measurement through a half-sandwich.*

Lemma 9.24 (Commuting past multiple completed measurements)**.** *For every $k \ge 2$,*

$$\widehat{G}^{x_1}_{g_1} \widehat{G}^{x_2}_{g_2} \cdots \widehat{G}^{x_k}_{g_k} \otimes I \approx_{\nu_4} \widehat{G}^{x_2}_{g_2} \cdots \widehat{G}^{x_k}_{g_k} \widehat{G}^{x_1}_{g_1} \otimes I,$$

where

$$\nu_4 = 426k^2 m\left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right).$$

Proof. Repeatedly move the leftmost factor across the remaining product, using 9.21 and 9.22 at each step and summing the resulting errors with 3.27. The paper's calculation gives

$$3k(4k\zeta + k\nu_3) \leq 426k^2 m \left(\gamma^{1/16} + \zeta^{1/16} + (d/q)^{1/16}\right).$$

□

Lemma 9.25 (Line-interpolation mismatch estimates). *If the tuple of completed slices does not interpolate to the sampled vertical-line value, then some active coordinate witnesses a line-value mismatch. Concretely, for every direction u and injective tuple $\mathbf{x} = (x_1, \dots, x_k)$, let $\widehat{H}^{\mathbf{x}}|_u$ be the vertical-line restriction of the pasted interpolation family. Then*

$$\mathrm{Defect}_\psi(\widehat{H}^{\mathbf{x}}|_u,\, B^u) \;\leq\; \mathrm{Bad}(\psi; u, \mathbf{x}) \;\leq\; \sum_{i=1}^{k} \mathrm{Defect}_\psi(\widehat{H}^{(i)}(u,\mathbf{x}),\, B^{(i)}(u,\mathbf{x})),$$

where $\mathrm{Defect}_\psi(A, B)$ *denotes the bipartite consistency defect* $\max(0,\, \langle\psi|A_{\mathrm{tot}} \otimes B_{\mathrm{tot}}|\psi\rangle - \sum_a \langle\psi|A_a \otimes B_a|\psi\rangle)$,

$$\mathrm{Bad}(\psi; u, \mathbf{x}) = \sum_f \langle\psi|\Big(\sum_{\substack{g_1,\dots,g_k \\ \exists i:\, g_i \neq \perp,\ g_i(u) \neq f(x_i)}} \widehat{H}^{\mathbf{x}}_{g_1,\dots,g_k} \Big) \otimes B^u_f|\psi\rangle,$$

and $\widehat{H}^{(i)}, B^{(i)}$ *are the one-point line comparison families (Lemma 9.27) restricted to coordinate* i*, and* $\mathrm{Bad}(\psi; u, \mathbf{x}) \leq 1$.

Lemma 9.26 (Averaging the line-interpolation defect). *Replacing distinct tuples by independent tuples costs the total-variation error k^2/q from Proposition 9.6. After averaging over u and $\mathbf{x} \sim \mathsf{Distinct}_k$, the expected bad mass and the expected pasted interpolation defect are bounded by the one-point sandwich error:*

$$\mathbb{E}_u\, \mathbb{E}_{\mathbf{x} \sim \mathsf{Distinct}_k}\; \mathrm{Bad}(\psi; u, \mathbf{x}) \;\leq\; k \cdot \nu_5 \;+\; \frac{k^2}{q},$$

and consequently

$$\mathbb{E}_u\, \mathbb{E}_{\mathbf{x} \sim \mathsf{Distinct}_k}\; \mathrm{Defect}_\psi(\widehat{H}^{\mathbf{x}}|_u,\, B^u) \;\leq\; k \cdot \nu_5 \;+\; \frac{k^2}{q},$$

where $\nu_5 = 43km(\varepsilon^{1/32} + \cdots + (d/q)^{1/32})$ (Lemma 9.27). Using $1/q \leq (d/q)^{1/32}$ and $m \geq 1$, this is further bounded by $44k^2 m(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32})$ as used in Lemma 9.28.

Lemma 9.27 (Consistency of the sandwiched measurement with one line value). *For any $1 \leq i \leq k$,*

$$\mathbb{E}_u \mathbb{E}_{x_1,\dots,x_k} \sum_{g_1,\dots,g_k \,:\, g_i \neq \perp} \sum_{a \neq g_i(u)} \langle\psi|\widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes B^u_{[f(x_i)=a]}|\psi\rangle \leq \nu_5,$$

where

$$\nu_5 = 43km\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right).$$

Proof. Summing out the coordinates to the right of i gives

$$\begin{aligned} &\mathbb{E}_u \mathbb{E}_{x_1,\dots,x_k} \sum_{g_1,\dots,g_k \,:\, g_i \neq \perp} \sum_{a \neq g_i(u)} \langle\psi|\widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes B^u_{[f(x_i)=a]}|\psi\rangle \\ &= \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_1,\dots,g_i \,:\, g_i \neq \perp} \sum_{a \neq g_i(u)} \langle\psi|\widehat{H}^{x_1,\dots,x_i}_{g_1,\dots,g_i} \otimes B^u_{[f(x_i)=a]}|\psi\rangle. \end{aligned} \tag{9.23}$$

Write $\widehat{G}^{x_{<i}}_{g_{<i}} = \widehat{G}^{x_1}_{g_1} \cdots \widehat{G}^{x_{i-1}}_{g_{i-1}}$. Then

$$
\begin{aligned}
(9.23) &= \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_1,\dots,g_i\,:\,g_i\neq\perp} \langle\psi|\widehat{G}^{x_{<i}}_{g_{<i}} \widehat{G}^{x_i}_{g_i} \widehat{G}^{x_i}_{g_i} (\widehat{G}^{x_{<i}}_{g_{<i}})^\dagger \otimes (I - B^u_{[f(x_i)=g_i(u)]})|\psi\rangle \\
&\approx_{\sqrt{\nu_4}} \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_1,\dots,g_i\,:\,g_i\neq\perp} \langle\psi|\widehat{G}^{x_i}_{g_i} \widehat{G}^{x_{<i}}_{g_{<i}} \widehat{G}^{x_i}_{g_i} (\widehat{G}^{x_{<i}}_{g_{<i}})^\dagger \otimes (I - B^u_{[f(x_i)=g_i(u)]})|\psi\rangle. \qquad (9.24)
\end{aligned}
$$

The Cauchy–Schwarz argument uses

$$
\mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_1,\dots,g_i} \langle\psi|((\widehat{G}^{x_{<i}}_{g_{<i}} \widehat{G}^{x_i}_{g_i} - \widehat{G}^{x_i}_{g_i} \widehat{G}^{x_{<i}}_{g_{<i}})(\widehat{G}^{x_i}_{g_i} (\widehat{G}^{x_{<i}}_{g_{<i}})^\dagger - (\widehat{G}^{x_{<i}}_{g_{<i}})^\dagger \widehat{G}^{x_i}_{g_i})) \otimes I|\psi\rangle, \qquad (9.25)
$$

which is at most ν_4 by Lemma 9.24. A second Cauchy–Schwarz step gives

$$
(9.24) \approx_{\sqrt{\nu_4}} \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_1,\dots,g_i\,:\,g_i\neq\perp} \langle\psi|\widehat{G}^{x_i}_{g_i} \widehat{G}^{x_{<i}}_{g_{<i}} (\widehat{G}^{x_{<i}}_{g_{<i}})^\dagger \widehat{G}^{x_i}_{g_i} \otimes (I - B^u_{[f(x_i)=g_i(u)]})|\psi\rangle. \qquad (9.26)
$$

Summing over $g_1, \dots, g_{i-1}$ collapses the middle factor to I, so 9.26 becomes

$$
\mathbb{E}_u \mathbb{E}_{x_1,\dots,x_i} \sum_{g_i\,:\,g_i\neq\perp} \langle\psi|\widehat{G}^{x_i}_{g_i} \otimes (I - B^u_{[f(x_i)=g_i(u)]})|\psi\rangle \le \nu_1
$$

by 9.2. Thus the total error is

$$
\nu_1 + 2\sqrt{\nu_4} \le 43km\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right).
$$

$\square$

9.0.5 Consistency of H with A

Lemma 9.28 (Consistency of the pasted measurement with the line measurement)**.** *The pasted submeasurement satisfies*

$$
H_{[h|_u=f]} \otimes I \simeq_{\nu_6} I \otimes B^u_f,
$$

where

$$
\nu_6 = 44k^2m\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right).
$$

Proof. The inconsistency with B^u expands as

$$
\begin{aligned}
&\mathbb{E}_u \sum_{f\neq f'} \langle\psi|H_{[h|_u=f']} \otimes B^u_f|\psi\rangle \\
&= \mathbb{E}_u \sum_h \sum_{f\neq h|_u} \langle\psi|H_h \otimes B^u_f|\psi\rangle \\
&= \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_k} \sum_h \sum_{f\neq h|_u} \langle\psi|H^{x_1,\dots,x_k}_h \otimes B^u_f|\psi\rangle \\
&= \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_k} \sum_h \sum_{w:|w|\ge d+1} \sum_{(g_1,\dots,g_k)=h_w} \sum_{f\neq h|_u} \langle\psi|\widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes B^u_f|\psi\rangle. \qquad (9.27)
\end{aligned}
$$

If $|w| \ge d+1$ and $f \neq h|_u$, then some active coordinate i must satisfy $g_i \neq \perp$ and $g_i(u) \neq f(x_i)$. Hence

$$
(9.27) \le \mathbb{E}_u \mathbb{E}_{x_1,\dots,x_k} \sum_{g_1,\dots,g_k} \sum_{\substack{f:\exists i,\ g_i\neq\perp,\\ g_i(u)\neq f(x_i)}} \langle\psi|\widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes B^u_f|\psi\rangle. \qquad (9.28)
$$

Passing from independent tuples to distinct tuples costs at most k^2/q by Lemma 9.6. A union bound over the offending coordinate then gives

$$(9.28) \le \sum_{i=1}^{k} \mathbb{E}_u \mathbb{E}_{y_1,\dots,y_k} \sum_{g_1,\dots,g_k\,:\,g_i\neq\perp} \sum_{a\neq g_i(u)} \langle\psi|\widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes B^u_{[f(y_i)=a]}|\psi\rangle \le k\nu_5.$$

Therefore

$$\frac{k^2}{q} + k\nu_5 \le 44k^2 m\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right) = \nu_6.$$

□

Corollary 9.29 (Consistency of H with A)**.** *On average over $(u,x) \sim \mathbb{F}_q^{m+1}$,*

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu} I \otimes A^{u,x}_a.$$

Proof. Lemma 9.28 implies

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu_6} I \otimes B^u_{[f(x)=a]}.$$

Applying 3.20 to this relation and 9.1 gives

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu_6+\sqrt{8m\varepsilon+4\delta}} I \otimes A^{u,x}_a.$$

Since

$$\sqrt{8m\varepsilon+4\delta} + \nu_6 \le 47k^2 m\left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right) \le \nu,$$

the claim follows. □

Corollary 9.30 (Consistency of the completed H with A)**.** *Let H_{meas} be the completion of H at the fallback outcome. On average over $(u,x) \sim \mathbb{F}_q^{m+1}$,*

$$(H_{\mathrm{meas}})_{[h(u,x)=a]} \otimes I \simeq_{\sigma} I \otimes A^{u,x}_a.$$

Proof. Corollary 9.29 gives

$$H_{[h(u,x)=a]} \otimes I \simeq_{\nu} I \otimes A^{u,x}_a.$$

Corollary 9.39 bounds the missing mass of H by

$$\kappa \cdot \left(1 + \frac{1}{100m}\right) + \nu + e^{-k/(80000m^2)}.$$

Adding this completion mass to the ν-bound gives

$$\nu + \kappa \cdot \left(1 + \frac{1}{100m}\right) + \nu + e^{-k/(80000m^2)} = \sigma.$$

□

9.0.6 Completeness of H

Definition 9.31 (Outcomes of a fixed type)**.** For a type $\tau \in \{0,1\}^k$, let $\mathsf{Outcomes}_\tau$ be the set of tuples $(g_1, \dots, g_k)$ such that $g_i \in \mathcal{P}(m,q,d)$ when $\tau_i = 1$ and $g_i = \perp$ when $\tau_i = 0$. If $x_1, \dots, x_k \in \mathbb{F}_q$, let $\mathsf{Global}_\tau(x)$ be the subset of $\mathsf{Outcomes}_\tau$ arising from restrictions of a single polynomial in $\mathcal{P}(m+1,q,d)$, and let $\overline{\mathsf{Global}_\tau(x)} = \mathsf{Outcomes}_\tau \setminus \mathsf{Global}_\tau(x)$.

Lemma 9.32 (Summing over all outcomes)**.** *If $x_1, \dots, x_k$ are sampled independently and uniformly from $\mathbb{F}_q$, then*

$$\langle \psi | H \otimes I | \psi \rangle \approx_{\nu_7} \mathbb{E}_{x_1,\dots,x_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Outcomes}_\tau} \langle \psi | \widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle,$$

where

$$\nu_7 = 46k^2 m \left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32} \right).$$

Proof. Let $(y_1, \dots, y_k) \sim \mathsf{Distinct}_k$. By definition,

$$\begin{aligned}
\langle \psi | H \otimes I | \psi \rangle &= \sum_h \langle \psi | H_h \otimes I | \psi \rangle \\
&= \mathbb{E}_{y_1,\dots,y_k} \sum_h \langle \psi | H^{y_1,\dots,y_k}_h \otimes I | \psi \rangle \\
&= \mathbb{E}_{y_1,\dots,y_k} \sum_h \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) = h_\tau} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle \\
&= \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Global}_\tau(y)} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle. \qquad (9.29)
\end{aligned}$$

We now remove the restriction to globally consistent tuples:

$$(9.29) \approx_{\frac{k^2}{q} + k\nu_5 + \frac{md}{q}} \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Outcomes}_\tau} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle. \qquad (9.30)$$

Since the right-hand side is at least the left-hand side, it suffices to bound the difference:

$$\begin{aligned}
(9.30) - (9.29) &= \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \overline{\mathsf{Global}_\tau(y)}} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle \\
&= \mathbb{E}_u \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_f \sum_{(g_1,\dots,g_k) \in \overline{\mathsf{Global}_\tau(y)}} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes B^u_f | \psi \rangle. \qquad (9.31)
\end{aligned}$$

We next insert an indicator recording consistency along the sampled line:

$$(9.31) \approx_{\frac{k^2}{q} + k\nu_5} \mathbb{E}_u \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_f \sum_{(g_1,\dots,g_k) \in \overline{\mathsf{Global}_\tau(y)}} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes B^u_f | \psi \rangle \cdot \mathbf{1}[[\,]\ \forall i \in \tau,\ f(y_i) = g_i(u)]. \qquad (9.32)$$

The discarded part is

$$\mathbb{E}_u \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_f \sum_{(g_1,\dots,g_k) \in \overline{\mathsf{Global}_\tau(y)}} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes B^u_f | \psi \rangle \cdot \mathbf{1}[[\,]\ \exists i \in \tau,\ f(y_i) \neq g_i(u)], \qquad (9.33)$$

and after switching to independent tuples and applying Lemma 9.27 coordinatewise, this contributes at most $k^2/q + k\nu_5$.

Let $\mathsf{Consistent}_\tau(g, y, u)$ denote the event that there exists a degree-d polynomial f with $f(y_i) = g_i(u)$ for all $i \in \tau$. Then

$$\mathbf{1}[[] \, \forall i \in \tau, \ f(y_i) = g_i(u)] \leq \mathbf{1}[[] \, \mathsf{Consistent}_\tau(g, y, u)],$$

so 9.32 is bounded by

$$\begin{aligned} &\mathbb{E}_u \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Global}_\tau(y)} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle \cdot \mathbf{1}[[] \, \mathsf{Consistent}_\tau(g, y, u)] \\ &= \mathbb{E}_{y_1,\dots,y_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Global}_\tau(y)} \langle \psi | \widehat{H}^{y_1,\dots,y_k}_{g_1,\dots,g_k} \otimes I | \psi \rangle \cdot \Pr_u[\mathsf{Consistent}_\tau(g, y, u)]. \end{aligned} \tag{9.34}$$

For a fixed globally inconsistent tuple, choose $d+1$ active coordinates and let h^* be the unique interpolant through them. Since the tuple is not globally consistent, some active coordinate i^* satisfies $g_{i^*} \neq h^*|_{y_{i^*}}$, and then Schwartz–Zippel gives

$$\Pr_u[\mathsf{Consistent}_\tau(g, y, u)] \leq \Pr_u[g_{i^*}(u) = h^*(u, y_{i^*})] \leq \frac{md}{q}.$$

Therefore 9.34 is at most md/q, which proves 9.30. A final application of Lemma 9.6 replaces distinct tuples by independent tuples, giving a total error of

$$2\frac{k^2}{q} + \frac{md}{q} + k\nu_5 \leq 46k^2 m \left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right) = \nu_7.$$

□

Definition 9.33 (Truncated type sums)**.** For $1 \leq \ell \leq k+1$ and a tail type $\tau_{\geq \ell} \in \{0,1\}^{k-\ell+1}$, define

$$S_{\tau_{\geq \ell}} = \sum_{\tau_{<\ell} : |\tau| \geq d+1} G^{|\tau_{<\ell}|} (I - G)^{(\ell-1) - |\tau_{<\ell}|}.$$

This records the contribution of all prefixes that can still be completed to total weight at least $d+1$ once the tail $\tau_{\geq \ell}$ is fixed.

Lemma 9.34 (Recurrence for truncated type sums)**.** *Each $S_{\tau_{\geq \ell}}$ is Hermitian, positive semidefinite, and bounded by I. Moreover, if $\tau_{>\ell}$ is a tail type of length $k - \ell$, then*

$$S_{\tau_{>\ell}} = \sum_{\tau_\ell \in \{0,1\}} S_{\tau_{\geq \ell}} G^{\tau_\ell} (I - G)^{1 - \tau_\ell}.$$

Proof. Every summand is a polynomial in the commuting positive operators G and $I - G$, so $S_{\tau_{\geq \ell}}$ is Hermitian and positive semidefinite. Summing over all prefixes gives

$$S_{\tau_{\geq \ell}} \leq (G + (I - G))^{\ell - 1} = I.$$

Splitting the next type bit according to whether $\tau_\ell = 1$ or $\tau_\ell = 0$ gives the stated recurrence. □

Lemma 9.35 (From the pasted sum to a polynomial in G)**.** *Let $|\psi_{\mathrm{bi}}\rangle$ be a bipartite state on $\iota \otimes \iota$. If $x_1, \dots, x_k$ are sampled independently and uniformly from $\mathbb{F}_q$, then*

$$\mathbb{E}_{x_1,\dots,x_k} \sum_{\tau : |\tau| \geq d+1} \sum_{(g_1,\dots,g_k) \in \mathsf{Outcomes}_\tau} \langle \psi_{\mathrm{bi}} | \widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k} \otimes I | \psi_{\mathrm{bi}} \rangle \approx_{\nu_8} \sum_{r=d+1}^{k} \binom{k}{r} \langle \psi_{\mathrm{bi}} | G^r (I - G)^{k-r} \otimes I | \psi_{\mathrm{bi}} \rangle,$$

where

$$\nu_8 = 46k^2 m \left(\gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right).$$

Proof. For a type $\tau \in \{0,1\}^k$, write $\tau_{<\ell}$, $\tau_{>\ell}$, $\tau_{\le\ell}$, and $\tau_{\ge\ell}$ for the obvious truncations, and similarly for tuples g. Also write

$$\widehat{G}^{x_{\ge\ell}}_{g_{\ge\ell}} = \widehat{G}^{x_\ell}_{g_\ell} \cdots \widehat{G}^{x_k}_{g_k}.$$

Then

$$\widehat{G}^{x_{\ge\ell}}_{g_{\ge\ell}} = \widehat{G}^{x_\ell}_{g_\ell} \widehat{G}^{x_{>\ell}}_{g_{>\ell}}, \tag{9.35}$$

and

$$\widehat{H}^{x_{\ge\ell}}_{g_{\ge\ell}} = \widehat{G}^{x_{\ge\ell}}_{g_{\ge\ell}} (\widehat{G}^{x_{\ge\ell}}_{g_{\ge\ell}})^\dagger. \tag{9.36}$$

Write $\mathsf{O}_\tau = \mathsf{Outcomes}_\tau$. The main iterative step is that for each $1 \le \ell \le k$,

$$\begin{aligned}
&\mathbb{E}_{x_{\ge\ell}} \sum_{\tau:\,|\tau|\ge d+1} \sum_{g_{\ge\ell}\in\mathsf{O}_{\tau_{\ge\ell}}} \langle\psi|\widehat{H}^{x_{\ge\ell}}_{g_{\ge\ell}} \otimes \big(G^{|\tau_{<\ell}|}(I-G)^{(\ell-1)-|\tau_{<\ell}|}\big)|\psi\rangle \\
\approx_{2\sqrt{2\zeta}+2\sqrt{\nu_4}}\ &\mathbb{E}_{x_{>\ell}} \sum_{\tau:\,|\tau|\ge d+1} \sum_{g_{>\ell}\in\mathsf{O}_{\tau_{>\ell}}} \langle\psi|\widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes \big(G^{|\tau_{\le\ell}|}(I-G)^{\ell-|\tau_{\le\ell}|}\big)|\psi\rangle.
\end{aligned} \tag{9.37}$$

Iterating 9.37 for $\ell = 1,\dots,k$ produces the Bernoulli polynomial

$$\sum_{r=d+1}^{k} \binom{k}{r} \langle\psi|G^r(I-G)^{k-r} \otimes I|\psi\rangle,$$

and the accumulated error is

$$k(2\sqrt{2\zeta}+2\sqrt{\nu_4}) \le 46k^2 m \left(\gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right) = \nu_8.$$

This corrects the displayed arithmetic in the paper: the iterated adjacent estimate contributes k copies of the $2\sqrt{\nu_4}$ term, and the definition of ν_4 already contains a factor k^2.

To prove 9.37, define for each $1 \le \ell \le k+1$ and each tail type $\tau_{\ge\ell}$,

$$S_{\tau_{\ge\ell}} = \sum_{\tau_{<\ell}\,:\,|\tau|\ge d+1} G^{|\tau_{<\ell}|}(I-G)^{(\ell-1)-|\tau_{<\ell}|}. \tag{9.38}$$

Then 9.37 may be rewritten as

$$\begin{aligned}
&\mathbb{E}_{x_{\ge\ell}} \sum_{\tau_{\ge\ell}} \sum_{g_{\ge\ell}\in\mathsf{O}_{\tau_{\ge\ell}}} \langle\psi|\widehat{H}^{x_{\ge\ell}}_{g_{\ge\ell}} \otimes S_{\tau_{\ge\ell}}|\psi\rangle \\
\approx_{2\sqrt{2\zeta}+2\sqrt{\nu_4}}\ &\mathbb{E}_{x_{>\ell}} \sum_{\tau_{>\ell}} \sum_{g_{>\ell}\in\mathsf{O}_{\tau_{>\ell}}} \langle\psi|\widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes S_{\tau_{>\ell}}|\psi\rangle.
\end{aligned} \tag{9.39}$$

Remark 9.36 (Lean proof-stage notation). In Lean, the type-restricted averaged submeasurement, the per-tail operator/mass, the aggregated stage quantity, and the final Bernoulli-tail scalar from this proof are recorded by the declarations above. Their indexing is shifted to the repository's 0-based convention: Lean stage ℓ is the paper's stage $\ell+1$.

The operators $S_{\tau_{\ge\ell}}$ are Hermitian and positive semidefinite, and

$$\begin{aligned}
S_{\tau_{\ge\ell}} &= \sum_{\tau_{<\ell}\,:\,|\tau|\ge d+1} G^{|\tau_{<\ell}|}(I-G)^{(\ell-1)-|\tau_{<\ell}|} \\
&\le \sum_{\tau_{<\ell}} G^{|\tau_{<\ell}|}(I-G)^{(\ell-1)-|\tau_{<\ell}|} \\
&= (G+(I-G))^{\ell-1} = I.
\end{aligned} \tag{9.40}$$

For any $\tau_\ell \in \{0,1\}$,

$$\mathbb{E}_{x_\ell} \sum_{g_\ell \in \mathsf{O}_{\tau_\ell}} \widehat{G}^{x_\ell}_{g_\ell} = G^{\tau_\ell}(I-G)^{1-\tau_\ell}. \tag{9.41}$$

Therefore, for every $\tau_{>\ell}$,

$$\begin{aligned} \sum_{\tau_\ell} S_{\tau_{\geq \ell}} \cdot \Big(\mathbb{E}_{x_\ell} \sum_{g_\ell \in \mathsf{O}_{\tau_\ell}} \widehat{G}^{x_\ell}_{g_\ell}\Big) &= \sum_{\tau_\ell} S_{\tau_{\geq \ell}} G^{\tau_\ell}(I-G)^{1-\tau_\ell} \\ &= S_{\tau_{>\ell}}. \end{aligned} \tag{9.42}$$

Moreover,

$$S_{\tau_{\geq \ell}} \cdot \Big(\mathbb{E}_{x_\ell} \sum_{g_\ell \in \mathsf{O}_{\tau_\ell}} \widehat{G}^{x_\ell}_{g_\ell}\Big) \cdot S_{\tau_{\geq \ell}} \leq \mathbb{E}_{x_\ell} \sum_{g_\ell \in \mathsf{O}_{\tau_\ell}} \widehat{G}^{x_\ell}_{g_\ell}, \tag{9.43}$$

because $S_{\tau_{\geq \ell}}$ commutes with G and $(I-G)$ and is bounded by I.

We now prove 9.39. Writing $\widehat{H}$ as a sandwich of $\widehat{G}$ operators and moving the rightmost $\widehat{G}^{x_\ell}_{g_\ell}$ to the second tensor factor gives

$$\begin{aligned} &\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{\geq \ell}}_{g_{\geq \ell}} \otimes S_{\tau_{\geq \ell}} | \psi \rangle \\ &= \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{\geq \ell}}_{g_{\geq \ell}} (\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger \widehat{G}^{x_\ell}_{g_\ell}) \otimes S_{\tau_{\geq \ell}} | \psi \rangle \\ &\approx_{\sqrt{2\zeta}} \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{\geq \ell}}_{g_{\geq \ell}} (\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger) \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell}) | \psi \rangle. \end{aligned} \tag{9.44}$$

The corresponding Cauchy–Schwarz bound is

$$\begin{aligned} &\Big| \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{\geq \ell}}_{g_{\geq \ell}} \otimes S_{\tau_{\geq \ell}})((\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger \otimes I)(\widehat{G}^{x_\ell}_{g_\ell} \otimes I - I \otimes \widehat{G}^{x_\ell}_{g_\ell}) | \psi \rangle \Big| \\ &\leq \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{\geq \ell}}_{g_{\geq \ell}} \otimes (S_{\tau_{\geq \ell}})^2 | \psi \rangle} \cdot \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_\ell}_{g_\ell} \otimes I - I \otimes \widehat{G}^{x_\ell}_{g_\ell})(\widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes I)(\widehat{G}^{x_\ell}_{g_\ell} \otimes I - I \otimes} \end{aligned}$$

The first square root is at most 1 because $S_{\tau_{\geq \ell}} \leq I$ and $\widehat{H}$ is a submeasurement, and the second is at most 2ζ by 9.21.

Next, commute the leftmost $\widehat{G}^{x_\ell}_{g_\ell}$ across the left half-sandwich:

$$\begin{aligned} (9.44) &= \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_\ell}_{g_\ell} \widehat{G}^{x_{>\ell}}_{g_{>\ell}} (\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger) \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell}) | \psi \rangle \\ &\approx_{\sqrt{\nu_4}} \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{>\ell}}_{g_{>\ell}} \widehat{G}^{x_\ell}_{g_\ell} (\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger) \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell}) | \psi \rangle. \end{aligned} \tag{9.46}$$

The associated Cauchy–Schwarz estimate is

$$\Big|\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | ([\widehat{G}^{x_{>\ell}}_{g_{>\ell}}, \widehat{G}^{x_\ell}_{g_\ell}] \otimes I)((\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell}))|\psi\rangle\Big|$$
$$\leq \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | [\widehat{G}^{x_{>\ell}}_{g_{>\ell}}, \widehat{G}^{x_\ell}_{g_\ell}][\widehat{G}^{x_{>\ell}}_{g_{>\ell}}, \widehat{G}^{x_\ell}_{g_\ell}]^\dagger \otimes I |\psi\rangle} \cdot \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes (\widehat{G}^{x_\ell}_{g_\ell} (S_{\tau_{\geq \ell}})^2 \widehat{G}^{x_\ell}_{g_\ell})|\psi\rangle}. \tag{9.47}$$

The first square root is bounded by ν_4 via Lemma 9.24, and the second by 1 since $S_{\tau_{\geq \ell}} \leq I$.

Continue commuting:

$$(9.46) \approx_{\sqrt{\nu_4}} \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{>\ell}}_{g_{>\ell}} (\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger \widehat{G}^{x_\ell}_{g_\ell}) \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell})|\psi\rangle. \tag{9.48}$$

This time the Cauchy–Schwarz step is

$$\Big|\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | (\widehat{G}^{x_{>\ell}}_{g_{>\ell}} \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell}))([(\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger, \widehat{G}^{x_\ell}_{g_\ell}] \otimes I)|\psi\rangle\Big|$$
$$\leq \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell} S_{\tau_{\geq \ell}})|\psi\rangle} \cdot \sqrt{\mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | [(\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger, \widehat{G}^{x_\ell}_{g_\ell}]^\dagger [(\widehat{G}^{x_{>\ell}}_{g_{>\ell}})^\dagger, \widehat{G}^{x_\ell}_{g_\ell}] \otimes I |\psi\rangle}. \tag{9.49}$$

By 9.43, the first square root is at most 1, and the second matches the first square root in 9.47, hence is at most ν_4.

Finally move the remaining $\widehat{G}^{x_\ell}_{g_\ell}$ to the second tensor factor:

$$(9.48) \approx_{\sqrt{2\zeta}} \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell} \widehat{G}^{x_\ell}_{g_\ell})|\psi\rangle. \tag{9.50}$$

The omitted Cauchy–Schwarz bound reuses the first square root from 9.49 and the second square root from 9.45. Since $\widehat{G}$ is projective,

$$\begin{aligned}
(9.50) &= \mathbb{E}_{x_{\geq \ell}} \sum_{\tau_{\geq \ell}} \sum_{(g_\ell, g_{>\ell}) \in \mathsf{O}_{\tau_{\geq \ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes (S_{\tau_{\geq \ell}} \widehat{G}^{x_\ell}_{g_\ell})|\psi\rangle \\
&= \mathbb{E}_{x_{>\ell}} \sum_{\tau_{>\ell}} \sum_{g_{>\ell} \in \mathsf{O}_{\tau_{>\ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes \Big(\sum_{\tau_\ell} S_{\tau_{\geq \ell}} \Big(\mathbb{E}_{x_\ell} \sum_{g_\ell \in \mathsf{O}_{\tau_\ell}} \widehat{G}^{x_\ell}_{g_\ell}\Big)\Big)|\psi\rangle \\
&= \mathbb{E}_{x_{>\ell}} \sum_{\tau_{>\ell}} \sum_{g_{>\ell} \in \mathsf{O}_{\tau_{>\ell}}} \langle \psi | \widehat{H}^{x_{>\ell}}_{g_{>\ell}} \otimes S_{\tau_{>\ell}} |\psi\rangle, && \text{(by 9.42)}
\end{aligned}$$

which is exactly 9.39. □

Lemma 9.37 (Context specializations for the pasted measurement)**.** *The standing pasting context specializes the consistency of H with A and the conversion from the pasted sum to the polynomial in G to the parameters and error bounds fixed in Definition 9.2.*

Lemma 9.38 (Matrix Chernoff bound for the Bernoulli polynomial)**.** *Let $0 < \theta < 1$ and integers $k, d > 0$ with $k \geq 2d/\theta$. Define*

$$F(X) = \sum_{r=d+1}^{k} \binom{k}{r} X^r (I - X)^{k-r}.$$

If X is Hermitian, $0 \leq X \leq I$, and $\langle\psi|X \otimes I|\psi\rangle \geq 1-\kappa$, then

$$\langle\psi|F(X) \otimes I|\psi\rangle \geq 1-\frac{\kappa}{1-\theta}-e^{-\theta^2 k/2}.$$

Proof. Let ρ be the reduced state of $|\psi\rangle$ on one prover's subsystem, and write the eigendecomposition $X = \sum_i \lambda_i |v_i\rangle\langle v_i|$. This defines a probability distribution $\mu(i) = \langle v_i|\rho|v_i\rangle$, with

$$\mathbb{E}_{i\sim\mu}\lambda_i = \mathrm{Tr}(X\rho) = \langle\psi|X \otimes I|\psi\rangle \geq 1-\kappa.$$

Equivalently, $\mathbb{E}_{i\sim\mu}(1-\lambda_i) \leq \kappa$, so Markov's inequality shows that most of the spectral mass of X lies on eigenvalues at least θ:

$$\Pr_{i\sim\mu}[\lambda_i > \theta] \geq 1-\frac{\kappa}{1-\theta}. \tag{9.51}$$

For a scalar $p \in [d/k, 1]$, the quantity

$$F(p) = \sum_{r=d+1}^{k} \binom{k}{r} p^r (1-p)^{k-r}$$

is the probability of observing at least $d+1$ successes in k Bernoulli(p) trials. Applying the scalar Chernoff bound eigenvalue-by-eigenvalue therefore yields

$$F(p) \geq 1-\exp\Big(-2\Big(p-\frac{d}{k}\Big)^2 k\Big). \tag{9.52}$$

Hence

$$\begin{aligned}
\langle\psi|F(X) \otimes I|\psi\rangle &= \mathrm{Tr}(F(X)\rho) \\
&= \sum_i F(\lambda_i)\langle v_i|\rho|v_i\rangle \\
&= \mathbb{E}_{i\sim\mu}F(\lambda_i) \\
&\geq \Pr_{i\sim\mu}[\lambda_i \geq \theta] \cdot F(\theta) \\
&\geq \Big(1-\frac{\kappa}{1-\theta}\Big) \cdot F(\theta) && \text{(by 9.51)} \\
&\geq \Big(1-\frac{\kappa}{1-\theta}\Big) \cdot \Big(1-\exp\Big(-2\Big(\theta-\frac{d}{k}\Big)^2 k\Big)\Big). && (9.53)
\end{aligned}$$

Since $(1-b)(1-c) \geq 1-b-c$ for $b, c \geq 0$, 9.53 implies

$$\langle\psi|F(X) \otimes I|\psi\rangle \geq 1-\frac{\kappa}{1-\theta}-\exp\Big(-2\Big(\theta-\frac{d}{k}\Big)^2 k\Big).$$

Finally, $k \geq 2d/\theta$ implies $d/k \leq \theta/2$, hence $(\theta–d/k)^2 \geq \theta^2/4$, which yields the claimed bound. ☐

Corollary 9.39 (Completeness of the pasted submeasurement)**.** *If $k \geq 400md$, then*

$$\langle\psi|H \otimes I|\psi\rangle \geq 1-\kappa\left(1+\frac{1}{100m}\right)-\nu-e^{-k/(80000m^2)}.$$

Proof. Lemma 9.32 gives

$$\langle\psi|H\otimes I|\psi\rangle \approx_{\nu_7} \mathbb{E}_{x_1,\dots,x_k} \sum_{\tau:\,|\tau|\geq d+1} \sum_{(g_1,\dots,g_k)\in\mathsf{Outcomes}_\tau} \langle\psi|\widehat{H}^{x_1,\dots,x_k}_{g_1,\dots,g_k}\otimes I|\psi\rangle.$$

Lemma 9.35 turns the right-hand side into

$$\sum_{r=d+1}^{k} \binom{k}{r} \langle\psi|G^r(I-G)^{k-r}\otimes I|\psi\rangle$$

up to error ν_8. Apply Lemma 9.38 with $\theta = 1/(200m)$. Since $k \geq 400md$, the hypothesis $k \geq 2d/\theta$ is satisfied, and the resulting lower bound is

$$1 - \frac{\kappa}{1-1/(200m)} - e^{-k/(80000m^2)} \geq 1 - \kappa\left(1+\frac{1}{100m}\right) - e^{-k/(80000m^2)}.$$

The paper then absorbs the approximation losses $\nu_7+\nu_8$ into ν. $\square$

Chapter 10

The main induction step

10.1 Restricting to one slice

Definition 10.1 (Appending a line at height x)**.** For $x \in \mathbb{F}_q$, let $\mathrm{append}_x(\ell)$ be the line in $\mathbb{F}_q^{m+1}$ obtained by appending the coordinate x to every point of a line $\ell \subseteq \mathbb{F}_q^m$. If $f\colon \ell \to \mathbb{F}_q$, let $\mathrm{append}_x(f)$ be the polynomial on $\mathrm{append}_x(\ell)$ defined by $\mathrm{append}_x(f)(u, x) = f(u)$.

Definition 10.2 (x-restricted strategy)**.** Let (ψ, A, B, L) be a symmetric strategy for the $(m + 1, q, d)$ low individual degree test. For each $x \in \mathbb{F}_q$, the x-restricted strategy (ψ, A^x, B^x, L^x) for the (m, q, d) test is defined by

$$(A^x)^u_a = A^{u,x}_a, \qquad (B^x)^\ell_f = B^{\mathrm{append}_x(\ell)}_{\mathrm{append}_x(f)}, \qquad (L^x)^\ell_f = L^{\mathrm{append}_x(\ell)}_{\mathrm{append}_x(f)}.$$

Lemma 10.3 (Restricted success probabilities)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m + 1, q, d)$ test. For each $x \in \mathbb{F}_q$, let (ψ, A^x, B^x, L^x) be the restricted strategy and let $\varepsilon_x, \delta_x, \gamma_x$ be its three failure probabilities. Then*

$$\mathbb{E}_x \varepsilon_x \le \frac{m+1}{m}\varepsilon, \qquad \mathbb{E}_x \delta_x \le \delta, \qquad \mathbb{E}_x \gamma_x \le \frac{m+1}{m}\gamma.$$

Proof. The self-consistency test restricts exactly to the slice at height x. For the axis-parallel and diagonal-line tests, conditioning on the sampled line not being parallel to the last coordinate identifies the test with the corresponding restricted game. □

Definition 10.4 (Average restricted errors)**.** The average restricted errors are the expectations over $x \in \mathbb{F}_q$ of the axis-parallel, self-consistency, and restricted diagonal-line failure probabilities of the x-restricted strategies.

Lemma 10.5 (Weighted restricted-probability bounds)**.** *The weighted bounds from Lemma 10.3 imply the restricted-probability data used in the successor step.*

10.2 The inductive construction

Theorem 10.6 (Self-improvement)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the (m, q, d) low individual degree test. Let $G \in \mathrm{PolySub}(m, q, d)$ be a submeasurement with the following property:*

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\nu I \otimes G_{[g(u)=a]}.$$

Let

$$\zeta = 3000m\left(\varepsilon^{1/32} + \delta^{1/32} + (d/q)^{1/32}\right).$$

Then there exists a projective submeasurement $H \in \mathrm{PolySub}(m,q,d)$ with the following properties:

1. *(Completeness): If $H = \sum_h H_h$, then*

$$\langle\psi|H \otimes I|\psi\rangle \geq (1-\nu) - \zeta.$$

2. *(Consistency with A): On average over $u \sim \mathbb{F}_q^m$,*

$$A_a^u \otimes I \simeq_\zeta I \otimes H_{[h(u)=a]}.$$

3. *(Strong self-consistency):*

$$H_h \otimes I \approx_\zeta I \otimes H_h.$$

4. *(Boundedness): There exists a positive semidefinite operator Z such that*

$$\langle\psi|Z \otimes (I-H)|\psi\rangle \leq \zeta$$

and for each $h \in \mathcal{P}(m,q,d)$,

$$Z \geq \mathbb{E}_u A_{h(u)}^u.$$

Proof. Complete G to a measurement by adjoining an extra outcome $\perp$. Apply Theorem 7.10 to the completed measurement, then discard the $\perp$ outcome. By Lemma 3.42, the completeness, consistency, strong self-consistency, and boundedness estimates transfer back to the original polynomial outcomes with an additive error from the missing mass $\langle\psi|(I-\sum_g G_g)\otimes I|\psi\rangle$. Since the output is projective, Lemma 3.36 converts the strong self-consistency statement to the usual bipartite formulation. $\square$

Remark 10.7 (Lean conditional declaration for induction self-improvement). The current Lean declaration for Theorem 10.6 assumes the helper self-consistency, orthonormalization, and final-field hypotheses that are still explicit in the Chapter 7 formalization. It also assumes a measurement whose underlying submeasurement is the given one. These are not hypotheses of the paper theorem; they are proof obligations to be discharged when the Chapter 7 statement matches the paper theorem.

Theorem 10.8 (Pasting)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m+1, q, d)$ low individual degree test. Let $\{G^x\}_{x\in\mathbb{F}_q}$ be projective submeasurements in* $\mathrm{PolySub}(m,q,d)$ *with the following properties:*

1. *(Completeness): If $G = \mathbb{E}_x \sum_g G_g^x$, then*

$$\langle\psi|G \otimes I|\psi\rangle \geq 1-\kappa.$$

2. *(Consistency with A): On average over $(u,x) \sim \mathbb{F}_q^{m+1}$,*

$$A_a^{u,x} \otimes I \simeq_\zeta I \otimes G^x_{[g(u)=a]}.$$

3. *(Strong self-consistency): On average over $x \sim \mathbb{F}_q$,*

$$G_g^x \otimes I \approx_\zeta I \otimes G_g^x.$$

4. *(Boundedness): There are positive semidefinite operators Z^x such that*

$$\mathbb{E}_x \langle \psi | (I - G^x) \otimes Z^x | \psi \rangle \leq \zeta$$

and for each $x \in \mathbb{F}_q$ and $g \in \mathcal{P}(m, q, d)$,

$$Z^x \geq \mathbb{E}_u A^{u,x}_{g(u)}.$$

Let $k \geq 400md$ and set

$$\nu = 100k^2 m \left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32}\right)$$

and

$$\sigma = \kappa \left(1 + \frac{1}{100m}\right) + 2\nu + e^{-k/(80000m^2)}.$$

Then there exists a measurement $H \in \mathrm{PolyMeas}(m + 1, q, d)$ with the following property:

1. *(Consistency with A): On average over $u \sim \mathbb{F}_q^{m+1}$,*

$$A_a^u \otimes I \simeq_\sigma I \otimes H_{[h(u)=a]}.$$

Proof. This is exactly Theorem 9.1, restated in the induction chapter because it is the pasting input used in the proof of Theorem 10.12. □

Definition 10.9 (Self-improvement bridge data for the successor step)**.** The successor step records the slice-wise self-improvement output in `SelfImprovementPackage`, and the answer-valued route records the analogous output in `AnswerSelfImprovementPackage`. The Lean-only `SliceBridgeInputs` packages are narrow wiring interfaces: they assume honest per-slice symmetric strategies and Section 7 bridge inputs, while averaged point-operator compatibility is derived structurally from point-measurement transport. They then convert the remaining data into the ordinary or answer-valued self-improvement package.

Definition 10.10 (Pasting data for the successor step)**.** The successor step forms the pasting input by combining the restricted probability estimates with the slice-wise self-improvement data.

Lemma 10.11 (Base case of main induction)**.** *Under the hypotheses of Theorem 10.12 with $m = 1$, there exists a measurement $G \in \mathrm{PolyMeas}(1, q, d)$ such that, on average over $u \sim \mathbb{F}_q$,*

$$A_a^u \otimes I \simeq_\varepsilon I \otimes G_{[g(u)=a]},$$

which is stronger than the bound required by Theorem 10.12 for $m = 1$.

Proof. For $m = 1$, there is only one axis-parallel line ℓ in $\mathbb{F}_q$, so the line measurement B^ℓ is already an element of $\mathrm{PolyMeas}(1, q, d)$. Because the strategy fails the axis-parallel lines test with probability at most ε, the consistency bound follows directly from the axis-parallel consistency hypothesis. □

Theorem 10.12 (Main induction)**.** *Let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the (m, q, d) low individual degree test. Let $k \geq 400\, md$ and set*

$$\nu = 1000k^2m^2 \left(\varepsilon^{1/1024} + \delta^{1/1024} + \gamma^{1/1024} + (d/q)^{1/1024}\right)$$

and

$$\sigma = m^2 \left(\nu + e^{-k/(80000m^2)}\right).$$

Then there exists a measurement $G \in \mathrm{PolyMeas}(m, q, d)$ such that, on average over $u \sim \mathbb{F}_q^m$,

$$A_a^u \otimes I \simeq_\sigma I \otimes G_{[g(u)=a]}.$$

Proof. In Lean, the base case and the internal successor-step assembly are now formalized through together with the stage data that mirror the four paper stages restrict → induct → self-improve → paste. For the answer-valued restriction route used in the successor branch of the main formal theorem, Lean now also exposes , which record the predecessor induction hypothesis for the restricted strategies without appealing to the public theorem ‘mainFormal‘. The weighted restricted-probability estimates are now public via , and the final public successor-step assembly is formalized as . The remaining gap, tracked by issue #931, is to prove the restricted-strategy self-improvement hypotheses needed to instantiate that assembly directly from higher-level applications, so the displayed theorem is not yet marked as formalized. We may assume $\varepsilon, \delta, \gamma, d/q \leq 1$, since otherwise $\nu \geq 1$ and the bound is trivial.

The proof is by induction on m. The base case $m = 1$ is Lemma 10.11.

For the step from m to $m + 1$, let (ψ, A, B, L) be an $(\varepsilon, \delta, \gamma)$-good symmetric strategy for the $(m + 1, q, d)$ test, and let $k \geq 400(m + 1)d$. For each $x \in \mathbb{F}_q$, let (ψ, A^x, B^x, L^x) be the x-restricted strategy, with failure parameters $\varepsilon_x, \delta_x, \gamma_x$. Apply the induction hypothesis to each restricted strategy with the same integer k and define

$$\nu_x = 1000k^2m^2 \left(\varepsilon_x^{1/1024} + \delta_x^{1/1024} + \gamma_x^{1/1024} + (d/q)^{1/1024}\right), \qquad \sigma_x = m^2 \left(\nu_x + e^{-k/(80000m^2)}\right).$$

This yields a measurement $G^x \in \mathrm{PolyMeas}(m, q, d)$ such that

$$(A^x)_a^u \otimes I \simeq_{\sigma_x} I \otimes G^x_{[g(u)=a]}.$$

Next define

$$\zeta_x = 3000m \left(\varepsilon_x^{1/32} + \delta_x^{1/32} + (d/q)^{1/32}\right).$$

Theorem 10.6 produces a projective submeasurement $\widehat{G^x} \in \mathrm{PolySub}(m, q, d)$ such that, for each x:

1. if $\widehat{G^x} = \sum_g \widehat{G^x_g}$, then
$$\langle\psi|\widehat{G^x} \otimes I|\psi\rangle \geq (1 - \sigma_x) - \zeta_x;$$
2. on average over $u \sim \mathbb{F}_q^m$,
$$(A^x)_a^u \otimes I \simeq_{\zeta_x} I \otimes \widehat{G^x}_{[g(u)=a]};$$
3. $$\widehat{G^x_g} \otimes I \approx_{\zeta_x} I \otimes \widehat{G^x_g};$$
4. there exists a positive semidefinite operator Z^x such that
$$\langle\psi|Z^x \otimes (I - \widehat{G^x})|\psi\rangle \leq \zeta_x$$
and for each $g \in \mathcal{P}(m, q, d)$,
$$Z^x \geq \mathbb{E}_u (A^x)_{g(u)}^u.$$

We now average the slice-dependent parameters. By concavity of $\alpha \mapsto \alpha^c$ for $c \le 1$ and Lemma 10.3,

$$\begin{aligned}
\mathbb{E}_x \nu_x &\le 1000k^2 m^2 \left((\mathbb{E}_x \varepsilon_x)^{1/1024} + (\mathbb{E}_x \delta_x)^{1/1024} + (\mathbb{E}_x \gamma_x)^{1/1024} + (d/q)^{1/1024} \right) \\
&\le 1000k^2 m^2 \left(\left(\frac{m+1}{m} \varepsilon \right)^{1/1024} + \delta^{1/1024} + \left(\frac{m+1}{m} \gamma \right)^{1/1024} + (d/q)^{1/1024} \right) \\
&\le 1000k^2 (m+1)^2 \left(\varepsilon^{1/1024} + \delta^{1/1024} + \gamma^{1/1024} + (d/q)^{1/1024} \right).
\end{aligned}$$

Call this final quantity ν. Then, defining

$$\sigma = m^2 \left(\nu + e^{-k/(80000m^2)} \right),$$

we have

$$\sigma \ge \mathbb{E}_x \sigma_x.$$

Similarly,

$$\begin{aligned}
\mathbb{E}_x \zeta_x &\le 3000m \left((\mathbb{E}_x \varepsilon_x)^{1/32} + (\mathbb{E}_x \delta_x)^{1/32} + (d/q)^{1/32} \right) \\
&\le 3000m \left(\left(\frac{m+1}{m} \varepsilon \right)^{1/32} + \delta^{1/32} + (d/q)^{1/32} \right) \\
&\le 3000(m+1) \left(\varepsilon^{1/32} + \delta^{1/32} + (d/q)^{1/32} \right).
\end{aligned}$$

Call this last quantity ζ. For later use,

$$\begin{aligned}
\zeta &= 3000(m+1) \left(\varepsilon^{1/32} + \delta^{1/32} + (d/q)^{1/32} \right) \\
&\le 1000k^2 (m+1)^2 \left(\varepsilon^{1/32} + \delta^{1/32} + (d/q)^{1/32} \right) \\
&\le 1000k^2 (m+1)^2 \left(\varepsilon^{1/1024} + \delta^{1/1024} + \gamma^{1/1024} + (d/q)^{1/1024} \right) \\
&= \nu. \qquad (10.1)
\end{aligned}$$

The averaged slice family $\{\widehat{G^x}\}_{x \in \mathbb{F}_q}$ therefore satisfies the four hypotheses of Theorem 10.8 with completeness parameter $\kappa = \sigma + \zeta$. To verify the new ν bound in that theorem, observe that $(3000)^{1/32} \le 2$, so

$$\zeta^{1/32} \le 2(m+1) \left(\varepsilon^{1/1024} + \delta^{1/1024} + (d/q)^{1/1024} \right),$$

and hence

$$\begin{aligned}
&100k^2 m \left(\varepsilon^{1/32} + \delta^{1/32} + \gamma^{1/32} + \zeta^{1/32} + (d/q)^{1/32} \right) \\
&\le 1000k^2 (m+1)^2 \left(\varepsilon^{1/1024} + \delta^{1/1024} + \gamma^{1/1024} + (d/q)^{1/1024} \right) \\
&= \nu.
\end{aligned}$$

Therefore Theorem 10.8 gives a pasted measurement $H \in \mathrm{PolyMeas}(m+1, q, d)$ such that, on average over $u \sim \mathbb{F}_q^{m+1}$,

$$A^u_a \otimes I \simeq_{\sigma^*} I \otimes H_{[h(u)=a]},$$

where

$$\sigma^* = (\sigma + \zeta) \left(1 + \frac{1}{100m} \right) + 2\nu + e^{-k/(80000m^2)}.$$

It remains to compare σ^* with the next-stage target. The pasting parameter in Theorem 10.8 is bounded by $\nu/5$ in this regime. Using this sharper estimate and (10.1), we get

$$\begin{aligned}\sigma^* &\le \left(m^2\left(\nu + e^{-k/(80000m^2)}\right) + \nu\right)\left(1 + \frac{1}{100m}\right) + \frac{2}{5}\nu + e^{-k/(80000m^2)} \\ &= \left((m^2+1)\left(1 + \frac{1}{100m}\right) + \frac{2}{5}\right)\nu + \left(m^2\left(1 + \frac{1}{100m}\right) + 1\right)e^{-k/(80000m^2)}.\end{aligned}$$

For every $m \ge 1$,

$$(m^2+1)\left(1 + \frac{1}{100m}\right) + \frac{2}{5} \le (m+1)^2, \qquad m^2\left(1 + \frac{1}{100m}\right) + 1 \le (m+1)^2.$$

Therefore

$$\sigma^* \le (m+1)^2\left(\nu + e^{-k/(80000m^2)}\right) \le (m+1)^2\left(\nu + e^{-k/(80000(m+1)^2)}\right),$$

which is exactly the bound required for dimension $m+1$. □

Remark 10.13 (Lean conditional declaration for the successor-step assembly). This auxiliary Lean statement records the five explicit successor-step ingredients used in the proof of Theorem 10.12: the weighted slice bounds feeding Lemma 10.3, the extracted slice-restriction data, the per-slice recursive witnesses, the corresponding self-improvement data, and the recursion-to-pasting assembly. These are not extra hypotheses in the paper theorem; they are intermediate proof obligations in the current formalization. Under those hypotheses the Lean statement returns a measurement $G \in \mathrm{PolyMeas}(m+1, q, d)$ satisfying the same consistency conclusion as Theorem 10.12.

One first applies Lemma 10.3 to the weighted slice bounds in order to extract the restricted failure profile, records the recursive slice witnesses, uses the explicit self-improvement data on that slice data, and then applies , which already returns the displayed error parameter directly.

10.3 Proof of the main formal theorem

Lemma 10.14 (Role-register symmetrization bridge)**.** *Let $(\psi, A^{\mathrm{A}}, B^{\mathrm{A}}, L^{\mathrm{A}}, A^{\mathrm{B}}, B^{\mathrm{B}}, L^{\mathrm{B}})$ be a projective strategy passing the (m, q, d) low individual degree test with probability at least $1-\varepsilon$. Introducing a two-dimensional role register on each side and taking $(\psi_{\mathrm{sym}}, A_{\mathrm{sym}}, B_{\mathrm{sym}}, L_{\mathrm{sym}})$ as in paper Section 3 yields a symmetric projective strategy that is $(3\varepsilon, 3\varepsilon, 3\varepsilon)$-good.*

Proof. Each of the three subtests (axis-parallel, self-consistency, diagonal) occurs with probability $1/3$, so the total failure probability at most ε implies each branch is at most 3ε. Block-diagonalizing the measurements over the role register preserves the axis-parallel and diagonal branch averages exactly, and the cross-register self-consistency branch of the symmetrized strategy equals the original point-agreement defect, which is itself bounded by 3ε. □

Remark 10.15 (Lean auxiliary data for role-register symmetrization). The formal proof of Theorem 2.12 also uses auxiliary bundled data containing the symmetrized strategy, its goodness proof, and its normalization proof. This auxiliary data supports the final assembly, not an additional hypothesis in Lemma 10.14.

Proposition 10.16 (Lean base-case handoff for the Section 3 proof)**.** *If $m = 1$, then for every k the role-register symmetrization of a projective strategy passing the low individual degree test with error ε admits a measurement G satisfying the base-case conclusion of Theorem 10.12 with parameters $(3\varepsilon, 3\varepsilon, 3\varepsilon)$.*

Proof. This is a Lean-only wrapper: combine Lemma 10.14 with the $m = 1$ clause of Theorem 10.12, formalized as . □

Definition 10.17 (Lean successor-boundary aliases for the Section 3 proof)**.** These Lean-only aliases name the successor-case data that Theorem 2.12 uses to invoke Remark 10.13. The public weighted-bound helper lemmas discharge the weighted restricted-axis and restricted-diagonal fields from the symmetrized passing strategy; the remaining external inputs are the recursive slice witnesses (together with their bridge-data structure and conversion theorem), the restricted-strategy self-improvement producer (together with its bridge-input packaging), and their bundled boundary package. The ordinary slice bridge records both Alice- and Bob-side compatibility with the restricted point, axis-parallel, and diagonal measurements; in particular, the formal lemmas extract the restricted point-agreement, axis-parallel, and diagonal branch bounds from the pinned slice's passing hypothesis. On the answer-register side, the slice bridge data now records both Alice- and Bob-side compatibility for the restricted point and axis-parallel measurements, together with the common zero-coordinate readout of the diagonal measurement on both registers. The symmetry proofs of the same-space slice, and any stronger full diagonal-family comparison, remain externally supplied by the recursive slice strategy.

Definition 10.18 (Lean base-case Step 6 hypotheses for the Section 3 proof)**.** The structure `MainFormalBaseProjectiveCompletionHypotheses` bundles the still-unformalized analytic hypotheses needed for the base case ($m = 1$) of Theorem 2.12: the locality-preserving repair witnesses used by orthonormalization, distinguished completion outcomes, and match-mass preservation for the orthonormalized projective submeasurements against the opposite unsymmetrized POVM. The auxiliary constructor `MainFormalPostRolePackageDiagonalOrthonormalizationInput.ofRepairInpu` is available to fill the spectral-truncation fields from the Section 5 spectral-truncation producer applied to the two unsymmetrized role measurements after left tensor placement. Thus the spectral half of the line-130 orthonormalization input has a constructive producer. The constructors `ofQXPLayerRepairWitnesses` and `ofLiftedQXPApproximations` also give constructive producers for the locality-preserving repair half: the former consumes the Section 9 QXP-layer repair witnesses, and the latter consumes explicit QXP-layer data together with the corresponding left-placed state-dependent approximation. The base bridge structures still accept a full orthonormalization input until a later assembly step routes them through these constructors. The line-130 cross consistency relation is reconstructed internally from the checked role residual and used directly in the completion step, so no separate diagonal strong self-consistency package remains in this bridge. The narrowed alias `MainFormalBaseBridgeHypotheses` omits the two distinguished outcomes; `baseProjectiveCompletionHypotheses_ofBaseBridge` fills both with the explicit zero polynomial. The assembly theorems `baseProjectiveCompletionResidual` and `baseProjectiveCompletionResidual_ofBaseBridge` turn these hypotheses (together with the checked base-case role residual from `MainFormalRolePackageResidual.ofBaseCase`, where the `params.m = 1` hypothesis is used) into the Step 6 witness residual consumed by the downstream cascade.

The sharper repaired base route is packaged separately by `MainFormalBaseRepairedBridgeHypotheses`. It keeps the same orthonormalization inputs but replaces the exact match-mass fields by the diagonal line-130 consistency input needed for completion. The generic theorem `mainFormal_ofRoleResidualAndRepaire` combines that diagonal completion input with the checked pre-completion line-169 repair from Remark 4.5, so the base branch of `mainFormal_ofRepairedBridge` retains the original final envelope rather than widening to the generic $\zeta_1+\sqrt{\zeta_2}$ scale. The theorem `baseMainFormal_ofRepairedBaseBridge` is the base-case wrapper of this role-residual form.

Proposition 10.19 (Lean successor handoff for the Section 3 proof)**.** *This Lean-only specialization applies Remark 10.13 to the role-register symmetrization used in Theorem 2.12. Given the* *bundled successor boundary, the Section 6 side condition $400\,md \le k$, and the matching base-universe field-model instance, it returns the symmetrized global polynomial measurement that the later unsymmetrization and completion transports consume. The base-universe field-model restriction is a current Lean API limitation inherited from the Section 6 wrapper, not a paper constraint; it should be relaxed once that wrapper is made universe-polymorphic.*

Proof. It is the direct application of after instantiating Lemma 10.14 at $(3\varepsilon, 3\varepsilon, 3\varepsilon)$ and unpacking the bundled successor-boundary fields. $\square$

Remark 10.20 (Successor projective-completion construction). The construction named here records the remaining successor-case construction needed after the Section 6 role-register hand-off. Starting from a projective strategy that passes the low individual degree test, the large-k bound $k \ge 400md$ from Section 6, the positivity condition $d > 0$, and the non-base condition $m \ne 1$, it is expected to produce the projective-completion residual consumed by the conditional theorem `mainFormal_ofRepairedBridge`. It is not an additional hypothesis of Theorem 2.12; it is the open construction that should replace the successor-branch residual currently tracked separately.

Remark 10.21 (Lean measurement-unsymmetrization bridge). Lean now isolates the Step 3 extraction from `inductive_step.tex` lines 84–108. The principal-block map `extractRoleBlock` turns a role-register POVM G into the two POVMs G^{A} and G^{B} on the original prover space and proves that point-evaluation postprocessing commutes with this extraction. The two round-trip lemmas identify the A and B blocks of the symmetrized point measurement with the original prover point measurements. The remaining analytic factor-two estimates (10.3) and (10.4) are named by `UnsymmetrizationBridgePackage`; no projectivity of arbitrary principal blocks is claimed, matching the paper's later projectivization step.

Definition 10.22 (Final error envelope)**.** Set

$$E = \varepsilon^{1/40000} + (d/q)^{1/40000} + e^{-k/(2560000m^2)}.$$

Theorem 10.23 (Basic facts about the final error envelope)**.** *If* $0 \le \varepsilon$*, then*

$$\nu_{\mathrm{main}} = 100000k^2m^4\left(\varepsilon^{1/40000} + (d/q)^{1/40000} + e^{-k/(2560000m^2)}\right) = 100000k^2m^4E,$$

and $E \ge 0$*.*

Proof. Both identities are by definition: the first unfolds ν_{main} and the second unfolds E. Nonnegativity of E follows summand-wise from `Real.rpow_nonneg` (using $0 \le \varepsilon$) and `Real.exp_nonneg`. $\square$

Definition 10.24 (Cascade parameters and standing hypotheses)**.** Assume $q > 0$, $k \ge 1$, $m \ge 1$, $0 \le \varepsilon \le 1$, and $d \le q$. Fix an incoming error parameter ν. Define

$$\sigma = m^2\left(\nu + e^{-k/(80000m^2)}\right), \qquad \zeta_1 = 2\sigma + 2\sqrt{3\varepsilon + 2\sigma} + md/q,$$

$$\zeta_2 = 200\zeta_1^{1/4} + 42\zeta_1^{1/8}, \qquad \zeta_3 = 6\zeta_1 + 6\zeta_2, \qquad \zeta_4 = 2\sigma + 2\sqrt{\zeta_1 + \zeta_3/2}.$$

The coefficient 42 is the Lean cascade scalar; it widens the paper's printed coefficient 40 so that the completion theorem's residual $2\zeta_1$ is absorbed while preserving the final error envelope.

Theorem 10.25 (Bound for the first cascade term)**.** *Under the standing hypotheses, if*

$$\nu \le 10000k^2m^2\left(\varepsilon^{1/1024} + (d/q)^{1/1024}\right),$$

then

$$\sigma \le 10000k^2m^4E.$$

Proof. Multiply the assumed bound by m^2, replace m^2 by m^4, coarsen the exponents from $1/1024$ to $1/40000$, and enlarge the exponential denominator from $80000m^2$ to $2560000m^2$. □

Theorem 10.26 (Cascade bounds for the later cascade terms)**.** *Under the standing hypotheses, assume*

$$0 \le \nu \le 10000k^2m^2\left(\varepsilon^{1/1024} + (d/q)^{1/1024}\right).$$

Define

$$\sigma = m^2\left(\nu + e^{-k/(80000m^2)}\right),$$

$$\zeta_1 = 2\sigma + 2\sqrt{3\varepsilon + 2\sigma} + md/q, \qquad \zeta_2 = 200\zeta_1^{1/4} + 42\zeta_1^{1/8},$$

$$\zeta_3 = 6\zeta_1 + 6\zeta_2, \qquad \zeta_4 = 2\sigma + 2\sqrt{\zeta_1 + \zeta_3/2}.$$

Then

$$\zeta_1, \zeta_2, \zeta_4 \le 100000k^2m^4E, \qquad \zeta_3 \le 2 \cdot 100000k^2m^4E.$$

Proof. The Lean declarations first derive the native bounds for $\zeta_1, \dots, \zeta_4$ from the cascade definitions using square-root and *rpow* monotonicity, with ζ_2 using the widened coefficient 42, and then coarsen those bounds to the common envelope E. The ζ_2 proof additionally uses $k \le k^2$, $m \le m^4$, and the nonnegativity of the earlier cascade variables. □

Theorem 10.27 (Consolidated error cascade)**.** *Assume the standing hypotheses and*

$$0 \le \nu \le 10000k^2m^2\left(\varepsilon^{1/1024} + (d/q)^{1/1024}\right).$$

Let $\sigma, \zeta_1, \zeta_2, \zeta_3$ be the cascade quantities defined above. Then

$$\sigma \le 100000k^2m^4E, \qquad \zeta_1, \zeta_2 \le 100000k^2m^4E, \qquad \zeta_3 \le 2 \cdot 100000k^2m^4E,$$

and the concrete quantity

$$\zeta_4 = 2\sigma + 2\sqrt{\zeta_1 + \zeta_3/2}$$

also satisfies

$$\zeta_4 \le 100000k^2m^4E.$$

Proof. This is the direct packaging of Theorem 10.25 and Theorem 10.26. The Lean declaration uses the explicit equalities naming $\sigma, \zeta_1, \zeta_2, \zeta_3$ as the corresponding cascade expressions. □

Remark 10.28 (Lean Step 5 expansion theorem)*.* For the Step 5 passage below, Lean now names the weighted polynomial-collision mass, proves its md/q bound by specializing the existing tensor Schwartz–Zippel helper, and proves the algebraic expansion/reindexing statement comparing the evaluated consistency defect with the full-polynomial defect plus that collision mass. The wrapper for Step 5 therefore no longer takes the expansion statement as an external hypothesis.

Remark 10.29 (Lean residual packages for the main-formal assembly)*.* The following Lean declarations are bookkeeping infrastructure for the final assembly residual: They split off the vacuous branch where the theorem error is at least 1, derive the Step 8 scalar hypotheses in the non-vacuous branch, and isolate the role-register Section 6 measurement behind a concrete role-package residual. Lean now records that every non-base parameter bundle has a predecessor via `Parameters.successorDecompositionOfNeOne`, transports the strategy and passing proof across that predecessor equality, and packages the two role-production branches as `MainFormalRolePackageBranchResidual`: the checked base handoff, an ordinary successor branch carrying `MainFormalSuccessorBoundary`, or an answer-valued successor branch carrying `MainFormalSuccessorAnswerBoundary`. The public large-k hypothesis supplies the Section 6

side condition $400m_{\mathrm{pred}}d \leq k$; the remaining role work is the successor-boundary package, the completion handoff, and the line-169 transport data. The ordinary successor-boundary package can now be assembled from recursive slice witnesses and self-improvement bridge inputs by `MainFormalRolePackageBranchResidual.successorOfBridgeInputs`; the answer-valued successor branch records the parallel routes. It can be assembled from answer recursive slice witnesses and an answer-valued self-improvement producer by `MainFormalRolePackageBranchResidual.answerSuccess`
from recursive slice witnesses and answer-side bridge inputs by `MainFormalRolePackageBranchResidual.answerSucc`
or from a predecessor induction package together with the bridge inputs by `MainFormalRolePackageBranchResidual.`
These constructors are structural and do not call `mainFormal_ofRepairedBridge`; the line-130 completion interfaces remain separate. The live completion residual carries the concrete role residual rather than a separate decorative branch witness; the branch residual types remain the documented base/successor supply paths. Once the role package is obtained, Lean proves the two factor-two unsymmetrization links from its symmetrized consistency estimate. The subsequent checked wrappers use the proved Section 5 Schwartz–Zippel theorem to convert evaluated pre-projective consistency to full-polynomial consistency at ζ_1, prove the paper line-116 triangle step, and reconstruct the line-156 $\approx_{\zeta_3}$ projectivization bridge from pre-projective consistency plus left/right completion closeness at ζ_2. Lean also derives the evaluated line-164 projective consistency, data-processes the two polynomial line-169 ζ_1 links to the evaluated line-172 links, and proves the two native ζ_4 point goals via `simeq-triangle-inequality`. After the #869 right-register completion transport, the remaining active residual is now the split `MainFormalCascadeRolePackageResidualLeftCompletionTransportResidual`: its first field is the concrete Section 6 role residual, and its post-role field asks only for Alice's line-146 completion closeness, Bob's left-register completion closeness as returned by orthonormalize-and-complete, and the construction-level match-mass monotonicity invariant needed for the exact polynomial-level ζ_1 transport links from paper line 169 and its role-reversed analogue. Lean reconstructs pre-projective full-polynomial consistency from the role residual and then uses the match-mass invariant to derive the two line-169 links; it also transports the Bob-side left-register completion estimate to the line-147 right-register form before applying the older wrappers. It no longer asks for an arbitrary role package or an arbitrary raw Section 6 witness as an independent field. It also no longer asks for the two factor-two role-block estimates, the unused Section 6 consistency field inside `UnsymmetrizationBridgePackage`, or the duplicated pre-projective consistency field inside the Section 6 handoff; those are rebuilt from the concrete role residual, the selected role package, and `hpass`. Lean then converts the resulting projective self-consistency to `MainFormalCascadeTransportTargets` using the projective converse of `simeq-to-approx`.

The exact line-169 transport fields are now derived from the stored match-mass monotonicity invariant rather than being independent residual data. As stated in `preliminaries.tex`, `prop:triangle-sub` gives a $\delta + \sqrt{\epsilon}$ loss, so applying it directly to $G^{\mathrm{A}} \simeq_{\zeta_1} G^{\mathrm{B}}$ and $G^{\mathrm{A}} \approx_{\zeta_2} Q^{\mathrm{A}}$ yields $Q^{\mathrm{A}} \simeq_{\zeta_1+\sqrt{\zeta_2}} G^{\mathrm{B}}$, not the printed ζ_1 estimate. The generic corrected route would degrade the final point error to scale $t^{1/65536}$, outside the current `mainFormalError` envelope. The sharper local pre-completion repair from Remark 4.5 avoids that widening and is the route used by the present base-case formal proof. No proof-level completeness tag is claimed here while `mainFormal_ofRepairedBridge` still has an open successor-case completion residual and `mainFormal` derives the paper theorem via a tracked open step (issue #1458).

Proof. Let

$$(\psi, A^{\mathrm{A}}, B^{\mathrm{A}}, L^{\mathrm{A}}, A^{\mathrm{B}}, B^{\mathrm{B}}, L^{\mathrm{B}})$$

be a projective strategy passing the (m, q, d) low individual degree test with probability at least $1 - \varepsilon$. Since the three subtests occur with probability $1/3$, this strategy is $(3\varepsilon, 3\varepsilon, 3\varepsilon)$-good.

Introduce a two-dimensional role register on each side and define the symmetrized state

$$|\psi_{\text{sym}}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_{\text{A}'}|1\rangle_{\text{B}'}|\psi\rangle_{\text{AB}} + |1\rangle_{\text{A}'}|0\rangle_{\text{B}'}|\psi_{\text{swap}}\rangle_{\text{AB}}\right).$$

Define the symmetrized point measurement by

$$(A_{\text{sym}})^u_a = |0\rangle\langle 0| \otimes A^{\text{A},u}_a + |1\rangle\langle 1| \otimes A^{\text{B},u}_a,$$

and similarly for B_{sym} and L_{sym}. This yields a symmetric $(3\varepsilon, 3\varepsilon, 3\varepsilon)$-good strategy. Set

$$\nu = 10000k^2m^2\left(\varepsilon^{1/1024} + (d/q)^{1/1024}\right), \qquad \sigma = m^2\left(\nu + e^{-k/(80000m^2)}\right).$$

Theorem 10.12 applied to the symmetrized strategy produces a measurement $G = \{G_g\} \in \text{PolyMeas}(m, q, d)$ such that

$$(A_{\text{sym}})^u_a \otimes I \simeq_\sigma I \otimes G_{[g(u)=a]}, \qquad G_{[g(u)=a]} \otimes I \simeq_\sigma I \otimes (A_{\text{sym}})^u_a. \tag{10.2}$$

Now unsymmetrize by defining

$$G^{\text{A}}_g = (\langle 0| \otimes I)\, G_g\, (|0\rangle \otimes I), \qquad G^{\text{B}}_g = (\langle 1| \otimes I)\, G_g\, (|1\rangle \otimes I).$$

These are POVMs on the original prover space. Since A_{sym} is a convex combination of $(|0\rangle\langle 0|\otimes I)$ and $(|1\rangle\langle 1|\otimes I)$ weighted by the test probabilities, the cross-terms $\langle 0|\cdot A_{\text{sym}}\cdot|1\rangle$ vanish. Expanding the consistency of G gives

$$G^{\text{A}}_{[g(u)=a]} \otimes I \simeq_{2\sigma} I \otimes A^{\text{B},u}_a, \tag{10.3}$$

$$I \otimes G^{\text{B}}_{[g(u)=a]} \simeq_{2\sigma} A^{\text{A},u}_a \otimes I. \tag{10.4}$$

Since the original strategy is $(3\varepsilon, 3\varepsilon, 3\varepsilon)$-good,

$$A^{\text{A},u}_a \otimes I \simeq_{3\varepsilon} I \otimes A^{\text{B},u}_a.$$

Therefore Lemma 3.29 yields

$$G^{\text{A}}_{[g(u)=a]} \otimes I \simeq_{2\sigma+2\sqrt{3\varepsilon+2\sigma}} I \otimes G^{\text{B}}_{[g(u)=a]}.$$

Expanding this consistency relation and applying Lemma 3.7 to replace point evaluations by polynomial evaluations (at cost md/q) gives

$$G^{\text{A}}_g \otimes I \simeq_{\zeta_1} I \otimes G^{\text{B}}_g, \tag{10.5}$$

where

$$\zeta_1 = 2\sigma + 2\sqrt{3\varepsilon + 2\sigma} + \frac{md}{q}.$$

The Lean Step 5 wrapper formalizes this expansion/reindexing step and combines it with the checked Schwartz–Zippel tensor bound .

Apply Lemma 4.8 to 10.5. This gives projective submeasurements $P^{\text{A}}, P^{\text{B}} \in \text{PolySub}(m, q, d)$ with

$$G^{\text{A}}_g \otimes I \approx_{100\zeta_1^{1/4}} P^{\text{A}}_g \otimes I, \qquad I \otimes G^{\text{B}}_g \approx_{100\zeta_1^{1/4}} I \otimes P^{\text{B}}_g.$$

Completing them by Lemma 3.42 yields projective measurements $Q^{\mathrm{A}}, Q^{\mathrm{B}} \in \mathrm{PolyMeas}(m, q, d)$ such that

$$G_g^{\mathrm{A}} \otimes I \approx_{\zeta_2} Q_g^{\mathrm{A}} \otimes I, \qquad (10.6)$$
$$I \otimes G_g^{\mathrm{B}} \approx_{\zeta_2} I \otimes Q_g^{\mathrm{B}},$$

where

$$\zeta_2 = 200\zeta_1^{1/4} + 42\zeta_1^{1/8}.$$

The paper prints the coefficient 40 in this display. The formal Lean cascade uses the widened coefficient 42 to absorb the literal completion error $2 \cdot (100\zeta_1^{1/4}) + 4\sqrt{100\zeta_1^{1/4}} + 2\zeta_1$: under the non-vacuous scalar regime $0 \le \zeta_1 \le 1$, the residual term $2\zeta_1$ is bounded by $2\zeta_1^{1/8}$.

By 10.5 and Lemma 3.21,

$$G_g^{\mathrm{A}} \otimes I \approx_{2\zeta_1} I \otimes G_g^{\mathrm{B}}.$$

Hence Lemma 3.27 gives

$$Q_g^{\mathrm{A}} \otimes I \approx_{\zeta_3} I \otimes Q_g^{\mathrm{B}},$$

where

$$\zeta_3 = 6\zeta_1 + 6\zeta_2.$$

Because both measurements are projective, Lemma 3.21 converts this to

$$Q_g^{\mathrm{A}} \otimes I \simeq_{\zeta_3/2} I \otimes Q_g^{\mathrm{B}}. \qquad (10.7)$$

By data processing,

$$Q_{[g(u)=a]}^{\mathrm{A}} \otimes I \simeq_{\zeta_3/2} I \otimes Q_{[g(u)=a]}^{\mathrm{B}}. \qquad (10.8)$$

Next, Lemma 3.20, applied to 10.5 and 10.6, implies

$$Q_g^{\mathrm{A}} \otimes I \simeq_{\zeta_1} I \otimes G_g^{\mathrm{B}}.$$

Data processing then gives

$$Q_{[g(u)=a]}^{\mathrm{A}} \otimes I \simeq_{\zeta_1} I \otimes G_{[g(u)=a]}^{\mathrm{B}}. \qquad (10.9)$$

Applying Lemma 3.29 to 10.4, 10.9, and 10.8 yields

$$A_a^{\mathrm{A},u} \otimes I \simeq_{\zeta_4} I \otimes Q_{[g(u)=a]}^{\mathrm{B}}, \qquad (10.10)$$

where

$$\zeta_4 = 2\sigma + 2\sqrt{\zeta_1 + \zeta_3/2}.$$

The same argument with the roles reversed gives

$$I \otimes A_a^{\mathrm{B},u} \simeq_{\zeta_4} Q_{[g(u)=a]}^{\mathrm{A}} \otimes I. \qquad (10.11)$$

Remark 10.29 records the Lean residual packages that separate the scalar cascade and the Step 5 Schwartz–Zippel handoff from the remaining projective-stage transport work. The scalar side discharges the paper's line 71–73 coarsening and the Step 8 hypotheses in the non-vacuous branch; the Step 5 wrapper now turns the evaluated estimate at line 116 into full-polynomial consistency at ζ_1. The remaining Lean residual is the completion-and-line-169 construction: supply the actual role-register measurement, Alice's line-146 completion estimate, Bob's left-register completion estimate, and the two exact polynomial-level ζ_1 links. Lean transports the Bob-side completion estimate to the line-147 right-register form, reconstructs the duplicated

Step 6 pre-projective consistency field from these inputs and `hpass`, then data-processes the line-169 links to line 172 and derives the two ζ_4 point goals and the $\zeta_3/2$ self-consistency target.

With E as in Definition 10.22, the quantities $\sigma, \zeta_1, \zeta_2, \zeta_3, \zeta_4$ form the cascade of Definition 10.24. Theorem 10.25 and Theorem 10.26 package the final inflation to the scale $100000k^2m^4E$. The underlying estimates, with the formal widened ζ_2, are:

$$\sigma \le 10000k^2m^4 \left(\varepsilon^{1/1024} + (d/q)^{1/1024} + e^{-k/(80000m^2)}\right).$$

Then

$$\zeta_1 \le 20204k^2m^4 \left(\varepsilon^{1/2048} + (d/q)^{1/2048} + e^{-k/(160000m^2)}\right),$$

and

$$\zeta_2 \le 2568km \left(\varepsilon^{1/16384} + (d/q)^{1/16384} + e^{-k/(1280000m^2)}\right).$$

Consequently,

$$\zeta_3 \le 150000k^2m^4 \left(\varepsilon^{1/16384} + (d/q)^{1/16384} + e^{-k/(1280000m^2)}\right),$$

and

$$\zeta_4 \le 40000k^2m^4 \left(\varepsilon^{1/32768} + (d/q)^{1/32768} + e^{-k/(2560000m^2)}\right).$$

Both $\zeta_3/2$ and ζ_4 are therefore bounded by

$$100000k^2m^4 \left(\varepsilon^{1/40000} + (d/q)^{1/40000} + e^{-k/(2560000m^2)}\right).$$

Hence 10.10, 10.11, and 10.7 give the three claimed conclusions, with Q^{A} and Q^{B} renamed as the theorem's G^{A} and G^{B}. □

Bibliography

[Sch80] Jacob T. Schwartz. Fast probabilistic algorithms for verification of polynomial identities. *Journal of the ACM*, 27(4):701–717, 1980.

[Zip79] Richard Zippel. Probabilistic algorithms for sparse polynomials. In *Proceedings of the 2nd International Symposium on Symbolic and Algebraic Manipulation*, pages 216–226, 1979.